

STATISTICS PAPER II · 2027 FORECAST

Statistical Inference — 2027 Forecast

405 forecast questions across Estimation and Hypothesis Testing, with exam shortcut and answer key

405

Forecast Questions

17

Topics Covered

FORECAST · AI-GENERATED

No problem is copied from a past paper. Every explanatory note is an AI-derived shortcut, independently generated and verified by the builder of this platform, not an official answer key.

Contents

Point Estimation	4 questions
Methods of Estimation	5 questions
Sufficiency and Completeness	5 questions
Optimal Estimation	6 questions
Interval Estimation	5 questions
Unbiasedness, Consistency, Efficiency, Sufficiency & Completeness — Foundations	40 questions
Cramér–Rao Lower Bound & Fisher Information	40 questions
Factorization Theorem, Sufficiency & Completeness — Worked Families	30 questions
Maximum Likelihood Estimation	50 questions
Rao–Blackwell, UMVUE & Method of Moments	40 questions
Confidence Intervals	50 questions
Hypothesis Testing	15 questions
Simple and Composite Hypotheses, Two Kinds of Error, Size and Power	1 questions
UMP, MLR, Unbiased and Similar Tests	1 questions
Sequential and Decision Theory	8 questions
Neyman–Pearson Lemma, UMP Tests & Power Functions	50 questions
Likelihood Ratio Tests, SPRT & Decision Theory	55 questions

Point Estimation

4 forecast questions in this topic

Estimation

Characteristics of a Good Estimator (Unbiasedness, Consistency, Efficiency, MSE)

Easy

Q1.

An estimator T_n of θ is said to be (weakly) consistent if

A. $\mathcal{P}(|T_n - \theta| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$, for every $\epsilon > 0$

B. $\text{Var}(T_n) = 0$ for every n

C. $E(T_n) = \theta$ for every n

D. $T_n = \theta$ exactly for all sufficiently large n

Correct answer: A

EXAM SHORTCUT

Consistency is convergence in probability of T_n to θ - a large-sample property, distinct from (and not implied by) unbiasedness.

Estimation

Characteristics of a Good Estimator (Unbiasedness, Consistency, Efficiency, MSE)

Easy

Q2.

For an estimator T of θ , the mean squared error $\text{MSE}(T) = E[(T - \theta)^2]$ decomposes as

A. $\text{Var}(T) - [\text{Bias}(T)]^2$

B. $\text{Var}(T)/\text{Bias}(T)$

C. $[\text{Var}(T)]^2 + \text{Bias}(T)$

D. $\text{Var}(T) + [\text{Bias}(T)]^2$

Correct answer: D

EXAM SHORTCUT

The bias-variance decomposition: $\text{MSE} = \text{Var}(T) + [\text{Bias}(T)]^2$.

Estimation

Characteristics of a Good Estimator (Unbiasedness, Consistency, Efficiency, MSE)

Medium

Q3.

Regarding minimization of MSE over ALL estimators (not just unbiased ones), which statement is correct?

A. The sample mean always has the smallest MSE among all estimators of the population mean

B. Minimizing MSE and requiring unbiasedness are equivalent conditions

C. The MLE always has the smallest MSE for every θ

D. **No single estimator achieves uniformly minimum MSE over all θ among all estimators - restricting to unbiased estimators is necessary for a meaningful 'best estimator' theory**

Correct answer: D

EXAM SHORTCUT

The trivial constant estimator $T \equiv \theta_0$ has $\text{MSE} = (\theta_0 - \theta)^2 = 0$ exactly at $\theta = \theta_0$, so no estimator can beat it everywhere - this is why UMVUE theory restricts attention to unbiased estimators.

Q4.

For iid observations from a standard Cauchy distribution, the sample mean $\bar{X}$ as an estimator of the location parameter

A. is inconsistent - $\bar{X}$ has exactly the same Cauchy distribution regardless of n

B. equals the maximum likelihood estimator

C. is unbiased with variance decreasing as $1/n$

D. is consistent and efficient

Correct answer: A

EXAM SHORTCUT

The Cauchy distribution has no finite mean or variance; remarkably, $\bar{X}$ for n iid standard Cauchy variables has EXACTLY the same standard Cauchy distribution as a single observation - it never concentrates.

Methods of Estimation

5 forecast questions in this topic

Estimation

Maximum Likelihood Estimation

Medium

Q1.

For $X_1, \dots, X_n$ iid $U(0, \theta)$, the maximum likelihood estimator of θ is

A. $X_{(1)}$, the smallest order statistic

B. $X_{(n)}$, the largest order statistic

C. $\frac{\bar{X}}{2}$

D. $\bar{X}$

Correct answer: B

EXAM SHORTCUT

Likelihood $= \theta^{-n}$ for $\theta \geq X_{(n)}$, strictly decreasing in θ - so the MLE is the smallest admissible θ , namely $X_{(n)}$.

Estimation

Maximum Likelihood Estimation

Hard

Q2.

For $X_1, \dots, X_n$ iid $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, the maximum likelihood estimator of θ

A. is uniquely $\bar{X}$

B. does not exist

C. ✓ is not unique - any value in $[X_{(n)} - \frac{1}{2}, X_{(1)} + \frac{1}{2}]$ maximizes the likelihood equally

D. is uniquely the midrange $(X_{(1)} + X_{(n)})/2$

Correct answer: C

EXAM SHORTCUT

The likelihood is constant ($=1$) for every θ satisfying $X_{(n)} - \frac{1}{2} \leq \theta \leq X_{(1)} + \frac{1}{2}$, so the MLE is non-unique - the midrange is merely one conventional choice among infinitely many.

Q3.

An iid sample from $U(a,b)$ has $\bar{X}=10$ and $\frac{1}{n} \sum (X_i - \bar{X})^2 = 12$. The method-of-moments estimates of a and b are

A. $a=0, b=20$ B. ~~$a=4, b=16$~~ C. $a=8, b=12$ D. $a=-2, b=22$

Correct answer: B

EXAM SHORTCUT

$a+b=2\bar{X}=20$; $b-a=2\sqrt{3 \times 12} = 2\sqrt{36} = 12$; solving gives $a=4, b=16$.

Q4.

For $X_1, \dots, X_n$ iid Exponential with rate θ (mean $1/\theta$), do the method-of-moments and maximum-likelihood estimators of θ coincide?

A. No, they differ by a factor of $n/(n-1)$ B. Only asymptotically, not for finite n C. No: MLE $= 1/\bar{X}$ but MoM $= \bar{X}$ D. ~~No~~ Yes, both equal $1/\bar{X}$

Correct answer: D

EXAM SHORTCUT

Both MLE and MoM for the exponential rate reduce to $\hat{\theta} = 1/\bar{X}$ - a rare case of exact coincidence for finite n .

Q5.

If $\hat{\theta}$ is the maximum likelihood estimator of θ , and g is any function, the MLE of $g(\theta)$ is

A. $g^{-1}(\hat{\theta})$ B. equal to $\hat{\theta}$ only if g is the identity functionC. undefined unless g is linearD. ~~$g(\hat{\theta})$~~ - the invariance property of MLEs

Correct answer: D

EXAM SHORTCUT

The MLE invariance property: for ANY function g (not just one-to-one ones, via the induced-likelihood extension), the MLE of $g(\theta)$ is simply $g(\hat{\theta})$.

Sufficiency and Completeness

5 forecast questions in this topic

Estimation

Sufficient Estimator and Factorization Theorem

Medium

Q1.

For $X_1, \dots, X_n$ iid with pdf $f(x, \theta) = e^{-(x-\theta)}$ for $x > \theta$ (shifted exponential), a sufficient statistic for θ is

A. $\sum x_i^2$

B. $X_{(1)}$, the minimum order statistic

C. $X_{(n)}$, the maximum order statistic

D. $\bar{X}$

Correct answer: B

EXAM SHORTCUT

Since the support $x > \theta$ depends on θ through the SMALLEST value needed, the binding constraint - and hence the sufficient statistic - is $X_{(1)}$.

Estimation

Sufficient Estimator and Factorization Theorem

Hard

Q2.

For X_1, X_2 iid $\text{Poisson}(\theta)$, why does $T = X_1 + 2X_2$ fail to be sufficient for θ , while $T' = X_1 + X_2$ succeeds?

A. ✓ The conditional distribution of (X_1, X_2) given $X_1 + X_2 = t$ is $\text{Binomial}(t, \frac{1}{2})$, free of θ ; the conditional distribution given $X_1 + 2X_2$ still involves θ

B. $X_1 + 2X_2$ need not be an integer

C. $X_1 + 2X_2$ has larger variance than $X_1 + X_2$

D. Sufficiency requires equal coefficients only for continuous distributions, not discrete ones

Correct answer: A

EXAM SHORTCUT

The joint pmf depends on the data only through $x_1 + x_2$ (the exponent of θ); a differently-weighted combination like $x_1 + 2x_2$ does not capture this and leaves θ in the conditional distribution.

Q3.

For $X \sim N(0, \theta)$ (mean known $=0$, θ the unknown variance), which statement is correct regarding completeness?

- A. X is complete for θ , but X^2 is not
- B. X alone is not complete for θ ; but X^2 is complete for θ
- C. Neither X nor X^2 is complete for θ
- D. Both X and X^2 are complete for θ

Correct answer: B

EXAM SHORTCUT

$g(X)=X$ has $E_\theta[X]=0$ for every θ without $X \equiv 0$ a.s., so X is not complete; but $X^2/\theta \sim \chi^2_1$ scaling makes X^2 complete.

Q4.

For iid observations from $U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, the range $R = X_{(n)} - X_{(1)}$ is

- A. ancillary - its distribution does not depend on θ at all
- B. distributed as $\text{Poisson}(\theta)$
- C. a complete sufficient statistic for θ
- D. an unbiased estimator of θ

Correct answer: A

EXAM SHORTCUT

Shifting θ shifts the whole sample by a constant, leaving the range unchanged in distribution - the textbook example of an ancillary statistic.

Q5.

For $X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$ with both parameters unknown, a complete sufficient statistic for (μ, σ^2) is

- A. $\bar{X}$ alone
- B. (sample median, sample range) jointly
- C. $(\bar{X}, S^2)$ jointly
- D. S^2 alone

Correct answer: C

EXAM SHORTCUT

The two-parameter normal family has $(\bar{X}, S^2)$ (equivalently $(\sum X_i, \sum X_i^2)$) as its complete sufficient statistic - neither component alone suffices for both parameters.

Optimal Estimation

6 forecast questions in this topic

Estimation

Rao-Blackwell Theorem and UMVUE Construction

Medium

Q1.

If T is sufficient for θ and U is any unbiased estimator of θ , then $\varphi(T) = E[U | T]$ satisfies

- A. $\varphi(T)$ always equals U
- B. $\varphi(T)$ is free of θ , unbiased for θ , with $\text{Var}(\varphi(T)) \leq \text{Var}(U)$ for every θ
- C. $\varphi(T)$ may still depend on θ
- D. $\varphi(T)$ is unbiased but always has strictly larger variance than U

Correct answer: B

EXAM SHORTCUT

This is exactly the Rao-Blackwell theorem: conditioning on a sufficient statistic never increases variance, and sufficiency guarantees the conditional expectation is a genuine (data-only) statistic.

Estimation

Rao-Blackwell Theorem and UMVUE Construction

Medium

Q2.

Rao-Blackwellizing an unbiased estimator using a statistic T produces the UMVUE if and only if T is

- A. complete only
- B. ancillary
- C. sufficient only
- D. ~~C~~complete and sufficient

Correct answer: D

EXAM SHORTCUT

Lehmann-Scheffe: completeness (in addition to sufficiency) is exactly what makes the Rao-Blackwellized estimator unique, and hence the UMVUE.

Estimation

Cramer-Rao Inequality, MVB and Fisher Information

Medium

Q3.

For $X \sim \text{Poisson}(\theta)$, based on a sample of size n , the Cramer-Rao lower bound for the variance of any unbiased estimator of θ is

- A. θ^2/n
- B. $1/(n\theta^2)$
- C. n/θ
- D. ~~C~~ θ/n

Correct answer: D

EXAM SHORTCUT

Single-observation Fisher information for Poisson is $I(\theta) = 1/\theta$; for n iid observations, $I_n(\theta) = n/\theta$, so CRLB $= \theta/n$.

Q4.

For a single observation from $N(\mu, \theta)$ with μ known and $\theta = \sigma^2$, the Fisher information $I(\theta)$ equals

- A. $1/\theta^2$
- B. $1/(2\theta^2)$
- C. $n/(2\theta^2)$
- D. $2\theta^2$

Correct answer: B

EXAM SHORTCUT

$\log f = -\frac{1}{2} \log(2\pi\theta) - (x-\mu)^2/(2\theta)$; differentiating twice and taking $E[\cdot]$ gives $I(\theta) = 1/(2\theta^2)$ for one observation.

Q5.

The Bhattacharya bound generalizes the Cramer-Rao bound by

- A. applying only to biased estimators
- B. ignoring the likelihood function entirely
- C. Incorporating higher-order derivatives of the log-likelihood to give a bound at least as sharp (tight) as the CRLB
- D. requiring the estimator to be sufficient

Correct answer: C

EXAM SHORTCUT

Bhattacharya bounds use second- and higher-order score derivatives, giving a sequence of bounds that are never looser than, and often strictly sharper than, the first-order Cramer-Rao bound.

Q6.

For $X_1, \dots, X_n$ iid $\text{Poisson}(\theta)$, with $T = \sum X_i$ complete sufficient, the UMVUE of θ^2 is

- A. $T(T-1)/n^2$
- B. T^2/n^2
- C. $T(T+1)/n^2$
- D. $T(T-1)/n$

Correct answer: A

EXAM SHORTCUT

Using the Poisson factorial-moment identity $E[T(T-1)] = (n\theta)^2$, dividing by n^2 gives an exactly unbiased estimator of θ^2 built from the complete sufficient T .

Interval Estimation

5 forecast questions in this topic

Estimation

Confidence Intervals, Pivots and Optimum Bounds

Medium

Q1.

For $X_1, \dots, X_n$ iid Exponential with mean θ , a pivotal quantity for constructing a confidence interval for θ is

A. $\bar{X} - \theta \sim \chi^2_1$

B. $n\bar{X}^2/\theta \sim F$

C. $2n\bar{X}/\theta \sim \chi^2_{2n}$

D. $\bar{X}/\theta \sim N(0,1)$

Correct answer: C

EXAM SHORTCUT

Sums of exponential variables relate to the gamma/chi-square family: $2n\bar{X}/\theta \sim \chi^2_{2n}$ exactly, for every n .

Estimation

Confidence Intervals, Pivots and Optimum Bounds

Medium

Q2.

A 95% confidence interval for μ (with $\sigma=4$ known), based on a sample of size n , is reported as (18.04, 21.96).

The sample mean and sample size are

A. $\bar{X}=20, n=16$

B. $\bar{X}=20, n=64$

C. $\bar{X}=20, n=4$

D. $\bar{X}=19, n=16$

Correct answer: A

EXAM SHORTCUT

Midpoint $= 20 = \bar{X}$; half-width $= 1.96 = 1.96(4)/\sqrt{n} \Rightarrow \sqrt{n} = 4 \Rightarrow n = 16$.

Estimation

Resampling: Bootstrap and Jackknife

Easy

Q3.

The bootstrap method estimates the sampling distribution of a statistic by

A. assuming normality and computing exact moments

B. requiring a known parametric family for the population

C. **resampling with replacement from the observed data**

D. resampling without replacement, taking exactly half the data each time

Correct answer: C

EXAM SHORTCUT

Bootstrap = resample WITH replacement from the empirical distribution (the observed sample) to approximate the true sampling distribution non-parametrically.

Q4.

The jackknife estimator of the bias of a statistic T_n is based on

- A. bootstrap resampling with replacement
- B. ~~leave-one-out~~ recomputation of the statistic**
- C. a single random split of the data into two halves
- D. repeating the full-sample estimate n times

Correct answer: B

EXAM SHORTCUT

Jackknife systematically deletes one observation at a time, recomputes $T_{(i)}$ each time, and uses the spread of these leave-one-out values to estimate bias (and variance).

Q5.

To halve the width of a 95% confidence interval for μ (with σ known), the sample size must be

- A. doubled
- B. increased by 50%
- C. ~~quadrupled~~**
- D. halved

Correct answer: C

EXAM SHORTCUT

CI width $\propto 1/\sqrt{n}$; halving the width requires $\sqrt{n}$ to double, i.e. n to quadruple.

Unbiasedness, Consistency, Efficiency, Sufficiency & Completeness — Foundations

40 forecast questions in this topic

Estimation

Unbiasedness

Medium

Q1.

An estimator $\hat{\theta}$ is unbiased if:

A. $MSE(\theta) = 0$

B. $E[\hat{\theta}] = \theta$ for all θ

C. $Var(\theta) = 0$

D. $\hat{\theta} \rightarrow \theta$ in probability

Correct answer: B

EXAM SHORTCUT

D is consistency, not unbiasedness; A implies both bias=0 AND var=0 simultaneously

Estimation

MSE Decomposition

Medium

Q2.

$MSE(\hat{\theta})$ equals:

A. $Var(\hat{\theta})$ only

B. $[Bias(\hat{\theta})]^2$ only

C. $Var(\hat{\theta}) + [Bias(\hat{\theta})]^2$

D. $Var(\hat{\theta}) - [Bias(\hat{\theta})]^2$

Correct answer: C

EXAM SHORTCUT

A holds only when estimator is unbiased; never subtract bias²

Estimation

Consistency

Medium

Q3.

$\hat{\theta}_n$ is consistent for θ if:

A. $\hat{\theta}_n \rightarrow \theta$ almost surely

B. $E[\hat{\theta}_n] \rightarrow \theta$ and $Var(\hat{\theta}_n) \rightarrow 0$

C. Both A and B

D. $P(|\hat{\theta}_n - \theta| > \epsilon) \rightarrow 0$ for every $\epsilon > 0$

Correct answer: D

EXAM SHORTCUT

B is sufficient (by Chebyshev) but not necessary; A is strong consistency (stricter)

Estimation

Efficiency

Medium

Q4.

The relative efficiency of $\hat{\theta}_1$ w.r.t. $\hat{\theta}_2$ is:

- A. $\text{Bias}(\theta_1)/\text{Bias}(\theta_2)$
- B. $\text{MSE}(\theta_1)/\text{MSE}(\theta_2)$
- C. $\text{Var}(\theta_1)/\text{Var}(\theta_2)$
- D. $\text{Var}(\theta_2)/\text{Var}(\theta_1)$

Correct answer: D

EXAM SHORTCUT

Efficiency of $\hat{\theta}_1 = \text{CRLB}/\text{Var}(\theta_1)$; relative efficiency = Var of denominator over numerator

Estimation

Sufficiency Definition

Medium

Q5.

A statistic $T(X)$ is sufficient for θ if:

- A. The conditional distribution of X given T does not depend on θ
- B. T has minimum variance among all unbiased estimators
- C. T attains the Cramér-Rao lower bound
- D. T is a complete statistic

Correct answer: A

EXAM SHORTCUT

Sufficiency is about conditional distribution, not variance; completeness is a separate concept

Estimation

Admissibility

Hard

Q6.

An estimator $\hat{\theta}$ is inadmissible if:

- A. There exists another estimator with smaller MSE for ALL θ
- B. It is not a function of sufficient statistic
- C. It is biased
- D. It does not attain the CRLB

Correct answer: A

EXAM SHORTCUT

Biased estimators can be admissible (e.g., James-Stein); CRLB non-attainment $\neq$ inadmissibility

Estimation

Ancillary Stat

Hard

Q7.

A statistic $A(X)$ is ancillary if:

- A. $E[A(X)] = 0$ for all θ
- B. It is independent of the sufficient statistic
- C. **Its distribution does not depend on θ**
- D. It has minimum variance

Correct answer: C

EXAM SHORTCUT

B is Basu's theorem consequence (ancillary $\perp$ complete sufficient), not the definition

Estimation

Completeness Def

Hard

Q8.

A statistic T is complete if:

- A. **$E_\theta[g(T)] = 0$ for all θ implies $g(T) = 0$ a.s.**
- B. T achieves minimum variance
- C. The family of distributions of T is full-rank exponential
- D. T is sufficient and unbiased

Correct answer: A

EXAM SHORTCUT

C is sufficient for completeness (exponential family with full rank) but not the definition itself

Estimation

Consistent Unbiased

Medium

Q9.

Sample mean $\bar{X}$ from $N(\mu, \sigma^2)$ is:

- A. **Unbiased and consistent for μ**
- B. Unbiased but inconsistent for μ
- C. Consistent but biased for μ
- D. Neither unbiased nor consistent

Correct answer: A

EXAM SHORTCUT

$E[\bar{X}] = \mu$ (unbiased); $\text{Var}(\bar{X}) = \sigma^2/n \rightarrow 0$ (consistent)

Estimation

Best Linear Unbiased

Medium

Q10.

By Gauss-Markov theorem, OLS estimator is BLUE meaning:

- A. Best (minimum variance) among all unbiased estimators
- B. Best among all estimators including biased ones
- C. Minimum MSE among linear estimators
- D. **Best (minimum variance) among all LINEAR unbiased estimators**

Correct answer: D

EXAM SHORTCUT

“Linear” is crucial; MLE may be better overall; Gauss-Markov does NOT require normality

Estimation

Asymptotic Efficiency

Hard

Q11.

An estimator is asymptotically efficient if:

- A. $\text{Var}(\hat{\theta}) = 1/(nI(\theta))$ exactly
- B. $\hat{\theta}$ is consistent
- C. $\hat{\theta}$ is unbiased for all n
- D. **$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1/I(\theta))$**

Correct answer: D

EXAM SHORTCUT

A is exact CRLB attainment (finite n); D is the asymptotic version; B alone is insufficient

Estimation

Pitman Closeness

Hard

Q12.

Pitman closeness criterion prefers $\hat{\theta}_1$ over $\hat{\theta}_2$ if:

- A. $\text{Bias}(\hat{\theta}_1) < \text{Bias}(\hat{\theta}_2)$
- B. $\text{MSE}(\hat{\theta}_1) < \text{MSE}(\hat{\theta}_2)$
- C. $\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2)$
- D. **$P(|\hat{\theta}_1 - \theta| < |\hat{\theta}_2 - \theta|) > 1/2$**

Correct answer: D

EXAM SHORTCUT

Pitman closeness can contradict MSE ordering — classic ISS trap

Q13.

If $\hat{\theta}$ is MLE of θ , then MLE of $g(\theta)$ is:

- A. $g(\theta)$ only if g is monotone
- B. Obtained by separate maximization
- C. $g(\theta)$ always
- D. $g(\theta)$ only if g is differentiable

Correct answer: C

EXAM SHORTCUT

Invariance of MLE holds for ANY function g , not just monotone or differentiable ones

Q14.

If a complete sufficient statistic exists, then:

- A. All unbiased estimators are functions of it
- B. There is at most one unbiased estimator based on it
- C. There is exactly one UMVUE
- D. The MLE is always unbiased

Correct answer: C

EXAM SHORTCUT

B is wrong (there can be many unbiased estimators); C follows from Lehmann- Scheffé theorem

Q15.

For i.i.d. sample of size n , $l(\theta)$ equals:

- A. $l_1(\theta)/n$
- B. $\sqrt{n} \cdot l_1(\theta)$
- C. $n \cdot l_1(\theta)$
- D. $l_1(\theta)$

Correct answer: C

EXAM SHORTCUT

Fisher information is ADDITIVE for independent observations; $\text{CRLB} = 1/(n l_1(\theta))$

Q16.

An estimator $\hat{\theta}$ is median-unbiased if:

A. $P(\hat{\theta} \leq \theta) = P(\hat{\theta} \geq \theta) = 1/2$

B. $\hat{\theta}$ has symmetric distribution

C. $E[\hat{\theta}] = \theta$

D. Median of $\hat{\theta}$ equals $E[\hat{\theta}]$

Correct answer: A

EXAM SHORTCUT

Mean-unbiasedness $\neq$ median-unbiasedness; for skewed distributions they differ

Q17.

CRLB requires regularity conditions including:

A. The distribution must be normal

B. The sample size must be large

C. θ must be a location parameter

D. **Support of $f(x; \theta)$ must NOT depend on θ**

Correct answer: D

EXAM SHORTCUT

For $U(0, \theta)$, $\text{Exp}(\theta)$ shifted — support depends on θ , so CRLB does NOT apply

Q18.

A minimal sufficient statistic is:

A. **Both A and B**

B. A sufficient statistic of smallest possible dimension

C. A function of every other sufficient statistic

D. The complete sufficient statistic

Correct answer: A

EXAM SHORTCUT

Minimal sufficient = coarsest sufficient partition; complete sufficient $\subseteq$ minimal sufficient (when it exists)

Q19.

A location equivariant estimator $\hat{\theta}$ satisfies:

- A. $E[\hat{\theta}] = \theta$
- B. $\hat{\theta}(cX) = \hat{\theta}(X)$ for all $c > 0$
- C. $\hat{\theta}(X + c) = \hat{\theta}(X) + c$ for all c
- D. $\hat{\theta}$ is a function of order statistics

Correct answer: C

EXAM SHORTCUT

B is scale equivariance; both together give affine equivariance

Q20.

If $\hat{\theta}_n \xrightarrow{p} \theta$ and g is continuous, then:

- A. $E[g(\hat{\theta}_n)] \rightarrow g(\theta)$
- B. $g(\hat{\theta}_n)$ may not be consistent
- C. $g(\hat{\theta}_n) \xrightarrow{p} g(\theta)$ only if g is monotone
- D. $g(\hat{\theta}_n) \xrightarrow{p} g(\theta)$

Correct answer: D

EXAM SHORTCUT

Continuous mapping theorem guarantees D; A requires uniform integrability (stronger condition)

Q21.

Shrinkage estimators like James-Stein are preferred because:

- A. They satisfy CRLB
- B. They are unbiased with smaller variance
- C. They introduce bias but reduce variance enough to lower total MSE
- D. They are consistent while OLS is not

Correct answer: C

EXAM SHORTCUT

James-Stein is BIASED; its advantage is total MSE reduction in $p \geq 3$ dimensions

Q22.

PDF of k -th order statistic $X(k)$ from $f(x)$ with CDF $F(x)$ is:

- A. $n! k!(n-k)! [F(x)]^k [1 - F(x)]^{n-k} f(x)$
- B. $n [1 - F(x)]^{n-1} f(x)$
- C. $n! (k-1)!(n-k)! [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x)$
- D. $n [F(x)]^{n-1} f(x)$

Correct answer: C

EXAM SHORTCUT

D is PDF of maximum ($k = n$); B is PDF of minimum ($k = 1$); check factorial carefully

Q23.

For a full-rank exponential family $f(x; \theta) = h(x) \exp[\eta(\theta)T(x) - B(\theta)]$:

- A. $T(x)$ is sufficient but not necessarily complete
- B. Completeness depends on sample size
- C. $T(x)$ is complete sufficient
- D. $T(x)$ is complete but not sufficient

Correct answer: C

EXAM SHORTCUT

Full-rank (open set in natural parameter space) is the key condition; curved exponential families may lack completeness

Q24.

By Basu's theorem, if T is complete sufficient and A is ancillary, then:

- A. T and A are independent
- B. A is a function of T
- C. T and A have the same distribution
- D. A is also sufficient

Correct answer: A

EXAM SHORTCUT

This is Basu's theorem — independence, not functional relationship; requires completeness

Estimation

Unbiased Estimator Existence

Hard

Q25.

For $U(0, \theta)$, an unbiased estimator of θ^{-1} based on $X(n)$:

A. ~~Is~~ $(n-1)/(n \cdot X_{(n)})$

B. Is $X_{(n)}$

C. Is $((n+1)/n) \cdot X_{(n)}$

D. Does not exist

Correct answer: A

EXAM SHORTCUT

$E[X_{(n)}] = n\theta/(n+1)$; by direct integration $E[1/X_{(n)}] = n/((n-1)\theta)$ for $n > 1$, so $(n-1)/(n \cdot X_{(n)})$ is unbiased for $1/\theta$.

Estimation

Lehmann Scheffe

Hard

Q26.

Lehmann-Scheffé theorem states:

A. Complete sufficient statistic is always MLE

B. Rao-Blackwell improvement always yields UMVUE

C. ~~Any unbiased estimator based on complete sufficient statistic is~~ UMVUE

D. UMVUE always attains CRLB

Correct answer: C

EXAM SHORTCUT

D is false (UMVUE may not attain CRLB); B requires completeness for uniqueness

Estimation

Fisher Consistency

Hard

Q27.

An estimator T_n is Fisher consistent if:

A. $T_n p \rightarrow \theta$ as $n \rightarrow \infty$

B. ~~The functional~~ $T(F\theta) = \theta$ where $F\theta$ is the true CDF

C. $E[T_n] = \theta$ for all n

D. T_n is a function of sufficient statistic

Correct answer: B

EXAM SHORTCUT

Fisher consistency is a property of the functional form, not a sequence property; different from ordinary consistency

Estimation

Superefficiency

Hard

Q28.

Hodges' superefficient estimator demonstrates:

- A. **An estimator can have asymptotic variance below CRLB at isolated points**
- B. CRLB can be beaten uniformly
- C. Superefficiency implies admissibility
- D. MLE is never asymptotically efficient

Correct answer: A

EXAM SHORTCUT

Superefficiency occurs only on a set of measure zero; Le Cam showed it cannot hold on a dense set

Estimation

Stein Paradox

Hard

Q29.

James-Stein estimator dominates sample mean in terms of MSE when:

- A. $p \geq 1$ (any dimension)
- B. **$p \geq 3$**
- C. $p \geq 2$
- D. $p \geq 4$

Correct answer: B

EXAM SHORTCUT

For $p = 1, 2$ sample mean is admissible; Stein paradox kicks in at $p \geq 3$

Estimation

Moment Estimator Consistency

Medium

Q30.

Method of moments estimators are generally:

- A. Consistent and efficient
- B. Neither consistent nor efficient
- C. Efficient but not consistent
- D. **Consistent but not necessarily efficient**

Correct answer: D

EXAM SHORTCUT

MOM estimators are consistent (by LLN) but typically less efficient than MLE

Q31.

For multiparameter case, CRLB for $g(\theta)$ involves:

- A. **Fisher information matrix $I(\theta)$ and gradient ∇g**
- B. Scalar Fisher information
- C. Only diagonal elements of $I(\theta)$
- D. Determinant of $I(\theta)$

Correct answer: A

EXAM SHORTCUT

$\text{Var}(g) \geq [\nabla g]^T I(\theta)^{-1} [\nabla g]$; off-diagonal elements matter

Q32.

For truncated normal $N(\mu, 1)$ truncated to $(0, \infty)$, MLE of μ :

- A. Is $\bar{X}$ (same as untruncated)
- B. Is the sample median
- C. **Requires solving a nonlinear equation**
- D. Does not exist

Correct answer: C

EXAM SHORTCUT

Truncation changes the likelihood; MLE satisfies $\bar{X} = \mu + \phi(\mu)/\Phi(\mu)$ (Mills ratio equation)

Q33.

The sample median is preferred over sample mean when:

- A. Distribution is symmetric
- B. **Distribution has heavy tails or outliers**
- C. Sample size is large
- D. Distribution is normal

Correct answer: B

EXAM SHORTCUT

For normal distribution, mean is more efficient; median's advantage is robustness to outliers

Estimation

Unbiased Variance Estimator

Medium

Q34.

$S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ is unbiased for σ^2 because:

A. $\bar{X}$ is independent of S^2 under normality

B. $E[\sum (X_i - \bar{X})^2] = (n-1)\sigma^2$

C. $E[\sum (X_i - \bar{X})^2] = n\sigma^2$

D. Both A and C

Correct answer: B

EXAM SHORTCUT

The $(n-1)$ denominator corrects for one degree of freedom lost by estimating μ with $\bar{X}$

Estimation

Jackknife Bias

Medium

Q35.

The jackknife bias estimate for $\hat{\theta}$ is:

A. $n(\hat{\theta} - \theta(\cdot))$

B. $\hat{\theta} - \theta(\cdot)$

C. $(n-1)(\hat{\theta}(\cdot) - \hat{\theta})$

D. $(n-1)(\hat{\theta} - \theta(\cdot))$

Correct answer: C

EXAM SHORTCUT

$\hat{\theta}(\cdot) = (1/n) \sum \hat{\theta}_i$ is mean of leave-one-out estimates; bias = $(n-1)(\hat{\theta}(\cdot) - \hat{\theta})$

Estimation

Bootstrap Principle

Medium

Q36.

Bootstrap estimates sampling distribution of $\hat{\theta}$ by:

A. Generating data from fitted parametric model only

B. **Resampling WITH replacement from observed data**

C. Analytical approximation using CLT

D. Resampling WITHOUT replacement from observed data

Correct answer: B

EXAM SHORTCUT

With replacement is key; without replacement is subsampling (different method); parametric bootstrap uses fitted model

Q37.

By delta method, asymptotic variance of $g(\bar{X})$ is:

A. $\sqrt{[g'(\mu)]^2 \cdot \sigma^2/n}$

B. $[g''(\mu)]^2 \cdot \sigma^2/n$

C. $[g(\mu)]^2 \cdot \sigma^2/n$

D. $g'(\mu) \cdot \sigma^2/n$

Correct answer: A

EXAM SHORTCUT

First derivative squared, not second; applies when $g'(\mu) \neq 0$

Q38.

Variance stabilizing transformation for $\text{Poisson}(\lambda)$ is:

A. $\log X$

B. $\sqrt{X}$

C. X^2

D. $1/X$

Correct answer: B

EXAM SHORTCUT

For Poisson, $\text{Var}(X) = \lambda = E[X]$; VST satisfies $g'(\mu) \propto 1/\sqrt{\mu}$, giving $g(\mu) = 2\sqrt{\mu}$; $\sqrt{X}$ is the answer

Q39.

For location family $f(x-\theta)$, which is ancillary?

A. Sample maximum $X(n)$

B. Sample variance S^2 (when θ is location only)

C. Sample range $X(n) - X(1)$

D. Sample mean $\bar{X}$

Correct answer: C

EXAM SHORTCUT

Range and all spacings are ancillary for location families; S^2 is ancillary only if scale is known

Q40.

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, $T = \sum X_i$ is complete because:

- A.** $E_p[g(T)] = 0$ for all $p \in (0, 1)$ implies $g \equiv 0$ (polynomial argument)
- B.** It is sufficient
- C.** It has binomial distribution
- D.** It is the MLE

Correct answer: A

EXAM SHORTCUT

Completeness proof: $E[g(T)] = \sum_{t=0}^n g(t) \binom{n}{t} p^t (1-p)^{n-t} = 0$ for all p implies all $g(t) = 0$.

Cramér–Rao Lower Bound & Fisher Information

40 forecast questions in this topic

Estimation

CRLB Formula

Medium

Q1.

For unbiased estimator $\hat{\theta}$, CRLB states:

A. $\text{Var}(\hat{\theta}) \geq n I(\theta)$

B. $\text{Var}(\hat{\theta}) \geq n I(\theta)$

C. $\text{Var}(\hat{\theta}) \geq I(\theta)$

D. $\text{Var}(\hat{\theta}) \geq I(\theta)$

Correct answer: A

EXAM SHORTCUT

For i.i.d. sample of size n : CRLB = $1/(n I(\theta))$ where I is single-observation Fisher info

Estimation

Fisher Info Normal

Medium

Q2.

For $X \sim N(\mu, \sigma^2)$ with σ^2 known, $I(\mu)$ equals:

A. $2/\sigma^4$

B. $1/(2\sigma^2)$

C. $1/\sigma^2$

D. σ^2

Correct answer: C

EXAM SHORTCUT

$\log f = -((x-\mu)^2)/(2\sigma^2) + C$; $\partial^2 \log f / \partial \mu^2 = -1/\sigma^2$; $I(\mu) = 1/\sigma^2$

Estimation

Fisher Info Bernoulli

Medium

Q3.

For $X \sim \text{Bernoulli}(p)$, $I(p)$ equals:

A. $1/p + 1/(1-p)$

B. $1/[p(1-p)]$

C. $p(1-p)$

D. $p/(1-p)$

Correct answer: B

EXAM SHORTCUT

$I(p) = E[(\partial \log f / \partial p)^2] = 1/[p(1-p)]$; CRLB for X is $p(1-p)/n$ — verify: $\text{Var}(X) = p(1-p)/n$ ✓ attains CRLB

Q4.

For $X \sim \text{Poisson}(\lambda)$, $I(\lambda)$ equals:

- A. $1/\lambda^2$
- B. $1/\lambda$
- C. λ^2
- D. λ

Correct answer: B

EXAM SHORTCUT

 $\log f = x \log \lambda - \lambda - \log(x!); \partial^2 \log f / \partial \lambda^2 = -x/\lambda^2; I(\lambda) = E[X]/\lambda^2 = 1/\lambda$

Q5.

CRLB for estimating $g(\theta)$ (unbiased) is:

- A. $[g'(\theta)]^2 I(\theta)$
- B. $(g'(\theta))/(nI(\theta))$
- C. $[g(\theta)]^2 nI(\theta)$
- D. $[g'(\theta)]^2 nI(\theta)$

Correct answer: D

EXAM SHORTCUT

Chain rule applies: $[g'(\theta)]^2$ in numerator; A forgets the n factor

Q6.

CRLB is attained if and only if:

- A. $\partial \log f / \partial \theta = A(\theta)[T(x) - g(\theta)]$ for some functions A, g
- B. The sample size is large
- C. The estimator is MLE
- D. The distribution is normal

Correct answer: A

EXAM SHORTCUT

This is the exact attainment condition; it implies $T(x)$ is UMVUE for $g(\theta)$; only exponential family members typically satisfy this

Estimation

CRLB Exponential

Medium

Q7.

For $X \sim \text{Exp}(\theta)$ (rate θ), CRLB for $\hat{\theta}$ (unbiased) is:

- A. n/θ^2
- B. θ/n
- C. $1/(n\theta^2)$
- D. θ^2/n

Correct answer: D

EXAM SHORTCUT

$l(\theta) = 1/\theta^2$; CRLB = θ^2/n ; MLE $1/\hat{X}$ attains this bound

Estimation

CRLB Uniform DNE

Hard

Q8.

For $X \sim U(0, \theta)$, CRLB for θ :

- A. Equals $4\theta^2/n$
- B. Equals $\theta^2/(3n)$
- C. Does not exist
- D. Equals θ^2/n

Correct answer: C

EXAM SHORTCUT

Support $(0, \theta)$ depends on θ ; regularity condition violated; CRLB framework inapplicable; UMVUE variance is $\theta^2/[n(n+2)]$

Estimation

Information Inequality Biased

Hard

Q9.

For biased estimator with bias $b(\theta)$, the generalized CRLB is:

- A. $\text{Var}(\hat{\theta}) \geq ([b'(\theta)]^2)/(nI(\theta))$
- B. $\text{MSE}(\hat{\theta}) \geq nI(\theta)$
- C. $\text{Var}(\hat{\theta}) \geq ([1+b'(\theta)]^2)/(nI(\theta))$
- D. $\text{Var}(\hat{\theta}) \geq nI(\theta)$

Correct answer: C

EXAM SHORTCUT

$[1 + b'(\theta)]^2$ not $[b'(\theta)]^2$; when $b(\theta) = 0$ (unbiased), reduces to standard CRLB

Estimation

Bhattacharya Bound

Hard

Q10.

Bhattacharya bounds improve upon CRLB by:

- A. Using larger sample sizes
- B. Restricting to exponential families
- C. Using sufficient statistics
- D. **Using higher-order score functions (derivatives of log-likelihood)**

Correct answer: D

EXAM SHORTCUT

Bhattacharya uses $\partial^k \log f / \partial \theta^k$ for $k = 1, 2, \dots$; first-order Bhattacharya = CRLB; higher orders give tighter bounds

Estimation

Bhattacharya Normal Theta Sq

Hard

Q11.

For $X \sim N(\theta, 1)$, estimating θ , Bhattacharya bound (2nd order) is:

- A. $2/n$
- B. **$4\theta^2/n + 2/n^2$**
- C. $4\theta^2/n - 2/n^2$
- D. $4\theta^2/n$

Correct answer: B

EXAM SHORTCUT

CRLB gives $4\theta^2/n$; 2nd-order Bhattacharya adds $2/n^2$ term; UMVUE $\bar{X}^2 - 1/n$ has variance $4\theta^2/n + 2/n^2$ — attains 2nd-order bound

Estimation

Fisher Info Gamma

Hard

Q12.

For $X \sim \Gamma(\alpha, \beta)$ with α known, $I(\beta)$ equals:

- A. β^2/α
- B. **α/β^2**
- C. α^2/β^2
- D. $1/(\alpha\beta^2)$

Correct answer: B

EXAM SHORTCUT

$E[X] = \alpha\beta$, $\text{Var}(X) = \alpha\beta^2$; $I(\beta) = \alpha/\beta^2$; CRLB for $\hat{\beta} = X/\alpha$ is $\beta^2/(n\alpha)$ ✓

Q13.

For location family $f(x-\theta)$, Fisher information is:

- A. $I(\theta) = \int [(f'(x))/(f(x))]^2 f(x) dx$ (independent of θ)
- B. $I(\theta)$ depends on θ
- C. $I(\theta) = \theta^2 \int [(f'(x))/(f(x))]^2 f(x) dx$
- D. $I(\theta) = 1/\sigma^2$ always

Correct answer: A

EXAM SHORTCUT

For location families, Fisher information is constant (does not depend on θ); this is a key property

Q14.

For scale family $(1/(\theta f))(x/\theta)$, $I(\theta)$ equals:

- A. θ^2 always
- B. $1/\theta^2$ times location Fisher information
- C. $\theta^2 \int [(f'(x))/(f(x))]^2 f(x) dx$
- D. $\theta^2 \int [1 + (xf'(x))/(f(x))]^2 f(x) dx$

Correct answer: D

EXAM SHORTCUT

Score for scale family: $\partial \log f / \partial \theta = -1/\theta - (x/\theta^2) f'(x/\theta)/f(x/\theta)$; careful substitution needed

Q15.

An unbiased estimator is efficient if:

- A. Its variance equals CRLB
- B. Both A and B (they are equivalent when CRLB is attained)
- C. Its variance is less than all other unbiased estimators
- D. It is consistent

Correct answer: B

EXAM SHORTCUT

When CRLB is attained, the estimator is UMVUE; A and C are equivalent in this context

Estimation

CRLB Cauchy

Hard

Q16.For $X \sim \text{Cauchy}(\theta, 1)$, $I(\theta)$ equals:

- A. Does not exist
- B. 2
- C. $1/2$
- D. 1

Correct answer: C**EXAM SHORTCUT**

$f(x; \theta) = \frac{1}{\pi[1+(x-\theta)^2]}$; $I(\theta) = \int \frac{4(x-\theta)^2}{\pi^2[1+(x-\theta)^2]^4} \cdot \pi[1+(x-\theta)^2] dx = 1/2$; sample mean is inconsistent for Cauchy

Estimation

Information Matrix Bivariate

Hard

Q17.For $N(\mu, \sigma^2)$ with both parameters unknown, Fisher information matrix is:

- A. $\text{diag}(1/\sigma^2, 2\sigma^4)$
- B. $\frac{1}{n} \text{diag}(1/\sigma^2, 1/(2\sigma^4))$
- C. $\text{diag}(n/\sigma^2, n/(2\sigma^4))$
- D. Non-diagonal matrix

Correct answer: B**EXAM SHORTCUT**

Off-diagonal is zero (score functions are uncorrelated); single-observation matrix; multiply by n for sample

Estimation

CRLB Beta Distribution

Hard

Q18.For $X \sim \text{Beta}(\theta, 1)$, $f(x; \theta) = \theta x^{\theta-1}$, $I(\theta)$ equals:

- A. θ
- B. θ^2
- C. $1/\theta$
- D. $1/\theta^2$

Correct answer: D**EXAM SHORTCUT**

$\log f = \log \theta + (\theta-1) \log x$; $\partial \log f / \partial \theta = 1/\theta + \log x$; $I(\theta) = \text{Var}(1/\theta + \log X) = \text{Var}(\log X) = 1/\theta^2$ (since $-\log X \sim \text{Exp}(\theta)$)

Q19.

For $X \sim \text{Exp}(\theta)$ truncated to $(0, c)$, CRLB:

- A. Is larger than θ^2/n (less information)
- B. Does not exist
- C. **Is smaller than θ^2/n (more information)**
- D. Is same as untruncated: θ^2/n

Correct answer: C

EXAM SHORTCUT

Truncation adds information (we know $X < c$); Fisher information increases; CRLB decreases

Q20.

The first-order Bhattacharya bound coincides with:

- A. **CRLB**
- B. Rao-Blackwell bound
- C. Chapman-Robbins bound
- D. Hammersley-Chapman-Robbins bound

Correct answer: A

EXAM SHORTCUT

Bhattacharya hierarchy: 1st order = CRLB; higher orders give tighter bounds; Chapman-Robbins is a different (non-differentiability-requiring) bound

Q21.

For $X \sim \text{Geometric}(p)$, $P(X = k) = (1-p)^{k-1}p$, $I(p)$ equals:

- A. **$-p^2(1-p)$**
- B. $p(1-p)$
- C. p^2
- D. $(1-p)^2$

Correct answer: A

EXAM SHORTCUT

$E[X] = 1/p$; $\text{Var}(X) = (1-p)/p^2$; $I(p) = 1/[p^2(1-p)]$; CRLB for $\hat{p} = 1/X$ is $p^2(1-p)/n$

Q22.

For $X \sim \text{NegBin}(r, p)$ with r known, $l(p)$ equals:

- A. $p^2(1-p)$
- B. $r^2 p^2(1-p)$
- C. $r p(1-p)$
- D. $\checkmark p^2(1-p)$

Correct answer: D

EXAM SHORTCUT

NegBin is exponential family; $l(p) = r/[p^2(1-p)]$; for $r=1$ reduces to geometric case

Q23.

In exponential family $f(x; \theta) = h(x) \exp[\eta(\theta)T(x) - B(\theta)]$, CRLB for $E\theta[T(X)]$ is attained by:

- A. $-T = (1)/(n) \sum T(X_i)$
- B. $T(X)$ itself
- C. $\checkmark$ All of the above
- D. MLE of $E\theta[T(X)]$

Correct answer: C

EXAM SHORTCUT

In exponential family, $\bar{T}$ is UMVUE for $E[T]$, equals MLE (by invariance), and attains CRLB — all three coincide

Q24.

For $X \sim \text{Laplace}(\theta, 1)$, $f(x; \theta) = (1)/(2e^{-|x-\theta|})$, $l(\theta)$ equals:

- A. 2
- B. Does not exist
- C. 1/2
- D. $\checkmark$ 1

Correct answer: D

EXAM SHORTCUT

 $\partial \log f / \partial \theta = \text{sgn}(x-\theta)$; $l(\theta) = E[\text{sgn}(X-\theta)^2] = 1$; sample median (not mean) is MLE and UMVUE

Q25.

For Weibull with shape k known and scale λ : $f(x; \lambda) = (k)/(\lambda)(x/\lambda)^{k-1}e^{-(x/\lambda)^k}$, $I(\lambda)$ equals:

- A. k^2/λ
- B. k^2/λ^2
- C. k/λ^2
- D. $1/\lambda^2$

Correct answer: B

EXAM SHORTCUT

Score: $\partial \log f / \partial \lambda = -k/\lambda + k(x/\lambda)^{k-1}/\lambda$; $I(\lambda) = k^2/\lambda^2$; CRLB = $\lambda^2/(nk^2)$

Q26.

If $\phi = g(\theta)$ and g is differentiable, then $I(\phi)$ equals:

- A. $I(\theta) \cdot [g'(\theta)]^2$
- B. $I(\theta)/[g'(\theta)]^2$
- C. $I(\theta) \cdot g'(\theta)$
- D. $I(\theta)/g'(\theta)$

Correct answer: B

EXAM SHORTCUT

Fisher information transforms inversely to variance; CRLB for $g(\theta)$ is $[g'(\theta)]^2/(nI(\theta))$ which equals $1/(nI(\phi))$ ✓

Q27.

For mixture $f(x; \theta) = \theta f_1(x) + (1 - \theta)f_2(x)$, $I(\theta)$ equals:

- A. $I_1(\theta) + I_2(\theta)$
- B. $\theta I_1(\theta) + (1 - \theta)I_2(\theta)$
- C. $\theta^2 I_1(\theta) + (1 - \theta)^2 I_2(\theta)$
- D. $\int [f_1(x) - f_2(x)]^2 \theta f_1(x) + (1 - \theta)f_2(x) dx$

Correct answer: D

EXAM SHORTCUT

Score = $[f_1(x) - f_2(x)]/f(x; \theta)$; $I(\theta) = E[\text{score}^2]$ gives D; B is wrong — mixture Fisher info is NOT weighted average

Estimation

CRLB Censored

Hard

Q28.

For Type-I right-censored exponential data, Fisher information compared to complete data:

A. **Is smaller (censoring loses information)**

B. Is larger (censoring adds information about censoring mechanism)

C. Is the same

D. Depends on censoring time

Correct answer: A

EXAM SHORTCUT

Censoring always loses information; $I_{\text{censored}} < I_{\text{complete}}$; CRLB is larger for censored data

Estimation

Bhattacharya Poisson

Hard

Q29.

For $X \sim \text{Poisson}(\lambda)$, estimating $e^{-2\lambda}$, CRLB equals:

A. **$4\lambda e^{-4\lambda/n}$**

B. $4\lambda^2 e^{-4\lambda/n}$

C. $e^{-4\lambda/n}$

D. $4e^{-4\lambda/n}$

Correct answer: A

EXAM SHORTCUT

$g(\lambda) = e^{-2\lambda}$; $g'(\lambda) = -2e^{-2\lambda}$; $I(\lambda) = 1/\lambda$; CRLB = $[g'(\lambda)]^2 / (nI(\lambda)) = 4e^{-4\lambda} \cdot \lambda/n$

Estimation

CRLB Pareto

Hard

Q30.

For Pareto $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, $I(\theta)$ equals:

A. **$1/\theta^2$**

B. θ^2

C. θ

D. $1/\theta$

Correct answer: A

EXAM SHORTCUT

$\log f = \log \theta - (\theta + 1) \log x$; $\partial \log f / \partial \theta = 1/\theta - \log x$; $I(\theta) = \text{Var}(\log X) = 1/\theta^2$ (since $\log X \sim \text{Exp}(\theta)$)

Q31.

For Inverse Gaussian $IG(\mu, \lambda)$ with λ known, $I(\mu)$ equals:

- A. $1/\mu^2$
- B. μ^3/λ
- C. λ/μ^3
- D. λ^2/μ^3

Correct answer: C

EXAM SHORTCUT

$I(\mu) = \lambda/\mu^3$; CRLB for $\hat{\mu} = \bar{X}$ is $\mu^3/(n\lambda)$; verify: $\text{Var}(\bar{X}) = \mu^3/(n\lambda)$ ✓ attains CRLB

Q32.

For $X \sim \text{Logistic}(\theta, 1)$, $I(\theta)$ equals:

- A. 3
- B. $3/\pi^2$
- C. $\pi/3$
- D. $\pi^2/3$

Correct answer: C

EXAM SHORTCUT

$f(x; \theta) = e^{-(x-\theta)/[1+e^{-(x-\theta)}]^2}$; $I(\theta) = 1/3$; CRLB = $3/n$; sample mean has variance $\pi^2/(3n) > 3/n$ — not efficient

Q33.

For $X \sim \text{Bin}(n, p)$ (single observation), $I(p)$ equals:

- A. $np(1-p)$
- B. $1/[p(1-p)]$
- C. $p(1-p)/n$
- D. $n/[p(1-p)]$

Correct answer: D

EXAM SHORTCUT

$\text{Binomial}(n, p) = \text{sum of } n \text{ Bernoulli}(p)$; $I_{\text{Bin}}(n, p) = n \cdot I_{\text{Bern}}(p) = n/[p(1-p)]$

Q34.

For $X \sim N(\mu, \Sigma)$ with Σ known, $I(\mu)$ (as matrix) equals:

A. Σ B. Σ^{-1} C. $|\Sigma|^{-1/p}$ D. $\text{tr}(\Sigma^{-1})/p$

Correct answer: B

EXAM SHORTCUT

Fisher information matrix for location in multivariate normal is Σ^{-1} ; CRLB matrix is Σ/n ; X attains it

Q35.

For curved exponential family (e.g., $N(\theta, \theta^2)$), CRLB:

A. Is attained by MLE asymptotically

B. **Is never attained exactly for finite n**

C. Is attained by method of moments

D. Does not exist

Correct answer: B

EXAM SHORTCUT

Curved exponential families generally cannot attain CRLB for finite n ; MLE is asymptotically efficient

Q36.

Data processing inequality for Fisher information states:

A. $I(T(\theta)) \leq I(X(\theta))$ for any statistic $T = T(X)$ B. $I(T(\theta)) \geq I(X(\theta))$ for sufficient T C. **$I(T(\theta)) = I(X(\theta))$ if and only if T is sufficient**D. $I(T(\theta)) = I(X(\theta))$ always

Correct answer: C

EXAM SHORTCUT

A is true in general; C is the equality condition (sufficiency); B is wrong — sufficient statistic preserves, not increases, information

Q37.

In sequential sampling with random sample size N , CRLB:

- A. Does not apply
- B. Is $1/(N \cdot I(\theta))$ with N fixed
- C. Is $1/(E[N] \cdot I(\theta))$
- D. Requires Wald's identity

Correct answer: C

EXAM SHORTCUT

For sequential procedures, expected Fisher information is $E[N] \cdot I(\theta)$; CRLB uses expected sample size

Q38.

For $X \sim N(0, \sigma^2)$, estimating σ^2 , CRLB equals:

- A. σ^4/n
- B. σ^2/n
- C. $4\sigma^4/n$
- D. $2\sigma^4/n$

Correct answer: D

EXAM SHORTCUT

$I(\sigma^2) = 1/(2\sigma^4)$; CRLB = $2\sigma^4/n$; $n \sum X_i^2$ is UMVUE and attains CRLB (since $\mu = 0$ known)

Q39.

Which distribution's MLE does NOT attain CRLB for finite n ?

- A. Bernoulli(p), estimating p
- B. $N(\mu, \sigma^2)$ with both unknown, estimating σ^2
- C. $N(\mu, \sigma^2)$ with σ^2 known, estimating μ
- D. Poisson(λ), estimating λ

Correct answer: B

EXAM SHORTCUT

S^2 (with $n-1$ denominator) is UMVUE but does NOT attain CRLB; $ML\hat{E} \sigma^2$ (with n denominator) is biased; neither attains CRLB

Q40.

Which condition is NOT required for CRLB?

- A. Support of $f(x; \theta)$ independent of θ
- B. $f(x; \theta)$ twice differentiable in θ
- C. Interchange of differentiation and integration
- D. **✗Distribution must be from exponential family**

Correct answer: D

EXAM SHORTCUT

CRLB applies to any regular family; exponential family is sufficient for attainment but NOT required for the bound itself

Factorization Theorem, Sufficiency & Completeness — Worked Families

30 forecast questions in this topic

Estimation

Factorization Theorem

Medium

Q1.

By Fisher-Neyman factorization theorem, $T(X)$ is sufficient for θ iff:

A. $f(x; \theta) = g(T(x); \theta) \cdot h(x)$

B. $f(x; \theta) = g(T(x); \theta) + h(x)$

C. $f(x; \theta) = g(x) \cdot h(T(x); \theta)$

D. $f(x; \theta) = g(T(x)) \cdot h(x; \theta)$

Correct answer: A

EXAM SHORTCUT

g depends on x ONLY through $T(x)$ and on θ ; h depends on x but NOT on θ ; product not sum

Estimation

Sufficient Stat Normal

Medium

Q2.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with both unknown, minimal sufficient statistic is:

A. All of the above are equivalent

B. $(\bar{X}, \sum X_i^2)$

C. $(\bar{X}, S^2)$

D. $(\sum X_i, \sum X_i^2)$

Correct answer: A

EXAM SHORTCUT

All three are equivalent (one-to-one functions of each other); minimal sufficient statistic is not unique in representation

Estimation

Sufficient Stat Uniform

Medium

Q3.

For $X_1, \dots, X_n \sim U(0, \theta)$, sufficient statistic is:

A. $\bar{X}$

B. $X_{(1)}$ (minimum)

C. $\sum X_i$

D. $X_{(n)}$ (maximum)

Correct answer: D

EXAM SHORTCUT

$f(x; \theta) = \theta^{-n} I(X_{(n)} \leq \theta)$; factorization gives $T = X_{(n)}$; $\bar{X}$ is NOT sufficient here

Estimation

Sufficient Stat Exponential

Medium

Q4.

For $X_1, \dots, X_n \sim \text{Exp}(\theta)$, sufficient statistic is:A. $\sum X_i$ B. $\bar{X}$ C. **Both B and C**D. $X_{(1)}$

Correct answer: C

EXAM SHORTCUT

 $\bar{X}$ and $\sum X_i$ are one-to-one functions of each other; both are sufficient; $X_{(1)}$ is NOT sufficient for rate parameter

Estimation

Sufficient Stat Cauchy

Hard

Q5.

For $X_1, \dots, X_n \sim \text{Cauchy}(\theta, 1)$, sufficient statistic is:A. $X_{(1)}$ and $X_{(n)}$ B. **$(X_{(1)}, X_{(2)}, \dots, X_{(n)})$ — all order statistics**C. $\bar{X}$

D. Sample median

Correct answer: B

EXAM SHORTCUT

Cauchy is NOT exponential family; likelihood does not factor to reduce dimension; full order statistics are minimal sufficient

Estimation

Complete Sufficient Uniform

Hard

Q6.

For $U(0, \theta)$, $X_{(n)}$ is complete sufficient. UMVUE of θ is:A. $n+1X_{(n)}$ B. **$(n+1)/n X_{(n)}$** C. $X_{(n)}$ D. $2X$

Correct answer: B

EXAM SHORTCUT

 $E[X_{(n)}] = n\theta/(n+1)$ (unbiased correction: $(n+1)/n X_{(n)}$); $2X$ is unbiased but NOT UMVUE (not function of complete sufficient stat)

Q7.

For $X_1, \dots, X_n \sim \text{Beta}(\theta, 1)$, $f(x; \theta) = \theta x^{\theta-1}$, sufficient statistic is:

- A. $\sum X_i$
- B. $\sum \log X_i$
- C. $\prod X_i$
- D. **Both B and C**

Correct answer: D

EXAM SHORTCUT

$\log f = n \log \theta + (\theta - 1) \sum \log x_i$; $T = \sum \log X_i$ is sufficient; $\prod X_i = \exp(\sum \log X_i)$ is equivalent

Q8.

For $X_1, \dots, X_n \sim U(\alpha, \beta)$ with both unknown, complete sufficient statistic is:

- A. (X, S^2)
- B. $X(n)$ alone
- C. $X(1)$ alone
- D. **$(X(1), X(n))$**

Correct answer: D

EXAM SHORTCUT

Both endpoints needed; $(X(1), X(n))$ is complete sufficient for (α, β) ; neither alone is sufficient

Q9.

For $X_1, \dots, X_n \sim f(x; \theta) = e^{-(x-\theta)}$, $x > \theta$, sufficient statistic is:

- A. $\bar{X}$
- B. $\sum X_i$
- C. **$X(1)$ (minimum)**
- D. $X(n)$ (maximum)

Correct answer: C

EXAM SHORTCUT

$f(x; \theta) = e^{-\sum (x_i - \theta)} I(X(1) > \theta) = e^{-n\theta} e^{-\sum x_i} I(X(1) > \theta)$; $T = X(1)$ is sufficient

Estimation

Completeness Uniform Both

Hard

Q10.

For $U(\theta, \theta+1)$, complete sufficient statistic is:

- A. No complete sufficient statistic exists
- B. $\sum X$
- C. $X(n) - X(1)$ (range)
- D. $(X(1), X(n))$

Correct answer: D

EXAM SHORTCUT

Range $X(n) - X(1)$ is ancillary (not sufficient); $(X(1), X(n))$ is minimal sufficient and complete

Estimation

Factorization Twoparam

Hard

Q11.

For $N(\mu, \sigma^2)$ with both unknown, joint likelihood factors through:

- A. $(\sum X_i, \sum X_i^2)$
- B. $(\bar{X}, \sum (X_i - \bar{X})^2)$
- C. $(\bar{X}, S^2)$ where S^2 uses $n-1$
- D. All of the above

Correct answer: D

EXAM SHORTCUT

All are equivalent representations of the same sufficient statistic; factorization theorem doesn't specify unique form

Estimation

Sufficient Not Complete

Hard

Q12.

An example of sufficient but NOT complete statistic:

- A. $X(n)$ for $U(0, \theta)$
- B. $(X(1), X(n))$ for $N(\mu, 1)$
- C. $\sum X$ for $N(\mu, 1)$
- D. $\sum X_i$ for $\text{Poisson}(\lambda)$

Correct answer: B

EXAM SHORTCUT

$(X(1), X(n))$ is sufficient for μ in $N(\mu, 1)$ but NOT complete; $E[X(n) + X(1) - 2\mu] = 0$ but $X(n) + X(1) - 2\mu \neq 0$ a.s.

Q13.

For $N(\mu, \sigma^2)$ with σ^2 known, by Basu's theorem:

A. $\bar{X}$ and S^2 are independent

B. $\bar{X}$ and sample range are independent

C. All order statistics are independent of $\bar{X}$

D. S^2 and $\bar{X}$ are independent (since $\bar{X}$ is complete sufficient and S^2 is ancillary)

Correct answer: B

EXAM SHORTCUT

$\bar{X}$ is complete sufficient for μ (with σ^2 known); range is ancillary for μ ; by Basu, they're independent

Q14.

For $X_1, \dots, X_n \sim \text{Pareto}(\theta)$, $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, sufficient statistic is:

A. $\prod X_i$

B. $\sum X_i$

C. $-\sum \log X_i$

D. $X_{(1)}$

Correct answer: C

EXAM SHORTCUT

$\log f = n \log \theta - (\theta + 1) \sum \log x_i$; $T = \sum \log X_i$ is sufficient; $\prod X_i = \exp(\sum \log X_i)$ equivalent

Q15.

For k -parameter exponential family with natural parameters $\eta_1, \dots, \eta_k$ varying freely, the natural sufficient statistic $(T_1, \dots, T_k)$ is:

A. Complete but not sufficient

B. Minimal sufficient only

C. $\checkmark$ Complete sufficient

D. Sufficient only

Correct answer: C

EXAM SHORTCUT

Full-rank exponential family: natural sufficient statistic is BOTH complete AND minimal sufficient; "full-rank" = natural parameter space has non-empty interior

Q16.

For $X_1, \dots, X_n \sim \text{Geometric}(p)$, sufficient statistic is:

A. ~~Both B and C~~

B. $\bar{X}$

C. $\sum X_i$

D. $X_{(1)}$

Correct answer: A

EXAM SHORTCUT

$\log f = \sum x_i \log(1-p) + n \log p$; $T = \sum X_i$ is sufficient; $\bar{X}$ is equivalent

Q17.

If T is sufficient for θ and U is independent of T , then:

A. U is ancillary

B. U contains no information about θ

C. (T, U) is sufficient for θ

D. ~~All of the above~~

Correct answer: D

EXAM SHORTCUT

Independence from sufficient statistic implies ancillarity; (T, U) is trivially sufficient; ancillary statistics carry no θ -information

Q18.

Lehmann-Scheffé method finds minimal sufficient statistic via:

A. Rao-Blackwell theorem

B. Maximizing likelihood over θ

C. Finding complete statistic

D. ~~Ratio $f(x; \theta)/f(y; \theta)$ constant in θ iff $T(x) = T(y)$~~

Correct answer: D

EXAM SHORTCUT

The ratio method: T is minimal sufficient iff the ratio of likelihoods is free of θ exactly when $T(x) = T(y)$

Q19.

For $X \sim \text{Bin}(n, p)$ with n known, X itself is complete sufficient because:

- A. $E p[g(X)] = \sum_{x=0}^n g(x) \binom{n}{x} p^x (1-p)^{n-x} = 0$ for all p implies $g=0$
- B. It is the MLE
- C. It is a function of order statistics
- D. It attains CRLB

Correct answer: A

EXAM SHORTCUT

Completeness proof uses the fact that a polynomial in $p/(1-p)$ that is identically zero must have all coefficients zero

Q20.

For Hypergeometric(N, M, n) with M unknown, sufficient statistic is:

- A. $\sum X_i$
- B. No reduction possible
- C. $\sum X_i$ (number of successes in sample)
- D. $X_{(1)}$

Correct answer: C

EXAM SHORTCUT

Hypergeometric is NOT exponential family but $\sum X_i$ is still sufficient; factorization applies directly

Q21.

If T is sufficient for θ and $\phi = g(T)$ is one-to-one, then:

- A. ϕ is also sufficient for θ
- B. ϕ is sufficient only if g is linear
- C. ϕ is sufficient only if g is monotone
- D. ϕ is not necessarily sufficient

Correct answer: A

EXAM SHORTCUT

One-to-one functions of sufficient statistics are sufficient; the factorization still holds through the bijection

Q22.

For location family $f(x-\theta)$ with $\theta \in \mathbb{R}$, the order statistics $(X(1), \dots, X(n))$ are:

A. Minimal sufficient but NOT complete

B. Complete sufficient

C. Sufficient but not minimal

D. Neither sufficient nor complete

Correct answer: A

EXAM SHORTCUT

For location families, order statistics are minimal sufficient but NOT complete; spacings $X(i+1) - X(i)$ are ancillary, providing non-trivial functions with zero expectation

Q23.

For Multinomial($n; p_1, \dots, p_k$) with $\sum p_i = 1$, sufficient statistic is:

A. $(n_1, \dots, n_{k-1})$ (cell counts, last determined by constraint)

B. $n_1/n, \dots, n_{k-1}/n$ (proportions)

C. All of the above

D. $(n_1, \dots, n_k)$

Correct answer: C

EXAM SHORTCUT

All are equivalent representations; $(n_1, \dots, n_{k-1})$ determines $n_k = n - \sum_{i=1}^{k-1} n_i$

Q24.

When $\theta = (\psi, \lambda)$ with λ nuisance, T is sufficient for ψ in the presence of λ if:

A. T is ancillary for λ

B. T is sufficient for $\theta = (\psi, \lambda)$ jointly

C. T is complete for ψ

D. Conditional distribution of X given T is free of ψ (not necessarily λ)

Correct answer: B

EXAM SHORTCUT

Joint sufficiency implies marginal sufficiency; conditional sufficiency (given λ) is a weaker concept

Estimation

Complete Stat Twosample

Hard

Q25.

For two independent samples $X_i \sim N(\mu_1, \sigma^2)$ and $Y_j \sim N(\mu_2, \sigma^2)$ with common σ^2 , complete sufficient statistic is:

- A. $(\bar{X}, \bar{Y}, S^2_p)$ where S^2_p is pooled variance
- B. $(\bar{X}, \bar{Y}, \sum (X_i - \bar{X})^2 + \sum (Y_j - \bar{Y})^2)$
- C. $(\sum X_i, \sum Y_j, \sum X_i^2 + \sum Y_j^2)$

D. All of the above

Correct answer: D

EXAM SHORTCUT

All are equivalent; three-dimensional sufficient statistic for three parameters (μ_1, μ_2, σ^2)

Estimation

Factorization Inverse Gaussian

Hard

Q26.

For $X \sim IG(\mu, \lambda)$ with μ known, sufficient statistic for λ is:

- A. $\sum (1/X_i - 1/\mu)$
- B. $\sum X_i$
- C. $\sum (X_i - \mu)^2 / X_i$
- D. $\sum X_i^2$

Correct answer: C

EXAM SHORTCUT

$\log f \propto \lambda \sum [(x_i - \mu)^2 / x_i]$; $T = \sum (X_i - \mu)^2 / X_i$ is sufficient for λ

Estimation

Completeness Truncated Family

Hard

Q27.

For Poisson(λ) truncated to $\{1, 2, 3, \dots\}$ (zero-truncated), $\sum X_i$ is:

- A. Complete but not sufficient
- B. Complete sufficient
- C. Sufficient but not complete
- D. Neither

Correct answer: B

EXAM SHORTCUT

Zero-truncated Poisson is still exponential family (with modified normalizing constant); $\sum X_i$ remains complete sufficient

Estimation

Sufficient Stat Regression

Hard

Q28.

In simple linear regression $Y_i = \alpha + \beta x_i + \epsilon_i$, $\epsilon_i \sim N(0, \sigma^2)$, sufficient statistic for $(\alpha, \beta, \sigma^2)$ is:

- A. $(\hat{\alpha}, \hat{\beta}, \hat{\sigma}^2)$ — OLS estimates
- B. $(\bar{Y}, \sum (x_i - \bar{x})Y_i, \text{RSS})$
- C. $(\sum Y_i, \sum x_i Y_i, \sum Y_i^2)$
- D. All of the above

Correct answer: D

EXAM SHORTCUT

All are equivalent three-dimensional sufficient statistics; OLS estimates are one-to-one functions of the natural sufficient statistics

Estimation

Basu Theorem Proof Idea

Hard

Q29.

Basu's theorem proof relies on:

- A. Completeness of T and ancillarity of A to show $P(A \in B | T = t) = P(A \in B)$
- B. Factorization theorem
- C. Independence of order statistics
- D. Rao-Blackwell theorem

Correct answer: A

EXAM SHORTCUT

Key step: $E_\theta[P(A \in B | T) - P_\theta(A \in B)] = 0$ for all θ ; by completeness, $P(A \in B | T) = P_\theta(A \in B)$ a.s., giving independence

Estimation

Factorization Logistic Regression

Hard

Q30.

In logistic regression with $P(Y_i = 1) = \frac{e^{\beta x_i}}{1 + e^{\beta x_i}}$, sufficient statistic for β is:

- A. $\sum x_i Y_i$
- B. $\sum Y_i$
- C. $(\sum Y_i, \sum x_i Y_i)$
- D. $\sum Y_i$

Correct answer: A

EXAM SHORTCUT

$\log L = \beta \sum x_i y_i - \sum \log(1 + e^{\beta x_i})$; $T = \sum x_i Y_i$ is sufficient for β (single parameter)

Maximum Likelihood Estimation

50 forecast questions in this topic

Estimation

MLE Normal Both

Medium

Q1.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, MLE of σ^2 is:

A. $\frac{1}{n+1} \sum (X_i - \bar{X})^2$

B. $\frac{1}{n} \sum (X_i - \mu)^2$

C. $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$

D. $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$

Correct answer: D

EXAM SHORTCUT

MLE uses n denominator (biased); unbiased estimator uses $n-1$; B is MLE only when μ is known

Estimation

MLE Uniform

Medium

Q2.

For $X_1, \dots, X_n \sim U(0, \theta)$, MLE of θ is:

A. $\bar{X}$

B. $X(n)$

C. $2\bar{X}$

D. $(n+1)/(n\bar{X})$

Correct answer: B

EXAM SHORTCUT

Likelihood $L(\theta) = \theta^{-n} I(\theta \geq X(n))$ is decreasing in θ ; minimized at $\theta = X(n)$; D is UMVUE not MLE

Estimation

MLE Shifted Exponential

Hard

Q3.

For $X_1, \dots, X_n \sim e^{-(x-\theta)}, x > \theta$, MLE of θ is:

A. $X(1)$

B. $\bar{X} - 1$

C. $X(1) - 1/n$

D. $\bar{X}$

Correct answer: A

EXAM SHORTCUT

Likelihood $L(\theta) = e^{-\sum (x_i - \theta)} I(\theta \leq X(1)) = e^{n\theta} e^{-\sum x_i} I(\theta \leq X(1))$; increasing in θ ; maximized at $\theta = X(1)$

Q4.

For $X_1, \dots, X_n \sim \text{Cauchy}(\theta, 1)$, MLE of θ :

- A. Is $X_{(n/2)}$ (middle order statistic)
- B. Is $\bar{X}$
- C. Is sample median
- D. **Has no closed form; requires numerical optimization**

Correct answer: D

EXAM SHORTCUT

Score equation $\sum 2(x_i - \theta) = 0$ has no closed form; sample median is a good starting point but MLE $\neq$ median exactly

Q5.

If $\hat{\lambda} = \bar{X}$ is MLE of λ for Poisson, MLE of $P(X=0) = e^{-\lambda}$ is:

- A. $1 - \bar{X}$
- B. **$e^{-\bar{X}}$**
- C. $(1/n) \sum I(X_i=0)$
- D. $e^{-\lambda}$ evaluated at UMVUE

Correct answer: B

EXAM SHORTCUT

By invariance of MLE: $\hat{e^{-\lambda}} = e^{-\hat{\lambda}} = e^{-\bar{X}}$; C is unbiased but not MLE; note $e^{-\bar{X}} \neq$ UMVUE

Q6.

Under regularity conditions, $\hat{\theta}_n$ satisfies:

- A. Both A and B
- B. **$\sqrt{n}(\hat{\theta}_n - \theta_0) \rightarrow \text{true value}$**
- C. $\hat{\theta}_n$ is unbiased for all n
- D. $\hat{\theta}_n \xrightarrow{a.s.} \theta_0$

Correct answer: B

EXAM SHORTCUT

MLE is consistent (in probability) under regularity; strong consistency (a.s.) requires additional conditions; C is generally false

Q7.

Under regularity conditions, $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} :$

A. $N(0, 1/I(\theta))$ B. **Both B and C (same thing for scalar θ)**C. $N(0, I(\theta)^{-1})$ D. $N(0, I(\theta))$ **Correct answer: B****EXAM SHORTCUT**

For scalar θ : $1/I(\theta)$ and $I(\theta)^{-1}$ are identical; for vector θ : $I(\theta)^{-1}$ is the matrix inverse

Q8.

When log-likelihood has multiple local maxima, MLE is:

A. The average of all local maxima

B. The local maximum closest to MOM estimate

C. Undefined

D. **The global maximum****Correct answer: D****EXAM SHORTCUT**

MLE is defined as the GLOBAL maximum; in practice, numerical methods may find local maxima — this is a computational challenge, not a definitional one

Q9.

For $X_1, \dots, X_n \sim N(\mu, 1)$ with constraint $\mu \geq 0$, MLE of μ is:

A. $\bar{X}$ alwaysB. $\bar{X}$ if $\bar{X} > 0$, undefined otherwiseC. $|\bar{X}|$ D. **$\max(\bar{X}, 0)$** **Correct answer: D****EXAM SHORTCUT**

Constrained MLE: if unconstrained MLE $\bar{X} \geq 0$, use it; if $\bar{X} < 0$, boundary value $\mu = 0$ maximizes constrained likelihood

Estimation

MLE Discrete Uniform

Hard

Q10.

For $X_1, \dots, X_n \sim \text{DiscreteUniform}\{1, 2, \dots, N\}$ with N unknown, MLE of N is:

- A. $X(n)$
- B. $\lceil 2X \rceil$
- C. $X(n) + 1$
- D. $2X - 1$

Correct answer: A**EXAM SHORTCUT**

$L(N) = N - nI(N \geq X(n))$; decreasing in N ; MLE = $X(n)$ (smallest valid N); note MLE is biased

Estimation

MLE Mixture EM

Hard

Q11.

EM algorithm for MLE in mixture models:

- A. **E-step: compute expected complete-data log-likelihood; M-step: maximize it**
- B. E-step: maximize; M-step: compute expectation
- C. Directly maximizes observed-data likelihood
- D. Requires complete data

Correct answer: A**EXAM SHORTCUT**

EM: E-step computes $Q(\theta|\theta(t)) = E[\log L_C | \text{observed}, \theta(t)]$; M-step maximizes Q ; guaranteed to increase observed likelihood

Estimation

MLE Exponential Family

Medium

Q12.

In exponential family, MLE satisfies:

- A. $E\theta[T(X)] = T$ (observed sufficient statistic equals expected)
- B. Score equation: $\sum \partial \log f / \partial \theta = 0$
- C. Neither; MLE requires numerical methods
- D. **Both A and B**

Correct answer: D**EXAM SHORTCUT**

In exponential family, score equation reduces to moment equation $E\theta[T] = T$; both characterizations are equivalent

Q13.

MLE may fail to be consistent when:

- A. Distribution is normal
- B. Likelihood is unimodal
- C. **Number of parameters grows with sample size (Neyman-Scott problem)**
- D. Sample size is large

Correct answer: C

EXAM SHORTCUT

Neyman-Scott: incidental parameters problem; MLE of variance in panel data with individual fixed effects is inconsistent

Q14.

For zero-truncated Poisson, $\hat{\lambda}$ satisfies:

- A. $\hat{\lambda} = \bar{X}$
- B. **$\bar{X} = \lambda / (1 - e^{-\lambda})$**
- C. $\hat{\lambda} = \bar{X} / (1 - e^{-\bar{X}})$
- D. $\hat{\lambda} = \bar{X} - 1$

Correct answer: B

EXAM SHORTCUT

Truncated mean $E[X|X > 0] = \lambda / (1 - e^{-\lambda})$; setting equal to $\bar{X}$ gives implicit equation; no closed form

Q15.

For $\Gamma(\alpha, \beta)$ with both unknown, MLE of α satisfies:

- A. **$\log \alpha - \psi(\alpha) = \log \bar{X} - \log \bar{X}$**
- B. $\hat{\alpha} = n / \sum \log X_i$
- C. $\hat{\alpha} = \bar{X} / \beta$
- D. No closed form exists; use Newton-Raphson

Correct answer: A

EXAM SHORTCUT

ψ is digamma function; A is the score equation for α ; D is correct in spirit but A specifies the equation; C gives α in terms of $\hat{\beta}$

Estimation

MLE Logistic

Hard

Q16.

For logistic regression, MLE:

A. ~~Both B and C~~**B.** Is always unique**C.** Requires iterative methods (IRLS or Newton-Raphson)**D.** Has closed form via OLS**Correct answer: A****EXAM SHORTCUT**

Logistic MLE: no closed form; IRLS converges to unique global maximum (loglikelihood is concave); uniqueness requires full rank design matrix

Estimation

MLE Censored Exponential

Hard

Q17.For Type-II censored sample (smallest r out of n observed) from $\text{Exp}(\theta)$, MLE of θ is:**A.** $r/\sum_{i=1}^r X_{(i)}$ **B.** $1/X_{(r)}$ **C. ~~$r/[\sum_{i=1}^r X_{(i)} + (n-r)X_{(r)}]$~~** **D.** $1/\bar{X}_r$ where $\bar{X}_r$ is mean of r observed**Correct answer: C****EXAM SHORTCUT**

Total time on test = $\sum_{i=1}^r X_{(i)} + (n-r)X_{(r)}$; MLE = $r/(\text{total time on test})$; A ignores the censored contribution.

Estimation

MLE Weibull Shape

Hard

Q18.For Weibull(k, λ) with λ known, MLE of k satisfies:**A.** $\hat{k} = 1/\log \bar{X}$ **B.** $\hat{k} = n/\sum \log X_i$ **C.** $\hat{k} = \bar{X}/\lambda$ **D. ~~$1/\hat{k} + (1/n)\sum \log X_i - (\sum X_i^{\hat{k}} \log X_i)/(\sum X_i^{\hat{k}}) = 0$~~** **Correct answer: D****EXAM SHORTCUT**

Score equation for Weibull shape is transcendental; B is MLE for Beta($\theta, 1$) not Weibull; requires numerical solution

Q19.

For Dirichlet($\alpha_1, \dots, \alpha_k$), MLE of α_j satisfies:

A. $-\psi(\alpha_j) - \psi(\alpha_0) = \log X_j$ where $\alpha_0 = \sum \alpha_j$

B. $\hat{\alpha}_j = X_j$

C. No MLE exists

D. $\hat{\alpha}_j = \bar{n} X_j$

Correct answer: A

EXAM SHORTCUT

ψ is digamma; system of equations must be solved simultaneously; B/D are MOM not MLE

Q20.

For $X_1, \dots, X_n \sim U(\theta - 1/2, \theta + 1/2)$, MLE of θ :

A. Is $X(1) + X(n)$

B. Is X

C. Is any value in $[X(n) - 1/2, X(1) + 1/2]$

D. Does not exist

Correct answer: C

EXAM SHORTCUT

Likelihood is constant for $\theta \in [X(n) - 1/2, X(1) + 1/2]$; MLE is NOT unique; $X(1) + X(n)$ is a natural choice but not the unique MLE

Q21.

Under regularity, MLE converges at rate:

A. n^{-1}

B. $n^{-1/3}$

C. $n^{-1/2}$ (parametric rate)

D. $(\log n)^{-1}$

Correct answer: C

EXAM SHORTCUT

Parametric MLE: $\hat{\theta} - \theta = O_p(n^{-1/2})$; nonparametric rates are slower; $n^{-1/2}$ is the hallmark of parametric efficiency

Estimation

MLE Profile Likelihood

Hard

Q22.

Profile likelihood for ψ (with nuisance λ) is:

- A. $L_p(\psi) = \max_{\lambda} L(\psi, \lambda)$
- B. $L_p(\psi) = L(\hat{\psi}, \lambda)$ where $\hat{\lambda}$ is marginal MLE
- C. $L_p(\psi) = L(\hat{\psi}, \hat{\lambda}_{\psi})$ where $\hat{\lambda}_{\psi} = \arg \max_{\lambda} L(\psi, \lambda)$
- D. $L_p(\psi) = \int L(\psi, \lambda) d\lambda$

Correct answer: C

EXAM SHORTCUT

Profile likelihood: maximize over nuisance at each fixed ψ ; A and C are equivalent; D is marginal likelihood (Bayesian); B uses wrong $\hat{\lambda}$

Estimation

MLE Second Order

Hard

Q23.

Bartlett correction improves likelihood ratio test by:

- A. Using F -distribution instead of χ^2
- B. Scaling $-2 \log \Lambda$ by factor $1 + b/n$ to achieve χ^2 distribution to order $O(n^{-2})$
- C. Adjusting degrees of freedom
- D. Adding b/n to the test statistic

Correct answer: B

EXAM SHORTCUT

Bartlett correction: divide $-2 \log \Lambda$ by $(1 + b/n)$; improves χ^2 approximation from $O(n^{-1})$ to $O(n^{-2})$ error

Estimation

MLE Penalized

Hard

Q24.

Penalized MLE (e.g., LASSO) maximizes:

- A. $\log L(\theta) + \lambda \|\theta\|_1$
- B. $\log L(\theta) - \lambda \|\theta\|_2^2$
- C. A only (LASSO uses L_1 penalty)
- D. $\log L(\theta) - \lambda \|\theta\|_1$

Correct answer: D

EXAM SHORTCUT

LASSO: L_1 penalty subtracted; Ridge: L_2^2 penalty subtracted; both are D-type (subtract penalty); C is correct but incomplete

Estimation

MLE Robust

Hard

Q25.

M-estimators generalize MLE by solving:

A. ~~Both A and C~~ (equivalent formulations)

B. $\min \sum \rho(X_i; \theta)$ for loss ρ

C. $\sum \partial \log f / \partial \theta = 0$ only

D. $\sum \psi(X_i; \theta) = 0$ for some function ψ

Correct answer: A

EXAM SHORTCUT

M-estimators: $\psi = \partial \rho / \partial \theta$; MLE is special case with $\rho = -\log f$; Huber Mestimator uses bounded ψ

Estimation

MLE Exponential Rate

Medium

Q26.

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$ (rate parametrization), MLE of λ is:

A. $1/X$

B. ~~Both B and C~~

C. $n / \sum X_i$

D. $-X$

Correct answer: B

EXAM SHORTCUT

$1/X = n / \sum X_i$ — identical; both are correct expressions for MLE of rate λ

Estimation

MLE Beta Both

Hard

Q27.

For $X \sim \text{Beta}(\alpha, \beta)$ with both unknown, MLE satisfies:

A. ~~$\psi(\alpha) - \psi(\alpha + \beta) = \log X$ and $\psi(\beta) - \psi(\alpha + \beta) = \log(1 - X)$~~

B. No MLE exists for Beta with both parameters unknown

C. $\hat{\alpha} / (\hat{\alpha} + \hat{\beta}) = X$

D. $\hat{\alpha} = X, \hat{\beta} = 1 - X$

Correct answer: A

EXAM SHORTCUT

System of two digamma equations; C gives one equation (moment condition) but not MLE; requires numerical solution

Estimation

MLE Nonidentifiable

Hard

Q28.

MLE fails when model is non-identifiable because:

- A. Score equation has no solution
- B. Fisher information is zero
- C. Likelihood is unbounded
- D. Multiple parameter values give same likelihood; MLE is not unique**

Correct answer: D

EXAM SHORTCUT

Non-identifiability: $L(\theta_1) = L(\theta_2)$ for $\theta_1 \neq \theta_2$; MLE set is not a singleton; Fisher information matrix is singular

Estimation

MLE Separated Data

Hard

Q29.

In logistic regression, MLE does not exist when:

- A. Response is binary
- B. Predictors are correlated
- C. Sample size is small
- D. Data is perfectly separated (one predictor perfectly predicts outcome)**

Correct answer: D

EXAM SHORTCUT

Perfect separation: likelihood is monotone increasing; $\text{MLE} \rightarrow \pm\infty$; Firth's penalized likelihood is a remedy

Estimation

MLE Consistency Proof

Hard

Q30.

Wald's proof of MLE consistency uses:

- A. Rao-Blackwell theorem
- B. Law of large numbers applied to log-likelihood ratio**
- C. Factorization theorem
- D. Central limit theorem

Correct answer: B

EXAM SHORTCUT

Key: $n(\log L(\theta)/(L)(\theta_0) - E_{\theta_0}[\log(f(X;\theta))/(f)(X;\theta_0)]) \leq 0$ by Jensen (Kullback-Leibler divergence ≥ 0)

Q31.

In Poisson regression with $\log \mu_i = x_i(T)(i\beta)$, score equation for β is:

- A. $\sum x_i Y_i = \sum x_i \mu_i$
- B. **All of the above**
- C. $X^T(Y - \mu) = 0$
- D. $\sum x_i (Y_i - \mu_i) = 0$

Correct answer: B

EXAM SHORTCUT

All three are equivalent forms of the Poisson GLM score equation; IRLS solves this iteratively

Q32.

For $\text{NegBin}(r, p)$ with r known, MLE of p is:

- A. **$r/(r + X)$**
- B. r/X
- C. $X/(r + X)$
- D. $1/X$

Correct answer: A

EXAM SHORTCUT

$E[X] = r(1 - p)/p$; solving $X = r(1 - \hat{p})/\hat{p}$ gives $\hat{p} = r/(r + X)$; C is MLE of $1 - p$

Q33.

Zero-inflated Poisson (ZIP) has $P(X = 0) = \pi + (1 - \pi)e^{-\lambda}$. MLE requires:

- A. Bayesian methods
- B. **EM algorithm (latent class indicator)**
- C. Method of moments only
- D. Direct score equations with closed form

Correct answer: B

EXAM SHORTCUT

ZIP is a mixture model; EM treats class membership as latent; E-step computes posterior probability of structural zero; M-step updates π and λ

Estimation

MLE Survival Kaplan Meier

Hard

Q34.

Kaplan-Meier estimator is:

- A. Bayesian estimator
- B. **MLE of survival function under right censoring (nonparametric)**
- C. Method of moments estimator
- D. MLE of exponential survival function

Correct answer: B**EXAM SHORTCUT**

KM is the nonparametric MLE of $S(t)$; it maximizes the nonparametric likelihood over all step functions; not restricted to parametric family

Estimation

MLE Information Matrix Equality

Hard

Q35.

Fisher information matrix equality states:

- A. $-E[\partial^2 \log L / \partial \theta \partial \theta^T] = E[(\partial \log L / \partial \theta)(\partial \log L / \partial \theta)^T]$
- B. $I(\theta) = \text{Var}(\partial \log L / \partial \theta)$
- C. $E[\partial \log L / \partial \theta] = 0$
- D. **Both A and B**

Correct answer: D**EXAM SHORTCUT**

C is the score equation property; A is the information matrix equality; B follows from C (since $E[\text{score}] = 0$, variance = second moment); all three are related

Estimation

MLE Restricted Parameter

Hard

Q36.

For $X \sim N(\theta, 1)$ with $\theta \in \{0, 1\}$ (discrete parameter space), MLE is:

- A. 0 always (conservative)
- B. **0 if $X < 1/2$, 1 if $X \geq 1/2$**
- C. $\bar{X}$ always
- D. $\bar{X}$ rounded to nearest integer

Correct answer: B**EXAM SHORTCUT**

Compare $L(0) = e^{-\bar{n}X^2/2}$ vs $L(1) = e^{-n(X-1)^2/2}$; $L(1) > L(0)$ iff $X > 1/2$

Estimation

MLE Curved Exponential Normal

Hard

Q37.

For $X_1, \dots, X_n \sim N(\theta, \theta^2)$ (curved exponential family), MLE of θ is:

A. $\bar{X}$ B. $\hat{\theta} = \frac{\sum X_i}{\sum X_i^2 + S^2} = 0$ approximatelyC. $\sqrt{\sum X_i^2 + S^2}$ D. S (sample SD)

Correct answer: B

EXAM SHORTCUT

Score equations for $\mu = \theta$ and $\sigma^2 = \theta^2$ must be combined; leads to quadratic in θ ; no simple closed form

Estimation

MLE Poisson UMVUE Comparison

Hard

Q38.

For Poisson(λ), MLE of $e^{-\lambda}$ is $e^{-\bar{X}}$. UMVUE is:

A. $e^{-\bar{X}}$ B. $(1/n) \sum I(X_i = 0)$ C. $e^{-\bar{X}(1 + 1/(2n))}$ (bias-corrected)D. $(1 - 1/n) \sum X_i$

Correct answer: D

EXAM SHORTCUT

UMVUE of $e^{-\lambda} = P(X = 0)$: use Rao-Blackwell with $T = \sum X_i$; $E[I(X_1 = 0) | T = t] = (1 - 1/n)t$; so UMVUE = $(1 - 1/n)T$

Estimation

MLE Asymptotic Variance

Medium

Q39.

Asymptotic variance of MLE $\hat{\theta}$ is:

A. $nI(\theta)$ B. $I(\theta)/n$ C. $1/(nI(\theta))$ D. $1/I(\theta)$

Correct answer: C

EXAM SHORTCUT

$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1/I(\theta))$; so $\text{AsyVar}(\hat{\theta}) = 1/(nI(\theta))$; D is for single observation

Estimation

MLE Score Test

Hard

Q40.

Rao's score test statistic is:

A. ~~All equivalent asymptotically~~B. $(\theta - \theta_0)^2 \frac{1}{nI(\theta_0)} \xrightarrow{d} \chi^2_1$ C. $[\partial \log L / \partial \theta | \theta = \theta_0]^2 \frac{1}{nI(\theta_0)} \xrightarrow{d} \chi^2_1$ under H_0 D. $-2 \log \Lambda \xrightarrow{d} \chi^2_1$

Correct answer: A

EXAM SHORTCUT

Wald (C), Score/Rao (A), and LRT (B) are asymptotically equivalent under H_0 ; they differ in finite samples and under alternatives

Estimation

MLE Exponential Mean

Medium

Q41.

For $X_1, \dots, X_n \sim \text{Exp}(\mu)$ (mean parametrization), MLE of μ is:A. ~~$\bar{X}$~~ B. $\bar{X}$ (same as B)C. $n / \sum X_i$ D. $1/\bar{X}$

Correct answer: A

EXAM SHORTCUT

Mean parametrization: $f(x; \mu) = (1/\mu)e^{-x/\mu}$; MLE = $\bar{X}$; rate parametrization: MLE = $1/\bar{X}$; know which parametrization is used

Estimation

MLE Multivariate Normal

Hard

Q42.

For $X_1, \dots, X_n \sim N_p(\mu, \Sigma)$ with both unknown, MLE of Σ is:A. ~~$(1/n) \sum (X_i - \bar{X})(X_i - \bar{X})^T$~~ B. $(n-1) \sum (X_i - \bar{X})(X_i - \bar{X})^T$ C. $(n-p) \sum (X_i - \bar{X})(X_i - \bar{X})^T$

D. Sample covariance matrix

Correct answer: A

EXAM SHORTCUT

MLE uses n denominator (biased); unbiased estimator uses $n-1$; $n-p$ appears in regression residual variance

Estimation

MLE Dirichlet Multinomial

Hard

Q43.

For Dirichlet-Multinomial model, MLE of Dirichlet parameters:

A. ~~Requires solving system of digamma equations iteratively~~

B. Does not exist

C. Has closed form via method of moments

D. Is simply cell proportions

Correct answer: A

EXAM SHORTCUT

Same structure as Dirichlet MLE; digamma equations; Minka's fixed-point iteration is commonly used

Estimation

MLE Nonparametric

Hard

Q44.

Nonparametric MLE of CDF F based on i.i.d. sample is:

A. Kernel density estimate

B. Smoothed empirical CDF

C. Normal CDF fitted to data

D. ~~Empirical CDF~~ $\hat{F}_n(x) = (1/n) \sum I(X_i \leq x)$

Correct answer: D

EXAM SHORTCUT

NPMLE of CDF is the empirical CDF (Glivenko-Cantelli); it maximizes the nonparametric likelihood $\prod f(X_i)$ over all distributions

Estimation

MLE Exponential Family Canonical

Hard

Q45.

In canonical exponential family, MLE $\hat{\eta}$ of natural parameter satisfies:

A. $\hat{\eta} = T$

B. $\hat{\eta}$ minimizes KL divergence from empirical distribution

C. $\nabla A(\eta) = T$ where A is log-partition function

D. ~~Both A and C~~

Correct answer: D

EXAM SHORTCUT

$\nabla A(\eta) = E[T(X)]$; setting equal to T gives MLE; this is equivalent to minimizing KL divergence (maximum entropy principle)

Q46.

Breakdown point of MLE for normal location parameter is:

- A. $1/4$
- B. $1/n$
- C. **0 (single outlier can make MLE arbitrarily large)**
- D. $1/2$ (50% contamination)

Correct answer: C

EXAM SHORTCUT

Sample mean (MLE for normal location) has breakdown point 0; one extreme outlier destroys it; median has breakdown point $1/2$

Q47.

Conditional MLE (conditioning on sufficient statistic for nuisance) is used to:

- A. Handle missing data
- B. Reduce computation
- C. Increase efficiency
- D. **Eliminate nuisance parameters from likelihood**

Correct answer: D

EXAM SHORTCUT

Conditional likelihood eliminates nuisance parameters; used in matched casecontrol studies, logistic regression with fixed effects

Q48.

Marginal MLE (integrating out random effects) is used in:

- A. Fixed effects models
- B. **Mixed effects models (REML is a variant)**
- C. Saturated models
- D. Nonparametric models

Correct answer: B

EXAM SHORTCUT

REML (Restricted MLE) maximizes marginal likelihood after integrating out fixed effects; used for variance component estimation in mixed models

Q49.

In Cox proportional hazards model, partial likelihood MLE:

- A. Estimates baseline hazard and regression coefficients jointly
- B. Requires parametric baseline hazard
- C. Estimates regression coefficients without specifying baseline hazard**
- D. Is equivalent to full MLE

Correct answer: C

EXAM SHORTCUT

Cox partial likelihood: baseline hazard cancels; only regression coefficients estimated; semiparametric efficiency theory applies

Q50.

Observed Fisher information $I(\theta) = -\partial^2 \log L / \partial \theta^2 |_{\theta}$ vs expected $I(\theta)$:

- A. Expected is always larger
- B. They are always equal
- C. Observed is always larger
- D. They are asymptotically equivalent; observed often preferred in practice**

Correct answer: D

EXAM SHORTCUT

Both are consistent for $I(\theta_0)$; Efron-Hinkley showed observed information is preferred for conditional inference; neither dominates in finite samples

Rao–Blackwell, UMVUE & Method of Moments

40 forecast questions in this topic

Estimation

Rao Blackwell Statement

Medium

Q1.

Rao-Blackwell theorem states: if T_0 is unbiased for $g(\theta)$ and S is sufficient, then $T^* = E[T_0|S]$:

A. **All of the above**

B. Is unbiased for $g(\theta)$

C. Has smaller or equal variance than T_0

D. Is a function of S

Correct answer: A

EXAM SHORTCUT

All three properties hold: unbiasedness (tower property), variance reduction (conditional variance formula), and T^* is a function of S by definition

Estimation

Rao Blackwell Example Bernoulli

Medium

Q2.

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, improve $T_0 = X_1$ (unbiased for p) using $S = \sum X_i$:

A. $E[X_1 | \sum X_i = t] = t(t-1)/[n(n-1)]$

B. $E[X_1 | \sum X_i = t] = 1/n$

C. **$E[X_1 | \sum X_i = t] = t/n$**

D. $E[X_1 | \sum X_i = t] = t^2/n^2$

Correct answer: C

EXAM SHORTCUT

$P(X_1 = 1 | \sum X_i = t) = ((n-1)/(t-1))/(n-1) = t/n$; result is X — the UMVUE

Estimation

UMVUE Poisson Zero

Hard

Q3.

For $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$, UMVUE of $P(X=0) = e^{-\lambda}$ is:

A. **$(1 - 1/n)^{\sum X_i}$**

B. $(1)/(n)^{\sum (X_i = 0)}$

C. e^{-X}

D. $e^{-\sum X_i/n}$

Correct answer: A

EXAM SHORTCUT

$E[I(X_1 = 0) | T = t]$ where $T = \sum X_i \sim \text{Poisson}(n\lambda)$: $P(X_1 = 0 | T = t) = (1 - 1/n)^t$; so UMVUE = $(1 - 1/n)^T$

Q4.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with μ known, UMVUE of σ^2 is:

A. $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$

B. $\frac{1}{n} \sum (X_i - \mu)^2$

C. $\frac{1}{n} \sum (X_i - \bar{X})^2$

D. $\frac{1}{n-1} \sum (X_i - \mu)^2$

Correct answer: B

EXAM SHORTCUT

When μ is KNOWN, $\sum (X_i - \mu)^2 / n$ is UMVUE (attains CRLB); when μ unknown, UMVUE is S^2 with $n-1$ denominator

Q5.

For $X_1, \dots, X_n \sim U(0, \theta)$, UMVUE of θ^2 is:

A. $4X^2/3$

B. $\frac{(n+1)(n+2)}{n^2} X^2_{(n)}$

C. $(n+2)/(nX^2) (n)$

D. $X^2_{(n)}$

Correct answer: B

EXAM SHORTCUT

$E[X(k)/(n)] = n\theta k / (n+1)$; for $k=2$: $E[X^2_{(n)}] = n\theta^2 / (n+2)$; UMVUE = $\frac{(n+1)(n+2)}{n^2} X^2_{(n)}$

Q6.

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$, UMVUE of λ^k is:

A. $(1/\bar{X})^k$

B. $\bar{X}^{-k}$

C. $\frac{\Gamma(n)}{\Gamma(n-k)} \cdot (\sum X_i)^k$

D. $n! / (n-k)! \cdot n^k \cdot \Gamma(X)^k$ (for $k < n$)

Correct answer: C

EXAM SHORTCUT

$T = \sum X_i \sim \Gamma(n, 1/\lambda)$; $E[T-k] = (\lambda^k \Gamma(n-k)) / (\Gamma(n))$; UMVUE of $\lambda^k = \frac{\Gamma(n)}{\Gamma(n-k)} T^{-k}$

Estimation

UMVUE Bernoulli P Sq

Hard

Q7.

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, UMVUE of p^2 is:

A. T^2/n^2

B. $\bar{X}^2$

C. $\frac{T(T-1)}{n(n-1)}$ where $T = \sum X_i$

D. $\bar{X}(X-1/n)$

Correct answer: C

EXAM SHORTCUT

 $E[X_1 X_2 | T = t] = \frac{t(t-1)}{n(n-1)}$ (hypergeometric argument); UMVUE = $\frac{T(T-1)}{[n(n-1)]}$; note $\bar{X}^2$ is biased for p^2

Estimation

UMVUE Poisson Lambda Sq

Hard

Q8.

For $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$, UMVUE of λ^2 is:

A. $\bar{X}^2$

B. $\frac{T(T-1)}{n^2}$ where $T = \sum X_i$

C. T^2/n^2

D. $\bar{X}^2 - \bar{X}/n$

Correct answer: B

EXAM SHORTCUT

 $E[T(T-1)] = E[T^2] - E[T] = n^2 \lambda^2 + n\lambda - n\lambda = n^2 \lambda^2$; so UMVUE = $T(T-1)/n^2 = \bar{X}^2 - \bar{X}/n$; D and B are equivalent

Estimation

Method Of Moments Gamma

Medium

Q9.

For $X \sim \Gamma(\alpha, \beta)$, MOM estimators using first two moments:

A. $\hat{\alpha} = S^2/\bar{X}^2, \hat{\beta} = \bar{X}/S^2$

B. $\hat{\alpha} = \bar{X}/S^2, \hat{\beta} = S^2$

C. $\hat{\alpha} = \bar{X}^2/S^2, \hat{\beta} = S^2/\bar{X}$

D. $\hat{\alpha} = \bar{X}, \hat{\beta} = S^2/\bar{X}$

Correct answer: C

EXAM SHORTCUT

 $E[X] = \alpha\beta, \text{Var}(X) = \alpha\beta^2; \beta = \text{Var}/E = S^2/\bar{X}; \alpha = E/\beta = \bar{X}^2/S^2$

Q10.For $X \sim \text{Beta}(\alpha, \beta)$, MOM estimators:

A. $\hat{\alpha} = X[X(1-X)/S^2 - 1]; \hat{\beta} = (1-X)[X(1-X)/S^2 - 1]$

B. $\hat{\alpha} = X; \hat{\beta} = 1 - X$

C. $\hat{\alpha} = X^2/S^2; \hat{\beta} = (1-X)^2/S^2$

D. $\hat{\alpha} = X/S^2; \hat{\beta} = (1-X)/S^2$

Correct answer: A**EXAM SHORTCUT**
 $E[X] = \alpha/(\alpha + \beta)$, $\text{Var}(X) = \alpha\beta/[(\alpha + \beta)^2(\alpha + \beta + 1)]$; let $m = X$, $v = S^2$; $\alpha = m[m(1-m)/v - 1]$
Q11.

Rao-Blackwell improvement is strict (strict inequality in variance) unless:

A. T_0 is the UMVUE

B. **Both A and C (equivalent)**

C. T_0 is the MLE

D. T_0 is already a function of S

Correct answer: B**EXAM SHORTCUT**

Variance reduction is strict unless T_0 is already a function of S ; if $T_0 = h(S)$, then $E[T_0|S] = T_0$ and no improvement occurs

Q12.

UMVUE exists if and only if:

A. The distribution is from exponential family

B. A complete sufficient statistic exists

C. An unbiased estimator exists

D. **Both A and B**

Correct answer: D**EXAM SHORTCUT**

Need both: (1) an unbiased estimator must exist, and (2) a complete sufficient statistic must exist; then Lehmann-Scheffé gives unique UMVUE

Q13.

Method of moments estimators are consistent because:

- A. They are functions of sufficient statistics
- B. **Sample moments converge to population moments by LLN**
- C. They attain CRLB
- D. They are always unbiased

Correct answer: B

EXAM SHORTCUT

LLN: $\bar{X}(k)/(p) \rightarrow E[X^k]$; continuous mapping theorem: MOM estimators (continuous functions of sample moments) are consistent

Q14.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, UMVUE of $|\mu|$:

- A. Does not exist (no unbiased estimator of $|\mu|$)
- B. $|X|$
- C. **$\sqrt{n} \phi(|X|/\sqrt{n}\sigma) - \sigma \phi(|X|/\sqrt{n}\sigma)/\sqrt{n}$ (complex expression)**
- D. $|X|$ corrected for bias

Correct answer: C

EXAM SHORTCUT

UMVUE of $|\mu|$ exists but has complex form involving normal CDF and PDF; $|X|$ is biased for $|\mu|$; this is a classic hard ISS question

Q15.

For $X_1, \dots, X_n \sim e^{-(x-\theta)}, x > \theta$, UMVUE of θ is:

- A. $\bar{X} - 1$
- B. $X_{(1)} + 1/n$
- C. **$X_{(1)} - 1/n$**
- D. $X_{(1)}$

Correct answer: C

EXAM SHORTCUT

$E[X_{(1)}] = \theta + 1/n$; unbiased: $X_{(1)} - 1/n$; $X_{(1)}$ is complete sufficient; so $X_{(1)} - 1/n$ is UMVUE

Estimation

MOM Negative Binomial

Hard

Q16.For $X \sim \text{NegBin}(r, p)$ with r known, MOM estimator of p :

- A. rX
- B. $r/(r+X)$
- C. $-X/(r+X)$
- D. $1/(1+X/r)$

Correct answer: B**EXAM SHORTCUT** $E[X] = r(1-p)/p$; solving: $\hat{p} = r/(r+X)$; same as MLE in this case; D is equivalent to B

Estimation

UMVUE Cauchy

Hard

Q17.For $X_1, \dots, X_n \sim \text{Cauchy}(\theta, 1)$, UMVUE of θ :

- A. Is the MLE
- B. Is the sample median
- C. Is $\bar{X}$
- D. Does not exist (no complete sufficient statistic)

Correct answer: D**EXAM SHORTCUT**

Cauchy is not exponential family; minimal sufficient statistic is all order statistics; no complete sufficient statistic exists; hence no UMVUE

Estimation

Rao Blackwell Poisson Indicator

Hard

Q18.For $X_1, \dots, X_n \sim \text{Poisson}(\lambda)$, improve $T_0 = I(X_1 = 1)$ using $T = \sum X_i$:

- A. $E[I(X_1=1)|T=t] = e^{-1/n}(1/n)^t/t!$ (Poisson conditional)
- B. $E[I(X_1=1)|T=t] = C(t,1)(1/n)(1-1/n)^{t-1}$
- C. $E[I(X_1=1)|T=t] = t/n$
- D. $E[I(X_1=1)|T=t] = t(1-1/n)^{t-1}/n$

Correct answer: B**EXAM SHORTCUT**Given $T=t$, $X_1 \sim \text{Bin}(t, 1/n)$; $P(X_1=1|T=t) = C(t,1)(1/n)(1-1/n)^{t-1} = t(1-1/n)^{t-1}/n$; D and B are equivalent.

Q19.

For $X_1, \dots, X_n \sim U(0, \theta)$, UMVUE of $P(X \leq c) = c/\theta$ (for fixed $c < \theta$) is:

- A. $\min(c/X(n), 1)$
- B. $cn/[(n+1)X(n)]$
- C. $c/X(n)$
- D. $\frac{n+1}{n} \cdot c/X(n)$ if $X(n) > c$, else 1

Correct answer: D

EXAM SHORTCUT

Need to handle case $X(n) \leq c$ (then $P(X \leq c) = 1$ certainly); UMVUE must be bounded in $[0, 1]$; careful case analysis required

Q20.

For $X \sim \text{LogNormal}(\mu, \sigma^2)$, MOM estimators using $E[X] = e^{\mu + \sigma^2/2}$ and $\text{Var}(X) = e^{2\mu + 2\sigma^2}(e^{\sigma^2} - 1)$:

- A. $\hat{\sigma}^2 = \log(1 + S^2/\bar{X}^2)$, $\hat{\mu} = \log \bar{X} - \hat{\sigma}^2/2$
- B. Both A and B (A is MOM, B is MLE)
- C. $\hat{\mu} = \log \bar{X}$, $\hat{\sigma}^2 = S^2_{\log}$
- D. $\hat{\mu} = \log \bar{X}$, $\hat{\sigma}^2 = S^2/\bar{X}^2$

Correct answer: B

EXAM SHORTCUT

MOM uses raw moments (A); MLE uses log-transformed data (B); they differ; C is also the UMVUE approach

Q21.

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, UMVUE of $p(1-p) = \text{Var}(X)$ is:

- A. Both B and C (equivalent)
- B. $T(n-T)/n(n-1)$ where $T = \sum X_i$
- C. $S^2 = n^{-1} \sum (X_i - \bar{X})^2$
- D. $\bar{X}(1 - \bar{X})$

Correct answer: A

EXAM SHORTCUT

$S^2 = (T(n-T))/(n)(n-1)$ for Bernoulli data; both B and C are the same expression; $\bar{X}(1 - \bar{X})$ uses n denominator (biased)

Q22.

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$, UMVUE of $S(t) = P(X > t) = e^{-\lambda t}$ is:

A. $(1/n) \sum I(X_i > t)$

B. $(1 - t/T)^{n-1} I(T > t)$ where $T = \sum X_i$

C. $e^{-\hat{t}\lambda}$ where $\hat{\lambda} = 1/\bar{X}$

D. $e^{-t/\bar{X}}$

Correct answer: B

EXAM SHORTCUT

$E[I(X_1 > t) | T = \sum X_i]$: given T , $X_1/T \sim \text{Beta}(1, n-1)$; $P(X_1 > t | T) = (1 - t/T)^{n-1} I(T > t)$

Q23.

For Weibull(k, λ) with both unknown, MOM requires:

A. $\hat{k} = 1/\log \bar{X}$, $\hat{\lambda} = \bar{e} \bar{X}$

B. **Solving two equations involving gamma functions:** $E[X] = \lambda \Gamma(1 + 1/k)$, $E[X^2] = \lambda^2 \Gamma(1 + 2/k)$

C. $\hat{k} = \bar{X}/S$, $\hat{\lambda} = \bar{X}$

D. Closed form exists: $\hat{k} = \sqrt{6S/(\bar{t}\bar{X})}$

Correct answer: B

EXAM SHORTCUT

Weibull MOM: two equations with gamma functions; ratio $E[X^2]/(E[X])^2 = \Gamma(1 + 2/k)/[\Gamma(1 + 1/k)]^2$ depends only on k ; solve numerically

Q24.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with both unknown, UMVUE of σ is:

A. **Both B and C (equivalent)**

B. S/c_4 where $c_4 = \sqrt{2/(n-1)} \cdot \Gamma(n/2)/\Gamma((n-1)/2)$

C. $S = \sqrt{S^2}$

D. $S \cdot \sqrt{(n-1)/2} \cdot \Gamma((n-1)/2)/\Gamma(n/2)$

Correct answer: A

EXAM SHORTCUT

$E[S] = \sigma c_4$; UMVUE of σ is S/c_4 ; $c_4 < 1$ so UMVUE $> S$; for large n , $c_4 \approx 1$

Q25.

For $X_1, \dots, X_n \sim \text{Geometric}(p)$, improve $T_0 = I(X_1 = 1)$ using $S = \sum X_i$:

- A. $E[I(X_1 = 1)|S = s] = n/s$
- B. $E[I(X_1 = 1)|S = s] = ((s-1)/(n-1))/(s/n)$ (negative hypergeometric)
- C. $E[I(X_1 = 1)|S = s] = (n-1)/(s-1)$
- D. $E[I(X_1 = 1)|S = s] = p$

Correct answer: A

EXAM SHORTCUT

Given $\sum X_i = s$, by symmetry $P(X_1 = 1|S = s) = P(X_j = 1|S = s)$ for all j ; sums to $n \cdot P(X_1 = 1|S = s)$; careful computation gives n/s

Q26.

For $X_1, \dots, X_n \sim \text{Pareto}(\theta)$, $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, UMVUE of θ is:

- A. $-n / \sum \log X_i$
- B. $-(n-1) / \sum \log X_i$
- C. $n / \sum \log X_i$
- D. $(n+1) / \sum \log X_i$

Correct answer: B

EXAM SHORTCUT

$T = \sum \log X_i$; $-\log X_i \sim \text{Exp}(\theta)$; $T \sim \Gamma(n, 1/\theta)$; $E[1/T] = \theta/(n-1)$; UMVUE of θ : $E[-(n-1)/T] = \theta$

Q27.

For $X \sim IG(\mu, \lambda)$, MOM estimators:

- A. $\hat{\mu} = 1/\bar{X}$; $\hat{\lambda} = X^2/S^2$
- B. $\hat{\mu} = \bar{X}$; $\hat{\lambda} = S^2/\bar{X}$
- C. $\hat{\mu} = \bar{X}$; $\hat{\lambda} = X^2/S^2$
- D. $\hat{\mu} = \bar{X}$; $\hat{\lambda} = X^3/S^2$

Correct answer: D

EXAM SHORTCUT

$E[X] = \mu$, $\text{Var}(X) = \mu^3/\lambda$; $\hat{\mu} = \bar{X}$; $\hat{\lambda} = \mu^3/S^2 = X^3/S^2$

Q28.

For Hypergeometric(N, M, n) with N known, UMVUE of M/N (population proportion) is:

- A. $(X/n) \cdot N/(N-1)$
- B. X/n is already UMVUE
- C. X/N
- D. X/n (sample proportion)

Correct answer: B

EXAM SHORTCUT

X/n is unbiased for M/N and is a function of the complete sufficient statistic X ; hence it is UMVUE

Q29.

For $N(\mu, 1)$ truncated to $(0, \infty)$, MOM estimator of μ satisfies:

- A. $\hat{\mu} = X \cdot \Phi(X)$
- B. $\hat{\mu} = X - 1$
- C. $X = \mu + \phi(\mu)/\Phi(\mu)$ (Mills ratio equation)
- D. $\hat{\mu} = X$

Correct answer: C

EXAM SHORTCUT

$E[X|X > 0] = \mu + \phi(\mu)/\Phi(\mu)$; setting equal to X gives implicit equation; same as MLE equation for truncated normal

Q30.

For $X_1, \dots, X_n \sim \text{Laplace}(\theta, 1)$, UMVUE of θ :

- A. Does not exist (no complete sufficient statistic)
- B. Is X
- C. Is a weighted average of order statistics
- D. Is sample median

Correct answer: A

EXAM SHORTCUT

Laplace is a location family; order statistics are minimal sufficient but NOT complete; no UMVUE exists; sample median is best equivariant

Q31.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, improve $T_0 = \bar{X}^2 - \sigma^2$ (unbiased for μ^2) using $S = \sum X_i$:

- A. $\bar{X}^2 - \sigma^2$
- B. $\bar{X}^2$
- C. $\bar{X}^2 + \sigma^2/n$
- D. $\bar{X}^2 - \sigma^2/n$

Correct answer: D

EXAM SHORTCUT

$E[X_1^2 - \sigma^2 | \bar{X}] = E[X_1^2 | \bar{X}] - \sigma^2$; $E[X_1^2 | \bar{X}] = \bar{X}^2 + \sigma^2(n-1)/n$; so result = $\bar{X}^2 + \sigma^2(n-1)/n - \sigma^2 = \bar{X}^2 - \sigma^2/n$ ✓

Q32.

For $X_1, \dots, X_n \sim \text{Bin}(m, p)$ with m known, UMVUE of $P(X = k) = \binom{m}{k} p^k (1-p)^{m-k}$ is:

- A. $\binom{m}{k} p^k (1-p)^{m-k}$ where $p = \bar{X}/m$
- B. $\binom{nm-T}{m-k} \binom{T-k}{k} / \binom{nm}{m}$ where $T = \sum X_i$
- C. $\binom{m}{k} \binom{nm-m-T-k}{nm-T} / \binom{nm}{m}$ where $T = \sum X_i$
- D. Does not exist in closed form

Correct answer: C

EXAM SHORTCUT

Hypergeometric-type conditional probability; $E[I(X_1 = k) | T = \sum X_i]$ computed via hypergeometric distribution; C is the correct form

Q33.

For Generalized Pareto $GP(\xi, \sigma)$, MOM estimators:

- A. $\hat{\xi} = 1/2 - X_2/S_2$, $\hat{\sigma} = X(1 + \hat{\xi})$
- B. $\hat{\xi} = 1 - X_2/(2S_2)$, $\hat{\sigma} = X(1 - \hat{\xi})$ (Hosking-Wallis)
- C. $\hat{\xi} = X/S$, $\hat{\sigma} = S$
- D. $\hat{\xi} = S/X - 1$, $\hat{\sigma} = X$

Correct answer: B

EXAM SHORTCUT

$E[X] = \sigma/(1 - \xi)$, $\text{Var}(X) = \sigma^2/[(1 - \xi)^2(1 - 2\xi)]$; solving gives Hosking-Wallis MOM; valid only for $\xi < 1/2$

Q34.

For $X_1, \dots, X_n \sim \text{Geometric}(p)$, UMVUE of p is:

- A. $n / \sum X_i$
- B. $1/(X-1)$
- C. $1/X$
- D. $(n-1)/(\sum X_i - 1)$

Correct answer: D

EXAM SHORTCUT

$T = \sum X_i \sim \text{NegBin}(n, p)$; $E[T] = n/p$; unbiased estimator: n/T is NOT exactly unbiased; correct UMVUE: $(n-1)/(T-1)$ since $E[(n-1)/(T-1)] = p$ for NegBin

Q35.

For $X_1, \dots, X_n \sim U(0, \theta)$, improve $T_0 = 2X_1$ (unbiased for θ) using $S = X(n)$:

- A. $E[2X_1 | X(n) = t] = (n+1)/(nt)$
- B. $E[2X_1 | X(n) = t] = 2t/n+1$
- C. $E[2X_1 | X(n) = t] = t$
- D. $E[2X_1 | X(n) = t] = ((n+1)t)/(n)$

Correct answer: A

EXAM SHORTCUT

Given $X(n) = t$: $E[X_1 | X(n) = t] = (1)/(nt) + (n-1) \cdot t/(2) = t(n+1)/(2n)$; so $E[2X_1 | X(n) = t] = ((n+1)t)/(n)$ ✓

Q36.

For compound Poisson where $N \sim \text{Poisson}(\lambda)$ and $Y = \sum_{i=1}^N X_i$ with $X_i \sim \text{Exp}(\mu)$, MOM for (λ, μ) :

- A. $\hat{\mu} = 2Y/S^2, \hat{\lambda} = Y \mu$
- B. $\hat{\mu} = \sqrt{2Y/S^2}, \hat{\lambda} = Y \mu$
- C. $\hat{\lambda} = Y \mu, \hat{\mu} = Y/S^2$ solve simultaneously
- D. $E[Y] = \lambda/\mu, \text{Var}(Y) = 2\lambda/\mu^2$ (use $E[Y] = \lambda E[X], \text{Var}(Y) = \lambda E[X^2]$)

Correct answer: A

EXAM SHORTCUT

$E[Y] = \lambda/\mu, \text{Var}(Y) = \lambda E[X^2] = \lambda \cdot 2/\mu^2$; ratio: $\text{Var}/E^2 = 2/\mu$; $\hat{\mu} = 2Y/S^2$; then $\hat{\lambda} = Y \mu$

Estimation

UMVUE Normal Quantile

Hard

Q37.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, UMVUE of $\xi_p = \mu + z_p \sigma$ (the p -th quantile) is:

A. $\bar{X} + z_p \sigma$ B. $\bar{X} + z_p S$ C. $\bar{X} + z_p \sigma \sqrt{1 + 1/n}$ D. $\bar{X} + z_p \sigma / \sqrt{n}$

Correct answer: A

EXAM SHORTCUT

$\bar{X}$ is UMVUE of μ ; $z_p \sigma$ is a constant; UMVUE of $\mu + z_p \sigma$ is $\bar{X} + z_p \sigma$; note σ is KNOWN here

Estimation

MOM Mixture Normal

Hard

Q38.

For mixture $\pi N(\mu_1, \sigma^2) + (1 - \pi)N(\mu_2, \sigma^2)$ with σ^2 known, MOM requires:

A. 2 moment equations

B. Only first moment equation

C. EM algorithm (MOM not applicable)

D. At least 3 moment equations for 3 unknowns (π, μ_1, μ_2)

Correct answer: D

EXAM SHORTCUT

3 unknown parameters need 3 equations; use $E[X]$, $E[X^2]$, $E[X^3]$; mixture MOM can have multiple solutions — EM preferred

Estimation

UMVUE Negative Binomial

Hard

Q39.

For $X_1, \dots, X_n \sim \text{NegBin}(r, p)$ with r known, UMVUE of p is:

A. $nr / \sum X_i$ B. $r / (\bar{X} + r)$ C. $(nr - 1) / (\sum X_i + nr - 1)$ D. $nr / (\sum X_i + nr)$

Correct answer: C

EXAM SHORTCUT

$T = \sum X_i \sim \text{NegBin}(nr, p)$; UMVUE of p : $E[(nr - 1) / (T + nr - 1)] = p$ (analogous to geometric case); careful with exact formula

Q40.

For $GEV(\mu, \sigma, \xi)$, MOM estimators:

- A. Require $\xi < 1/2$ (finite variance)
- B. Are always less efficient than MLE
- C. ~~Exist~~ only for $\xi < 1/3$ (finite third moment required)
- D. Exist in closed form for all ξ

Correct answer: C

EXAM SHORTCUT

GEV moments: $E[X]$ exists for $\xi < 1$; $\text{Var}(X)$ for $\xi < 1/2$; third moment for $\xi < 1/3$; MOM using first 3 moments requires $\xi < 1/3$

Confidence Intervals

50 forecast questions in this topic

Estimation

CI Normal Mean

Medium

Q1.

100(1 - α)% CI for μ when σ^2 known is:

A. $\bar{X} \pm z_{\alpha/2} S / \sqrt{n}$

B. $\bar{X} \pm t_{\alpha/2, n-1} S / \sqrt{n}$

C. $\bar{X} \pm z_{\alpha/2} \sigma$

D. $\bar{X} \pm z_{\alpha/2} \sigma / \sqrt{n}$

Correct answer: D

EXAM SHORTCUT

Known σ : use $z_{\alpha/2}$; unknown σ : use t_{n-1} ; never mix z with S or t with σ

Estimation

CI Normal Variance

Medium

Q2.

100(1 - α)% CI for σ^2 is:

A. $[(n-1)S^2 / (\chi^2_{\alpha/2, n-1}) / ((n-1)S^2 / \chi^2_{1-\alpha/2, n-1})]$

B. $[S^2 \pm z_{\alpha/2} \sqrt{2\sigma^4/n}]$

C. $[(n-1)S^2 / (\chi^2_{1-\alpha/2, n-1}) / ((n-1)S^2 / \chi^2_{\alpha/2, n-1})]$

D. $[(nS^2 / (\chi^2_{\alpha/2, n}) / (nS^2 / \chi^2_{1-\alpha/2, n})]$

Correct answer: C

EXAM SHORTCUT

χ^2 CI: smaller χ^2 value in denominator gives LARGER bound; lower bound uses LARGER χ^2 quantile; $(n-1)$ df not n

Estimation

CI Proportion

Medium

Q3.

Wald CI for proportion p is:

A. $\hat{p} \pm z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}$

B. $\hat{p} \pm z_{\alpha/2} / \sqrt{n}$

C. $\hat{p} \pm t_{\alpha/2, n-1} \sqrt{\hat{p}(1-\hat{p})/n}$

D. $\hat{p} \pm z_{\alpha/2} \sqrt{p(1-p)/n}$

Correct answer: A

EXAM SHORTCUT

Wald uses estimated $\hat{p}$ in SE; Wilson CI is preferred for small n ; D uses true p (unknown); never use t for proportions

Q4.

For symmetric unimodal distributions, shortest CI is:

A. Highest density region (HDR)

B. One-sided CI

C. Depends on sample size

D. Equal-tails CI

Correct answer: A

EXAM SHORTCUT

For symmetric distributions, equal-tails = HDR; for skewed (e.g., χ^2), HDR is shorter than equal-tails; equal-tails is NOT always shortest

Q5.

A pivotal quantity $Q(X, \theta)$ has distribution:

A. That depends on θ

B. That is always χ^2

C. That is always normal

D. That does not depend on θ

Correct answer: D

EXAM SHORTCUT

Pivot: distribution free of θ ; examples: $(\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)$; $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$

Q6.

For $X_1, \dots, X_n \sim \text{Exp}(\lambda)$, exact CI for λ uses pivot:

A. $\sqrt{n}(\bar{X} - 1/\lambda) \sim N(0, 1/\lambda^2)$

B. $2\lambda \sum X_i \sim \chi^2_{2n}$

C. $n\lambda \bar{X} \sim \chi^2_n$

D. $\lambda \sum X_i \sim \chi^2_n$

Correct answer: B

EXAM SHORTCUT

$2\lambda \sum X_i \sim \chi^2_{2n}$; sum of n independent χ^2_2 is χ^2_{2n} ; CI: $[\chi^2_{1-\alpha/2, 2n}/(2 \sum X_i), \chi^2_{\alpha/2, 2n}/(2 \sum X_i)]$

Q7.

For $X_1, \dots, X_n \sim U(0, \theta)$, exact one-sided CI for θ based on $X(n)$ is:

- A. $[X(n)/\alpha^{1/n}, \infty)$
- B. $[X(n), X(n) \cdot \alpha^{-1/n}]$
- C. $[X(n) \cdot \alpha^{1/n}, \infty)$
- D. $[X(n), X(n) \cdot \alpha^{1/n}]$

Correct answer: B

EXAM SHORTCUT

$U = X_{(n)}/\theta \sim \text{Beta}(n, 1)$; $P(U \geq \alpha^{1/n}) = 1 - \alpha$; so $P(\theta \leq X_{(n)}/\alpha^{1/n}) = 1 - \alpha$; one-sided CI: $[X_{(n)}, X_{(n)} \cdot \alpha^{-1/n}]$.

Q8.

Fiducial interval (Fisher) differs from confidence interval in that:

- A. **It treats θ as random given observed data**
- B. It is always shorter
- C. It requires prior distribution
- D. It has different coverage probability

Correct answer: A

EXAM SHORTCUT

Fiducial: treats parameter as having distribution induced by data (without prior); numerically equals CI for location families; philosophically different from frequentist CI

Q9.

A 95% Bayesian credible interval differs from 95% frequentist CI in that:

- A. **$P(\theta \in \text{CI} | X) = 0.95$ (posterior probability)**
- B. It is always wider
- C. Coverage is 95% in repeated sampling
- D. $P(\theta \in \text{CI}) = 0.95$ (prior probability)

Correct answer: A

EXAM SHORTCUT

Bayesian: posterior probability statement about θ ; frequentist: coverage in repeated sampling; C is the frequentist interpretation; A is Bayesian

Q10.

Wilson CI for proportion p is centered at:

- A. $\hat{p} + z^2 \alpha/2 / (2n)$ (adjusted center)
- B. p (true proportion)
- C. $\hat{p}$
- D. $1/2$

Correct answer: A

EXAM SHORTCUT

Wilson CI: add $z^2/(2n)$ pseudo-observations; center shifts toward $1/2$; better coverage than Wald for small n or extreme p

Q11.

Prediction interval for future observation $X_{n+1} \sim N(\mu, \sigma^2)$ with σ^2 known is:

- A. $\bar{X} \pm z\alpha/2\sigma$
- B. $\bar{X} \pm t_{\alpha/2, n-1} S \sqrt{1 + 1/n}$
- C. $\bar{X} \pm z\alpha/2\sigma/\sqrt{n}$
- D. $\bar{X} \pm z\alpha/2\sigma\sqrt{1 + 1/n}$

Correct answer: D

EXAM SHORTCUT

Prediction interval wider than CI: $X_{n+1} - \bar{X} \sim N(0, \sigma^2(1 + 1/n))$; B is correct when σ unknown; D when σ known

Q12.

A tolerance interval $[L, U]$ with content p and confidence $1 - \alpha$ means:

- A. $P(L \leq X \leq U) = p$
- B. $P(P(L \leq X \leq U) \geq p) \geq 1 - \alpha$
- C. $P(L \leq X_{n+1} \leq U) = p$
- D. $P(L \leq \mu \leq U) = 1 - \alpha$

Correct answer: B

EXAM SHORTCUT

Tolerance interval: probabilistic statement about coverage of future population; A is exact (not achievable for continuous distributions); D is CI for mean; C is prediction interval

Estimation

CI Bootstrap Percentile

Medium

Q13.

Bootstrap percentile CI uses:

A. $\sqrt{\theta^*(*)/((\alpha/2), \hat{\theta}^*) (1-\alpha/2)}$ — quantiles of bootstrap distributionB. $[\hat{\theta}^* - \theta^*/((1-\alpha/2), \hat{\theta}^* - \theta^*) (\alpha/2)]$ — reflectedC. $\hat{\theta} \pm t_{\alpha/2, n-1} SE_{boot}$ D. $\hat{\theta} \pm z_{\alpha/2} SE_{boot}$

Correct answer: A

EXAM SHORTCUT

Percentile method: A; Basic/Reverse percentile: B; Normal bootstrap: D; BCa (bias-corrected accelerated) is most accurate

Estimation

CI Bootstrap BCa

Hard

Q14.

BCa bootstrap CI corrects for:

A. Only bias

B. Bias and skewness of bootstrap distribution

C. Only skewness

D. Non-normality of original data

Correct answer: B

EXAM SHORTCUT

BCa = Bias-Corrected and accelerated; bias correction $\hat{z}_0$ and acceleration $\hat{a}$ both applied; most accurate bootstrap CI method

Estimation

CI Jackknife SE

Medium

Q15.

Jackknife standard error estimate is:

A. $\sqrt{(n-1)/(n) \sum (\theta(i) - \hat{\theta}(\cdot))^2}$ B. $\sqrt{1/(n) \sum (\theta(i) - \hat{\theta}(\cdot))^2}$ C. $\sqrt{(n-1)/(n^2) \sum (\theta(i) - \hat{\theta}(\cdot))^2}$ D. $\sqrt{(n-1)/(n) \sum (\theta(i) - \hat{\theta})^2}$

Correct answer: A

EXAM SHORTCUT

Jackknife SE = $\sqrt{(n-1)/(n) \sum_{i=1}^n (\theta(i) - \hat{\theta}(\cdot))^2}$; the $(n-1)/n$ factor (not $1/n$) is crucial

Estimation

CI Optimum Unbiased

Hard

Q16.

Optimum confidence interval (shortest unbiased) for μ in $N(\mu, \sigma^2)$ with σ^2 unknown is:

- A. z-interval using S
- B. Shortest t -interval (not equal-tails)
- C. One-sided t -interval
- D. **Equal-tails t -interval**

Correct answer: D

EXAM SHORTCUT

For t -distribution (symmetric), equal-tails interval IS the shortest unbiased interval; for asymmetric distributions (e.g., χ^2), equal-tails is NOT shortest

Estimation

CI Ratio Two Means

Hard

Q17.

CI for ratio μ_1/μ_2 of two normal means uses:

- A. Fieller's theorem
- B. Log transformation then back-transform
- C. **All of the above are valid approaches**
- D. Delta method approximation

Correct answer: C

EXAM SHORTCUT

Fieller's theorem gives exact CI; delta method gives approximate CI; log transformation is another approximation; all are used in practice

Estimation

CI Nonparametric Median

Hard

Q18.

Distribution-free CI for population median uses:

- A. Sample median $\pm z_{\alpha/2} \cdot \pi/(2n)$ (asymptotic)
- B. Bootstrap only
- C. $\bar{X} \pm z_{\alpha/2} S/\sqrt{n}$
- D. **Order statistics $X(r)$ and $X(s)$ with coverage from binomial distribution**

Correct answer: D

EXAM SHORTCUT

Exact nonparametric CI for median: $[X(r), X(s)]$ where r, s chosen so $P(r \leq \text{Bin}(n, 1/2) \leq s-1) \geq 1 - \alpha$; A is asymptotic approximation

Q19.

Bonferroni simultaneous CIs for k parameters at family-wise level $1 - \alpha$ use individual level:

- A. $1 - k\alpha$
- B. $(1 - \alpha)^{1/k}$
- C. $1 - \alpha/2$
- D. $1 - \alpha/k$

Correct answer: D

EXAM SHORTCUT

Bonferroni: use α/k per interval; B is Šidák correction (exact for independent tests); both are conservative

Q20.

Double bootstrap (bootstrap-t) CI:

- A. Applies bootstrap twice to reduce bias
- B. Is equivalent to BCa
- C. Uses bootstrap to estimate the distribution of $(\hat{\theta} - \theta)/SE$
- D. Uses t -distribution with bootstrap SE

Correct answer: C

EXAM SHORTCUT

Bootstrap-t: estimate distribution of studentized statistic via nested bootstrap; computationally expensive but second-order accurate; different from BCa

Q21.

Profile likelihood CI for ψ is:

- A. $\hat{\psi} \pm z_{\alpha/2} / \sqrt{nl(\hat{\psi})}$
- B. $\{\psi: \ell p(\psi) \geq \ell p(\hat{\psi}) - z_{\alpha/2}^2 / 2\}$
- C. Both A and B (equivalent since $\chi^2_{\alpha,1} = z_{\alpha/2}^2$)
- D. $\{\psi: -2[\ell p(\hat{\psi}) - \ell p(\psi)] \leq \chi^2_{\alpha,1}\}$

Correct answer: C

EXAM SHORTCUT

$\chi^2_{0.05,1} = 3.841 = 1.96^2$; D and B are equivalent; profile likelihood CI is second order accurate unlike Wald CI

Q22.

One-sided 95% upper CI for λ in $\text{Exp}(\lambda)$ based on n observations:

- A. $\lambda \leq z_{0.05}/(\bar{X}/n)$
- B. $\lambda \leq \chi^2_{0.05, 2n} / \sum X_i$
- C. $\lambda \leq \chi^2_{0.95, 2n} / (2 \sum X_i)$
- D. $\lambda \leq \chi^2_{0.05, 2n} / (2 \sum X_i)$

Correct answer: D

EXAM SHORTCUT

$P(2\lambda \sum X_i \leq \chi^2_{0.05, 2n}) = 0.05$; upper CI: $\lambda \leq \chi^2_{0.05, 2n} / (2 \sum X_i)$; note $\chi^2_{0.05}$ is LOWER tail quantile giving UPPER bound for λ

Q23.

Exact CI for Poisson rate λ based on observed count x is:

- A. $[\bar{X} \pm z_{\alpha/2} \sqrt{\bar{X}/n}]$
- B. $[\chi^2_{\alpha/2, 2x} / (2n), \chi^2_{1-\alpha/2, 2(x+1)} / (2n)]$
- C. $[\chi^2_{1-\alpha/2, 2x} / (2n), \chi^2_{\alpha/2, 2(x+1)} / (2n)]$
- D. $[x/n \pm z_{\alpha/2} \sqrt{x/n}]$

Correct answer: B

EXAM SHORTCUT

Exact Poisson CI: lower = $\chi^2_{\alpha/2, 2x} / 2$ (for rate per unit), upper = $\chi^2_{1-\alpha/2, 2(x+1)} / 2$; divide by n for rate; careful with which tail goes where

Q24.

Fisher's z-transformation CI for correlation ρ :

- A. $\hat{\rho} \pm z_{\alpha/2} / \sqrt{n}$
- B. $[\tanh(\tanh^{-1}(\hat{\rho}) \pm z_{\alpha/2} / \sqrt{n-3})]$
- C. $\hat{\rho} \pm z_{\alpha/2} \sqrt{(1 - \hat{\rho}^2)^2 / (n-1)}$
- D. $\hat{\rho} \pm z_{\alpha/2} \sqrt{(1 - \hat{\rho}^2) / (n-2)}$

Correct answer: B

EXAM SHORTCUT

Fisher Z: $Z = \tanh^{-1}(\hat{\rho}) = (1/2) \log \frac{1+\hat{\rho}}{1-\hat{\rho}}$; $Z \sim N(\tanh^{-1}(\rho), 1/(n-3))$; back-transform via $\tanh$

Q25.

CI for $p_1 - p_2$ (two independent proportions):**A. ✗A for CI; B for hypothesis test****B. $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\hat{p}(1 - \hat{p})(1/n_1 + 1/n_2)}$ where $\hat{p}$ is pooled**

C. Both A and B are valid CIs

D. $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\hat{p}_1(1 - \hat{p}_1)/n_1 + \hat{p}_2(1 - \hat{p}_2)/n_2}$ **Correct answer: A****EXAM SHORTCUT**For CI: use unpooled SE (A); for hypothesis test $H_0: p_1 = p_2$: use pooled SE (B); common exam trap to mix them

Q26.

CI for slope β in simple linear regression:**A. $\hat{\beta} \pm t_{\alpha/2, n-2} \cdot S / \sqrt{S_{xx}}$ where $S_{xx} = \sum (x_i - \bar{x})^2$** **B. $\hat{\beta} \pm t_{\alpha/2, n-2} \cdot S_{\beta}$ where $S_{\beta}^2 = S^2 / S_{xx}$** C. $\hat{\beta} \pm z_{\alpha/2} \cdot \sigma / \sqrt{S_{xx}}$ **D. $\hat{\beta} \pm t_{\alpha/2, n-1} \cdot S / \sqrt{S_{xx}}$** **Correct answer: B****EXAM SHORTCUT**A and B are equivalent; $n-2$ df (not $n-1$); $S^2 = \text{RSS}/(n-2)$; C uses known σ

Q27.

Parametric bootstrap CI:

A. Is equivalent to nonparametric bootstrap

B. Uses analytical formula with bootstrap SE

C. Resamples from empirical distribution

D. ✗Resamples from fitted parametric model $f(x; \hat{\theta})$ **Correct answer: D****EXAM SHORTCUT**Parametric bootstrap: simulate from $f(x; \hat{\theta})$; nonparametric: resample from data; they differ when model is misspecified

Q28.

Jackknife bias-corrected estimator is:

A. **All of the above (equivalent)**B. $\hat{\theta} + \text{biasjack}$ where $\text{bias} = (n-1)(\hat{\theta}(\cdot) - \hat{\theta})$ C. $\hat{n}\hat{\theta} - (n-1)\hat{\theta}(\cdot)$ D. $\hat{\theta} - (n-1)(\hat{\theta}(\cdot) - \hat{\theta})$ **Correct answer: A****EXAM SHORTCUT**Jackknife bias estimate = $(n-1)(\hat{\theta}(\cdot) - \hat{\theta})$; bias-corrected = $\hat{\theta} - \hat{\text{bias}} = \hat{\theta} - (n-1)(\hat{\theta}(\cdot) - \hat{\theta}) = \hat{n}\hat{\theta} - (n-1)\hat{\theta}(\cdot)$; all equivalent

Q29.

Subsampling CI differs from bootstrap in that:

A. It is valid under weaker conditions than bootstrap

B. It requires correct specification of convergence rate

C. It samples WITHOUT replacement (subsets of size $b < n$)D. **All of the above****Correct answer: D****EXAM SHORTCUT**Subsampling: without replacement, size $b \rightarrow \infty$ with $b/n \rightarrow 0$; valid even when bootstrap fails (e.g., for maximum of i.i.d. uniform); requires knowing rate

Q30.

Scheffé simultaneous CI for all linear contrasts $c^T\beta$ in regression:A. $\hat{c^T\beta} \pm t_{\alpha/2, n-p} \cdot \text{SE}$ B. $\hat{c^T\beta} \pm \sqrt{F_{\alpha, 1, n-p}} \cdot \text{SE}$ C. $\hat{c^T\beta} \pm z_{\alpha/2} \cdot \text{SE}$ D. **$\hat{c^T\beta} \pm \sqrt{p \cdot F_{\alpha, p, n-p}} \cdot \text{SE}$** **Correct answer: D****EXAM SHORTCUT**Scheffé: $\sqrt{p \cdot F}$ critical value covers ALL contrasts simultaneously; Bonferroni covers only pre-specified contrasts; p = number of parameters

Q31.

Bootstrap CI is NOT valid when:

- A. ✗Statistic is the sample maximum from $U(0, \theta)$**
- B. Distribution is skewed
- C. Sample size is small
- D. Distribution has heavy tails

Correct answer: A

EXAM SHORTCUT

Bootstrap fails for non-regular statistics like sample maximum (convergence rate n^{-1} not $n^{-1/2}$); subsampling or m -out-of- n bootstrap needed

Q32.

Likelihood ratio CI has the property of being:

- A. Exact for all distributions
- B. Always equal to Wald CI
- C. ✗Transformation invariant (equivariant under reparametrization)**
- D. Always shorter than Wald CI

Correct answer: C

EXAM SHORTCUT

LR CI is transformation invariant (key advantage over Wald); Wald CI is NOT invariant; LR CI is generally more accurate than Wald in finite samples

Q33.

For bootstrap CI, recommended number of bootstrap replications B :

- A. $B=100$ is always sufficient
- B. $B=n$ (same as sample size)
- C. $B=\infty$ (theoretical limit)
- D. ✗ $B \geq 1000$ for percentile CI; $B \geq 2000$ for BCa**

Correct answer: D

EXAM SHORTCUT

Practical guidelines: $B=1000$ for SE and percentile; $B=2000-5000$ for BCa; more replications reduce Monte Carlo error in CI endpoints

Q34.

Clopper-Pearson exact CI for p based on x successes in n trials:

- A. $[p \pm z_{\alpha/2} \sqrt{p(1-p)/n}]$
- B. **Both A and C (equivalent via Beta-F relationship)**
- C. $[F_{\alpha/2, 2x, 2(n-x+1)} \cdot x / (n-x+1 + F \cdot x), \dots]$
- D. $[B_{\alpha/2; x, n-x+1}, B_{1-\alpha/2; x+1, n-x}]$ where B is Beta quantile

Correct answer: B

EXAM SHORTCUT

Clopper-Pearson uses Beta distribution; equivalent F-distribution form also exists; both are exact (conservative); A is Wald (approximate)

Q35.

Prediction interval for new response Y_0 at x_0 in regression:

- A. **$\hat{Y}_0 \pm t_{\alpha/2, n-2} \sqrt{1 + 1/n + (x_0 - \bar{x})^2 / S_{xx}}$**
- B. $\hat{Y}_0 \pm t_{\alpha/2, n-2} \sqrt{1/n}$
- C. $\hat{Y}_0 \pm z_{\alpha/2} \sigma \sqrt{1 + 1/n + (x_0 - \bar{x})^2 / S_{xx}}$
- D. $\hat{Y}_0 \pm t_{\alpha/2, n-2} \sqrt{1/n + (x_0 - \bar{x})^2 / S_{xx}}$

Correct answer: A

EXAM SHORTCUT

Prediction interval: extra 1 under square root (for new observation variance); D is CI for mean response; A is wider than D

Q36.

CI for σ_1^2 / σ_2^2 (two independent normal samples):

- A. $[S_1^2 / S_2^2 \pm z_{\alpha/2} \sqrt{2/(n_1 - 1) + 2/(n_2 - 1)}]$
- B. $[S_1^2 / (S_2^2 \cdot F_{\alpha/2, n_1-1, n_2-1}), S_1^2 \cdot F_{\alpha/2, n_2-1, n_1-1} / S_2^2]$
- C. **Both A and C (equivalent)**
- D. $[(S_1^2 / S_2^2) / F_{1-\alpha/2, n_1-1, n_2-1}, (S_1^2 / S_2^2) \cdot F_{1-\alpha/2, n_2-1, n_1-1}]$

Correct answer: C

EXAM SHORTCUT

$F_{1-\alpha/2, m, n} = 1 / F_{\alpha/2, n, m}$; both forms are equivalent; note df order reversal in the reciprocal

Estimation

CI Bootstrap Studentized

Hard

Q37.

Studentized bootstrap (bootstrap-t) CI is:

- A. Exact
- B. Less accurate than percentile bootstrap
- C. **Second-order accurate (error $O(n^{-1})$)**
- D. First-order accurate (error $O(n^{-1/2})$)

Correct answer: C**EXAM SHORTCUT**

Bootstrap-t is second-order accurate; percentile and basic bootstrap are first-order; BCa is also second-order; studentized bootstrap requires nested bootstrap for SE

Estimation

CI Jackknife Failure

Hard

Q38.

Jackknife fails (is inconsistent) for:

- A. Sample variance
- B. **Sample median (non-smooth functional)**
- C. Sample mean
- D. Regression coefficients

Correct answer: B**EXAM SHORTCUT**

Jackknife fails for non-smooth statistics like sample median, sample quantiles; bootstrap is preferred; jackknife works for smooth functionals

Estimation

CI Simultaneous Tukey

Hard

Q39.

Tukey's HSD simultaneous CI for pairwise differences $\mu_i - \mu_j$ in one-way ANOVA uses:

- A. Normal distribution
- B. **Studentized range distribution $q_{\alpha, k, N-k}$**
- C. t -distribution with Bonferroni correction
- D. F -distribution

Correct answer: B**EXAM SHORTCUT**

Tukey uses studentized range (maximum of pairwise t -statistics); exact for balanced designs; Bonferroni is conservative; Tukey is less conservative

Estimation

CI Bootstrap Bias Correction

Hard

Q40.

Bootstrap bias estimate for $\hat{\theta}$ is:

A. $E[\hat{\theta}] - \theta$

B. $\hat{\theta} - \theta$

C. $\bar{\theta} - \hat{\theta}$ where $\bar{\theta}$ is mean of bootstrap estimates

D. $\hat{\theta} - \theta$

Correct answer: C

EXAM SHORTCUT

Bootstrap bias = $E[\hat{\theta}] - \hat{\theta}$; bias-corrected estimate = $2\hat{\theta} - \bar{\theta}$; B has wrong sign

Estimation

CI Fiducial Location

Hard

Q41.

For location family $f(x - \theta)$, fiducial distribution of θ given $X = x$ is:

A. Normal distribution always

B. Posterior with flat prior

C. $f(x - \theta) \cdot \sqrt{n}$ (scaled)

D. Both A and B (coincide for location families)

Correct answer: D

EXAM SHORTCUT

For location families, fiducial = posterior with flat (improper) prior = confidence distribution; all three coincide — key result

Estimation

CI Tolerance Normal

Hard

Q42.

For $N(\mu, \sigma^2)$ with both unknown, tolerance interval covering at least proportion p with confidence $1 - \alpha$:

A. $\bar{X} \pm z_p \sigma$

B. $\bar{X} \pm z_p / 2S \sqrt{1 + 1/n}$

C. $\bar{X} \pm z_p S$

D. $\bar{X} \pm k(n, p, \alpha)S$ where k is tolerance factor from tables

Correct answer: D

EXAM SHORTCUT

Tolerance factor k accounts for estimation of both μ and σ ; $k > z_p$ always; $k \rightarrow z_p$ as $n \rightarrow \infty$; C undercovers

Estimation

CI Bootstrap Smoothed

Hard

Q43.

Smoothed bootstrap adds noise to resampled values to:

- A. Reduce computation time
- B. Handle missing data
- C. **Improve CI coverage for discrete distributions**
- D. Increase bias

Correct answer: C

EXAM SHORTCUT

Smoothed bootstrap: add small random noise (bandwidth h) to discrete resamples; treats discrete data as continuous; improves coverage for discrete statistics

Estimation

CI Confidence Distribution

Hard

Q44.

A confidence distribution $C(\theta)$ satisfies:

- A. **All of the above**
- B. $C(\theta_0) \sim U(0, 1)$ under $H_0: \theta = \theta_0$
- C. $P(\theta \in [C^{-1}(\alpha/2), C^{-1}(1 - \alpha/2)]) = 1 - \alpha$
- D. It is a proper probability distribution over θ

Correct answer: A

EXAM SHORTCUT

Confidence distribution unifies CI, p -values, and point estimates; $C(\theta_0)$ is the p -value for $H_0: \theta \leq \theta_0$; modern framework

Estimation

CI Bootstrap Accelerated

Hard

Q45.

The acceleration constant $\hat{a}$ in BCa bootstrap measures:

- A. Bias of $\hat{\theta}$
- B. Sample size effect
- C. Skewness of bootstrap distribution
- D. **Rate of change of SE with respect to θ**

Correct answer: D

EXAM SHORTCUT

$\hat{a} = \frac{\sum (\theta(\cdot) - \theta(i))^3}{6 [\sum (\theta(\cdot) - \theta(i))^2]^3/2}$ (jackknife-based); measures how SE changes with θ (influence function skewness)

Estimation

CI Regression Simultaneous

Hard

Q46.

Working-Hotelling simultaneous CI band for regression line $E[Y|x] = \alpha + \beta x$:

A. $\hat{Y}(x) \pm z_{\alpha/2} \cdot S \sqrt{1/n + (x - \bar{x})^2 / S_{xx}}$

B. $\hat{Y}(x) \pm \sqrt{2F_{\alpha, 2, n-2}} \cdot S \sqrt{1/n + (x - \bar{x})^2 / S_{xx}}$

C. $\hat{Y}(x) \pm t_{\alpha/2, n-2} \cdot S \sqrt{1/n + (x - \bar{x})^2 / S_{xx}}$

D. $\hat{Y}(x) \pm F_{\alpha, 2, n-2} \cdot S \sqrt{1/n + (x - \bar{x})^2 / S_{xx}}$

Correct answer: B

EXAM SHORTCUT

Working-Hotelling: $\sqrt{2F}$ critical value for ALL x simultaneously; C is pointwise CI; $\sqrt{2F_{0.05, 2, n-2}} > t_{0.025, n-2}$ so band is wider

Estimation

CI Bootstrap Monte Carlo Error

Hard

Q47.

Monte Carlo error in bootstrap CI endpoint $\hat{\theta}(\cdot)$ (α is approximately:)

A. $\sqrt{\alpha(1-\alpha)/B}$; σ^* where σ^* is bootstrap SD

B. $\alpha / \sqrt{B}$

C. $\hat{\sigma}^* / \sqrt{B}$

D. $1 / \sqrt{B}$

Correct answer: A

EXAM SHORTCUT

SE of sample quantile: $\sqrt{p(1-p)/f(\xi_p)}$; for bootstrap distribution, $f(\xi_p) \approx 1/\sigma^*$; so $SE \approx \sigma^* \sqrt{\alpha(1-\alpha)/B}$

Estimation

CI Empirical Likelihood

Hard

Q48.

Empirical likelihood CI (Owen) is:

A. Requires parametric model

B. Equivalent to bootstrap percentile CI

C. **Nonparametric, transformation invariant, Bartlett correctable**

D. Always wider than bootstrap CI

Correct answer: C

EXAM SHORTCUT

Empirical likelihood: nonparametric analog of likelihood ratio CI; shares properties of parametric LR CI (transformation invariance, Bartlett correction); Wilks theorem holds

Q49.

Shortest CI for σ^2 (not equal-tails) requires finding χ^2_L, χ^2_U satisfying:

- A. χ^2_U/χ^2_L minimized
- B. Equal tail probabilities
- C. $1/\chi^2_L - 1/\chi^2_U$ minimized subject to $P(\chi^2_L \leq \chi^2_{n-1} \leq \chi^2_U) = 1 - \alpha$
- D. $\chi^2_{1-\alpha/2, n-1} - \chi^2_{\alpha/2, n-1}$ minimized subject to coverage $1 - \alpha$

Correct answer: C

EXAM SHORTCUT

CI for σ^2 : length $\propto 1/\chi^2_L - 1/\chi^2_U$; minimize this subject to coverage; NOT equal tails; equal-tails is standard but not shortest

Q50.

Double bootstrap corrects coverage error of bootstrap CI from:

- A. $O(n^{-1})$ to $O(n^{-3/2})$
- B. $O(1)$ to $O(n^{-1/2})$
- C. Exact to approximate
- D. $O(n^{-1/2})$ to $O(n^{-1})$

Correct answer: D

EXAM SHORTCUT

Single bootstrap CI: coverage error $O(n^{-1/2})$; double bootstrap: $O(n^{-1})$; studentized bootstrap also achieves $O(n^{-1})$; computationally expensive

Hypothesis Testing

15 forecast questions in this topic

Hypothesis Testing

Simple and Composite Hypotheses, Two Kinds of Error, Size and Power

Easy

Q1.

The power of a test at an alternative $\theta \in H_1$ is defined as

- A. $\beta(\theta) = P_{\theta}(\text{accept } H_0)$
- B. $\alpha = P(\text{reject } H_0 / H_0 \text{ true})$
- C. $1 - \beta(\theta) = P_{\theta}(\text{reject } H_0)$
- D. $1 - \alpha$

Correct answer: C

EXAM SHORTCUT

Power is the probability of correctly rejecting H_0 when the alternative is true - i.e. 1 minus the Type II error probability at that alternative.

Hypothesis Testing

Neyman-Pearson Lemma and Best Critical Region

Medium

Q2.

X is a single observation from $\text{Exponential}(\theta)$ with pdf $f(x, \theta) = \theta e^{-\theta x}$, $x > 0$. To test $H_0: \theta = 1$ against $H_1: \theta = 2$, the test rejects H_0 if $X < 0.5$. The size and power of this test are respectively

- A. 0.50, 0.50
- B. 0.63, 0.39
- C. 0.39, 0.39
- D. 0.39, 0.63

Correct answer: D

EXAM SHORTCUT

Size = $P(X < 0.5 / \theta = 1) = 1 - e^{-0.5} \approx 0.39$; power = $P(X < 0.5 / \theta = 2) = 1 - e^{-1} \approx 0.63$.

Hypothesis Testing

Neyman-Pearson Lemma and Best Critical Region

Medium

Q3.

For $X \sim \text{Poisson}(\theta)$, testing $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$ with $\theta_1 > \theta_0$, the Neyman-Pearson best critical region is of the form

- A. $X \leq c$
- B. $X = c$ exactly
- C. $|X - \theta_0| \geq c$
- D. $X \geq c$

Correct answer: D

EXAM SHORTCUT

Likelihood ratio $f(x; \theta_0) / f(x; \theta_1) = e^{\theta_1 - \theta_0} (\theta_0 / \theta_1)^x$ is decreasing in x since $\theta_0 / \theta_1 < 1$; small ratio (reject H_0) means large x .

Q4.

A family with Monotone Likelihood Ratio (MLR) in a statistic T guarantees the existence of a UMP test for

- A. two-sided hypotheses, $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$
- B. only point-null versus point-alternative tests
- C. **one-sided hypotheses, e.g. $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$**
- D. any hypothesis, one- or two-sided, without exception

Correct answer: C**EXAM SHORTCUT**

MLR guarantees a UMP test exists for ONE-SIDED composite alternatives; for two-sided alternatives no UMP test generally exists, and one instead seeks a UMP-unbiased (UMPU) test.

Q5.

$X \sim \text{Binomial}(10, p)$. To test $H_0: p = 0.5$ vs $H_1: p > 0.5$ at nominal level $\alpha = 0.05$ using the NON-randomized test 'reject H_0 if $X \geq 9$ ', the EXACT size of this test is

- A. 0.0547
- B. 0.1074
- C. **0.0107**
- D. 0.05 exactly

Correct answer: C**EXAM SHORTCUT**

$$P(X \geq 9 \mid p = 0.5) = \frac{\binom{10}{9} + \binom{10}{10}}{2^{10}} = \frac{11}{1024} \approx 0.0107$$
 - discreteness prevents hitting $\alpha = 0.05$ exactly without randomizing.

Q6.

$X \sim \text{Poisson}(\lambda)$. To test $H_0: \lambda = 2$ vs $H_1: \lambda = 3$, the randomized test $\varphi(x) = 1$ if $x \geq 4$; $\varphi(x) = 0.4$ if $x = 3$; $\varphi(x) = 0$ if $x \leq 2$ has size and power respectively approximately

- A. 0.442, 0.215
- B. **0.215, 0.442**
- C. 0.323, 0.577
- D. 0.180, 0.224

Correct answer: B**EXAM SHORTCUT**

Size $= P(X \geq 4 \mid \lambda = 2) + 0.4P(X = 3 \mid \lambda = 2)$; power is the same formula evaluated at $\lambda = 3$.

Q7.

Randomized tests are needed in hypothesis testing primarily because

- A. continuous distributions require randomization to define a critical region
- B. **for discrete distributions, no non-randomized critical region can generally achieve an arbitrary prescribed size α exactly**
- C. they always have strictly higher power than non-randomized tests
- D. randomization removes the need to specify H_1

Correct answer: B

EXAM SHORTCUT

Discreteness of the sample space (Binomial, Poisson, etc.) means the achievable exact sizes form a finite set of values; randomizing at one boundary point lets us hit any α in between exactly.

Q8.

The likelihood ratio test statistic for testing $H_0: \theta \in \Theta_0$ against the general alternative $\theta \in \Theta$ is defined as

- A. **$\lambda = \frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)}$, and H_0 is rejected for small λ**
- B. λ is always ≥ 1
- C. $\lambda = \frac{\sup_{\theta \in \Theta} L(\theta)}{\sup_{\theta \in \Theta_0} L(\theta)}$, and H_0 is rejected for small λ
- D. $\lambda = L(\theta_0)$ alone, ignoring the alternative

Correct answer: A

EXAM SHORTCUT

LRT statistic $\lambda \in (0,1]$ compares the best-fit likelihood restricted to H_0 against the best-fit likelihood over the whole parameter space; small λ signals the data fit H_0 poorly relative to the unrestricted fit.

Q9.

Under H_0 , as $n \rightarrow \infty$, $-2 \log \lambda$ (where λ is the LRT statistic) asymptotically follows

- A. t_{n-1}
- B. $N(0,1)$
- C. an F distribution
- D. **χ^2_r , where r is the number of restrictions imposed by H_0**

Correct answer: D

EXAM SHORTCUT

Wilks' theorem: $-2 \log \lambda \rightarrow d\chi^2_r$ under H_0 , with $r = \dim(\Theta) - \dim(\Theta_0)$, the number of independent restrictions H_0 imposes.

Q10.

A critical region C is called 'similar' with respect to a nuisance parameter if

- A. it has size exactly 0.5
- B. **Its size is the same constant for every value of the nuisance parameter, throughout H_0**
- C. it is necessarily randomized
- D. it contains only sample points that resemble each other

Correct answer: B**EXAM SHORTCUT**

A similar region has rejection probability that does NOT vary with the nuisance parameter anywhere in H_0 - essential when H_0 is composite due to an unknown nuisance parameter.

Q11.

A test φ is said to be UNBIASED for testing H_0 against H_1 if

- A. it has power 1 at every alternative
- B. **Its power at every alternative in H_1 is at least as large as its size**
- C. its power equals exactly α everywhere
- D. it never commits a Type II error

Correct answer: B**EXAM SHORTCUT**

Unbiasedness of a test requires the power function to be $\geq \alpha$ throughout H_1 and $\leq \alpha$ throughout H_0 - the test should never be 'more likely' to reject under a false alternative than under some point of H_0 itself.

Q12.

The power function of a test φ , $\beta(\theta) = E_\theta[\varphi(X)]$, is defined

- A. as the constant $1-\alpha$ everywhere
- B. only at a single point, $\theta = \theta_0$
- C. only for $\theta \in H_1$
- D. **For every θ in the full parameter space, both under H_0 and H_1**

Correct answer: D**EXAM SHORTCUT**

The power function $\beta(\theta) = P_\theta(\text{reject } H_0)$ is defined over the ENTIRE parameter space: under H_0 it equals the (nominal) size, and under H_1 it equals the power.

Q13.

$X \sim N(\mu, 25)$. A sample of size $n=25$ is used to test $H_0: \mu=50$ against $H_1: \mu=53$ at $\alpha=0.05$ (one-sided, using $z_{0.05}=1.645$). The power of the test is approximately

- A. 0.99
- B. 0.05
- C. 0.50
- D. **0.91**

Correct answer: D

EXAM SHORTCUT

Critical value $\bar{x}=50+1.645(5/5)=51.645$; power $=P(\bar{X}>51.645/\mu=53)=P(Z>-1.355)\approx 0.91$.

Q14.

For $X \sim N(\mu, \sigma^2)$ with σ^2 known, testing $H_0: \mu \leq \mu_0$ against $H_1: \mu > \mu_0$, the UMP test rejects H_0 when

- A. $|\bar{X} - \mu_0| > z_{\alpha/2} \sigma / \sqrt{n}$
- B. $\bar{X} < \mu_0 - z_{\alpha} \sigma / \sqrt{n}$
- C. $\bar{X} = \mu_0$
- D. **$\bar{X} > \mu_0 + z_{\alpha} \sigma / \sqrt{n}$**

Correct answer: D

EXAM SHORTCUT

The standard one-sided UMP test for a normal mean (variance known) rejects for large $\bar{X}$, with the threshold set by the UPPER-tail critical value z_{α} .

Q15.

For testing equality of two normal population variances, $H_0: \sigma_1^2 = \sigma_2^2$, using independent samples of sizes n_1, n_2 , the statistic S_1^2/S_2^2 under H_0 follows

- A. **F_{n_1-1, n_2-1}**
- B. $t_{n_1+n_2-2}$
- C. $\chi^2_{n_1+n_2-2}$
- D. $N(0,1)$

Correct answer: A

EXAM SHORTCUT

The ratio of two independent sample variances (each scaled chi-square) from normal populations with equal true variance follows the F distribution with numerator and denominator df equal to each sample's error df.

Simple and Composite Hypotheses, Two Kinds of Error, Size and Power

1 forecast questions in this topic

Hypothesis Testing

Simple and Composite Hypotheses, Two Kinds of Error, Size and Power

Medium

Q1.

For a composite null hypothesis H_0 , the size of a test with critical region C is defined as

- A.** $\sup_{\theta \in H_0} P_{\theta}(C)$
- B.** the average of $P_{\theta}(C)$ over H_0
- C.** the Type II error probability
- D.** $\inf_{\theta \in H_0} P_{\theta}(C)$

Correct answer: A

EXAM SHORTCUT

For a composite null, size is the SUPREMUM (worst case) of the rejection probability over all $\theta \in H_0$ - guaranteeing the Type I error never exceeds this bound anywhere in H_0 .

UMP, MLR, Unbiased and Similar Tests

1 forecast questions in this topic

Hypothesis Testing

UMP, MLR, Unbiased and Similar Tests

Medium

Q1.

For testing $H_0:\theta=\theta_0$ vs $H_1:\theta\neq\theta_0$ in a one-parameter exponential family, since no UMP test exists in general, one instead seeks a

- A. test with power exactly 0.5 everywhere
- B. $\checkmark$ UMP Unbiased (UMPU) test**
- C. test most powerful against a single chosen alternative only
- D. randomized test only

Correct answer: B

EXAM SHORTCUT

The standard resolution to 'no UMP test exists for two-sided alternatives' is to restrict attention to UNBIASED tests and find the best one within that class - the UMPU test.

Sequential and Decision Theory

8 forecast questions in this topic

Hypothesis Testing

Wald's SPRT, OC and ASN Functions

Medium

Q1.

In Wald's SPRT for testing H_0 vs H_1 with Type I error α and Type II error β , the approximate stopping bounds A (lower) and B (upper) for the likelihood ratio satisfy

A. $A=\alpha, B=\beta$

B. ✓ $A \approx \frac{\beta}{1-\alpha}, B \approx \frac{1-\beta}{\alpha}$

C. $A \approx \frac{\alpha}{1-\beta}, B \approx \frac{1-\alpha}{\beta}$

D. $A \approx \frac{1-\beta}{\alpha}, B \approx \frac{\beta}{1-\alpha}$

Correct answer: B

EXAM SHORTCUT

Wald's bounds: continue sampling while $A < LR < B$, with $A \approx \beta/(1-\alpha)$ (lower) and $B \approx (1-\beta)/\alpha$ (upper) - getting these two the wrong way round is the classic slip.

Hypothesis Testing

Wald's SPRT, OC and ASN Functions

Medium

Q2.

In SPRT, the Operating Characteristic function $L(\theta)$ gives the probability of accepting H_0 . At $\theta=\theta_0$ (i.e. under H_0), $L(\theta_0)$ approximately equals

A. ✓ $1-\alpha$

B. $1-\beta$

C. α

D. β

Correct answer: A

EXAM SHORTCUT

$L(\theta_0)$ is the probability of (correctly) accepting H_0 when H_0 is true, which is 1 minus the Type I error rate, i.e. $1-\alpha$.

Q3.

The Average Sample Number (ASN) of Wald's SPRT, compared to a fixed-sample-size Neyman-Pearson test achieving the SAME α, β , is generally

- A. always larger than the fixed-sample test
- B. **✗smaller - SPRT is more efficient on average, often needing roughly half the expected sample size**
- C. exactly equal to the fixed sample size
- D. infinite

Correct answer: B**EXAM SHORTCUT**

SPRT's key advantage (Wald-Wolfowitz optimality) is achieving the same (α, β) error rates with a SMALLER expected sample size than any fixed-sample test.

Q4.

In decision-theoretic estimation, the risk function $R(\theta, \delta)$ of a decision rule δ is defined as

- A. the maximum possible loss over all θ
- B. **✗ $E_{\theta}[L(\theta, \delta(X))]$**
- C. $\text{Var}_{\theta}[\delta(X)]$
- D. $L(\theta, \delta(X))$ for a single realization of X

Correct answer: B**EXAM SHORTCUT**

Risk is the EXPECTED loss, averaged over the sampling distribution of X at each fixed θ - not the loss for one particular realization.

Q5.

A decision rule that minimizes the MAXIMUM risk over all θ is called

- A. admissible
- B. UMVUE
- C. Bayes
- D. **✗minimax**

Correct answer: D**EXAM SHORTCUT**

'Minimax' literally means minimizing the maximum (worst-case) risk across the parameter space - a conservative, distribution-free criterion.

Q6.

A decision rule δ is INADMISSIBLE if

- A. it has the single highest risk at exactly one value of θ
- B. it is unbiased
- C. **There exists δ' with $R(\theta, \delta') \leq R(\theta, \delta)$ for every θ , and strictly less for at least one θ**
- D. it is a randomized rule

Correct answer: C

EXAM SHORTCUT

Inadmissibility means another rule DOMINATES it: never worse anywhere, and strictly better somewhere - so there is no reason to ever use the dominated rule.

Q7.

Under squared error loss, the Bayes estimator of θ given data x is

- A. the maximum likelihood estimate
- B. the prior mean, ignoring the data
- C. the posterior mode
- D. **the posterior mean $E[\theta \mid x]$**

Correct answer: D

EXAM SHORTCUT

For squared error loss specifically, the Bayes estimator (minimizer of posterior expected loss) is exactly the posterior mean - a standard, frequently-tested result.

Q8.

Under Wald's SPRT, if θ is the true parameter value (with $\theta_0 \neq \theta_1$ and finite, nonzero boundaries A, B), the probability that the procedure eventually terminates with a decision is

- A. α
- B. **1 - the SPRT terminates with probability 1**
- C. β
- D. 0

Correct answer: B

EXAM SHORTCUT

A fundamental SPRT result: as long as $\theta_0 \neq \theta_1$, the procedure terminates (reaches a decision) with probability 1, regardless of the true θ .

Neyman–Pearson Lemma, UMP Tests & Power Functions

50 forecast questions in this topic

Hypothesis Testing

NP Lemma Statement

Medium

Q1.

Neyman-Pearson lemma: Most powerful test of size α for $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1$ rejects when:

- A. $X > k$ always
- B. $\Lambda = f(x; \theta_0)/f(x; \theta_1) > k$
- C. $\Lambda = f(x; \theta_1)/f(x; \theta_0) > k$ where k chosen so $P_{H_0}(\Lambda > k) = \alpha$
- D. $p\text{-value} < \alpha$

Correct answer: C

EXAM SHORTCUT

Ratio is f_1/f_0 (alternative over null); reject for LARGE ratio; B has ratio inverted; k determined by size constraint

Hypothesis Testing

UMP Exponential Family

Hard

Q2.

For one-parameter exponential family with monotone likelihood ratio in $T(x)$, UMP test of $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$:

- A. Rejects for large $T(x)$
- B. Rejects for small $T(x)$
- C. Rejects for $|T(x)| > k$
- D. Depends on direction of monotonicity

Correct answer: D

EXAM SHORTCUT

If LR is increasing in T : reject for large T ; if decreasing: reject for small T ; must check direction

Hypothesis Testing

UMP Existence Two Sided

Hard

Q3.

UMP test for two-sided $H_1: \theta \neq \theta_0$ against one-sided $H_0: \theta = \theta_0$:

- A. Generally does NOT exist
- B. Exists only for normal distribution
- C. Exists when distribution is symmetric
- D. Always exists for exponential families

Correct answer: A

EXAM SHORTCUT

UMP for two-sided alternatives generally fails; the test that is MP against $\theta_1 > \theta_0$ differs from MP against $\theta_1 < \theta_0$; UMPU tests exist instead

Q4.

For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known, MP test of $H_0: \mu = \mu_0$ vs $H_1: \mu = \mu_1 > \mu_0$:

A. Reject if $\bar{X} > \mu_1$

B. **Reject if $\bar{X} > c$ where $c = \mu_0 + z_{\alpha}\sigma/\sqrt{n}$**

C. Reject if $\bar{X} < c$

D. Reject if $|\bar{X} - \mu_0| > c$

Correct answer: B

EXAM SHORTCUT

LR = $\exp[n(\mu_1 - \mu_0)\bar{X}/\sigma^2 - \text{const}]$; increasing in $\bar{X}$ (since $\mu_1 > \mu_0$); reject for large $\bar{X}$; c from size constraint

Q5.

Power function $\beta(\theta)$ of a test is:

A. $P(\text{accept } H_0)$ for $\theta \in H_1$

B. $P(\text{Type I error})$ under H_0

C. **$P(\text{reject } H_0)$ for all θ**

D. $1 - P(\text{Type II error})$ only under H_1

Correct answer: C

EXAM SHORTCUT

Power function defined for ALL θ ; under H_0 : $\beta(\theta) = \alpha$ (size); under H_1 : $\beta(\theta) = \text{power}$; D is incomplete

Q6.

For testing $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$ (normal, σ known), critical region is:

A. $\{\bar{X} < \mu_0 - z_{\alpha}\sigma/\sqrt{n}\}$

B. **$\{|\bar{X} - \mu_0| > z_{\alpha/2}\sigma/\sqrt{n}\}$**

C. $\{\bar{X} > \mu_0 + z_{\alpha}\sigma/\sqrt{n}\}$

D. $\{(\bar{X} - \mu_0)^2 > z^2_{\alpha}\sigma^2/n\}$

Correct answer: B

EXAM SHORTCUT

Two-sided test: B and D are equivalent; C is one-sided right; A is one-sided left; for two-sided, use $z_{\alpha/2}$ not z_{α}

Q7.

A critical region C is similar of size α if:

A. $P_{\theta}(X \in C) = \alpha$ for all $\theta \in H_0$

B. $P_{\theta}(X \in C) = \alpha$ for all θ

C. $P_{\theta}(X \in C) \geq \alpha$ for all $\theta \in H_1$

D. $P_{\theta}(X \in C) \leq \alpha$ for all $\theta \in H_0$

Correct answer: A

EXAM SHORTCUT

Similar region: EXACTLY α for all $\theta \in H_0$ (not just $\leq \alpha$); level α test has $\leq \alpha$; similar tests are useful for composite H_0

Q8.

A test is unbiased if:

A. Type I error equals Type II error

B. $\beta(\theta) = \alpha$ for all θ

C. Power equals size

D. $\beta(\theta) \geq \alpha$ for all $\theta \in H_1$ and $\beta(\theta) \leq \alpha$ for all $\theta \in H_0$

Correct answer: D

EXAM SHORTCUT

Unbiased: power under H_1 is at least as large as size; ensures test is not “worse than random”; B is similar test; A is unrelated

Q9.

A randomized test with critical function $\phi(x)$ means:

A. Randomly choose between two tests

B. Randomly permute data

C. **Reject with probability $\phi(x) \in [0, 1]$**

D. Use random sample size

Correct answer: C

EXAM SHORTCUT

Randomized test: $\phi(x) = P(\text{reject} | X = x)$; needed to achieve exact size α for discrete distributions; $\phi(x) \in \{0, 1\}$ for non-randomized

Q10.For $X \sim \text{Poisson}(\lambda)$, MP test of $H_0: \lambda = \lambda_0$ vs $H_1: \lambda = \lambda_1 > \lambda_0$:

- A. Reject for $X = 0$
- B. **Reject for large X (i.e., $X \geq c$)**
- C. Reject for $|X - \lambda_0| > c$
- D. Reject for small X

Correct answer: B**EXAM SHORTCUT**LR = $(\lambda_1/\lambda_0)^x e^{-(\lambda_1 - \lambda_0)}$; increasing in x (since $\lambda_1 > \lambda_0$); reject for large X ; randomization at boundary for exact size**Q11.**For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with μ known, UMP test of $H_0: \sigma^2 \leq \sigma_0^2$ vs $H_1: \sigma^2 > \sigma_0^2$:

- A. Reject for small $\sum (X_i - \mu)^2$
- B. **Reject for large $\sum (X_i - \mu)^2$**
- C. Reject for large $|X - \mu|$
- D. UMP does not exist

Correct answer: B**EXAM SHORTCUT** $\sum (X_i - \mu)^2 / \sigma^2 \sim \chi^2_n$; MLR in $\sum (X_i - \mu)^2$; reject for large values; μ known is crucial**Q12.**For $H_0: \mu = 0$ vs $H_1: \mu = \delta > 0$ with σ known, power of z-test at level α is:

- A. **Both A and C (equivalent)**
- B. $\Phi(-z_\alpha + \delta/\sigma)$
- C. $1 - \Phi(z_\alpha - \delta/\sigma)$
- D. $\Phi(z_\alpha - \delta/\sigma)$

Correct answer: A**EXAM SHORTCUT**Power = $P(X > z_\alpha \sigma / \sqrt{n} | \mu = \delta) = P(Z > z_\alpha - \delta/\sigma) = 1 - \Phi(z_\alpha - \delta/\sigma) = \Phi(\delta/\sigma - z_\alpha)$; B and C equivalent

Q13.

For $X_1, \dots, X_n \sim \text{Exp}(\theta)$, MP test of $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1 < \theta_0$ (smaller rate = larger mean):

A. ~~Both A and C (equivalent)~~

B. Reject for large $\bar{X}$

C. Reject for small $\sum X_i$

D. Reject for large $\sum X_i$

Correct answer: A

EXAM SHORTCUT

LR = $(\theta_1/\theta_0)^n \exp[-(\theta_1 - \theta_0) \sum x_i]$; since $\theta_1 < \theta_0$: $-(\theta_1 - \theta_0) > 0$; LR increasing in $\sum x_i$; reject for large $\sum X_i$

Q14.

For $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$ in exponential family with MLR in $T(x)$:

A. UMP does not exist for composite H_0

B. UMP test rejects for $T(x) < c$

C. UMP rejects for $|T(x)| > c$

D. ~~UMP test rejects for $T(x) > c$ where c satisfies $\sup_{\theta \leq \theta_0} P_\theta(T > c) = \alpha$~~

Correct answer: D

EXAM SHORTCUT

MLR property: test that is MP for each simple $\theta_1 > \theta_0$ is the same test; this common test is UMP; size achieved at $\theta = \theta_0$ (boundary)

Q15.

When $P_{H_0}(\Lambda > k) = \alpha_1 < \alpha < \alpha_2 = P_{H_0}(\Lambda \geq k)$, the NP test randomizes at $\Lambda = k$ with probability:

A. $\gamma = \alpha/\alpha_2$

B. $\gamma = \alpha$

C. $\gamma = (\alpha_2 - \alpha)/(\alpha_2 - \alpha_1)$

D. ~~$\gamma = (\alpha - \alpha_1)/(\alpha_2 - \alpha_1)$~~

Correct answer: D

EXAM SHORTCUT

Randomization probability γ chosen so total size = $\alpha_1 + \gamma(\alpha_2 - \alpha_1) = \alpha$; solving gives D; needed for discrete distributions

Q16.

For UMP test in exponential family with MLR, power function $\beta(\theta)$ is:

- A. Constant
- B. **Monotone increasing in θ (for one-sided $H_1: \theta > \theta_0$)**
- C. Monotone decreasing
- D. Non-monotone (U-shaped)

Correct answer: B

EXAM SHORTCUT

MLR implies monotone power function; this is the key property that makes the test UMP; power increases as θ moves further into H_1

Q17.

A test is most powerful of level α (not size α) if:

- A. It minimizes Type II error
- B. $\beta(\theta_0) = \alpha$ exactly
- C. **$\sup_{\theta \in H_0} \beta(\theta) \leq \alpha$ and power is maximized**
- D. Power equals $1 - \beta$ for all $\theta \in H_1$

Correct answer: C

EXAM SHORTCUT

Level α : size $\leq \alpha$ (inequality); size α : size = α (equality); MP of level α may have size $< \alpha$ (conservative)

Q18.

For $X \sim U(0, \theta)$, testing $H_0: \theta = 1$ vs $H_1: \theta = 2$, MP test:

- A. Rejects if $X < c$
- B. Always rejects (since H_1 support contains H_0 support)
- C. Rejects if $X > 1$ (impossible under H_0)
- D. **Rejects if $X > c$ for some $c \in (0, 1)$**

Correct answer: D

EXAM SHORTCUT

$f(x; 2)/f(x; 1) = 1/2$ for $x \in (0, 1)$ and ∞ for $x \in (1, 2)$; always reject if $X > 1$; for $X \leq 1$, LR = $1/2$ (constant); randomize to achieve exact size α

Q19.

For $X_1, \dots, X_n \sim \text{Bernoulli}(p)$, UMP test of $H_0: p \leq p_0$ vs $H_1: p > p_0$:

A. UMP does not exist

B. **Reject for large $\sum X_i$**

C. Reject for $|\sum X_i - np_0| > c$

D. Reject for small $\sum X_i$

Correct answer: B

EXAM SHORTCUT

Bernoulli is exponential family with MLR in $\sum X_i$; UMP rejects for large $\sum X_i$; critical value from $\text{Bin}(n, p_0)$ distribution

Q20.

The power envelope is:

A. Power of the UMP test

B. Maximum achievable power at each $\theta \in H_1$ over all level- α tests

C. **Both A and B when UMP exists**

D. Power of the likelihood ratio test

Correct answer: C

EXAM SHORTCUT

Power envelope = upper bound on power; when UMP exists, it achieves the envelope; when UMP doesn't exist, no single test achieves the envelope everywhere

Q21.

NP lemma directly applies to:

A. Simple H_0 vs composite H_1

B. Composite H_0 vs composite H_1

C. **Simple H_0 vs simple H_1 only**

D. Any hypothesis testing problem

Correct answer: C

EXAM SHORTCUT

NP lemma: both hypotheses must be simple (point hypotheses); extension to composite uses UMP theory (MLR, exponential family); LRT handles general composite

Q22.

For composite $H_0: \theta \in \Theta_0$, similar tests of size α are constructed using:

- A. Rao-Blackwell theorem
- B. Bootstrap
- C. **Neyman structure: $E\theta[\phi(X)|T(X)] = \alpha$ for sufficient statistic T under H_0**
- D. Likelihood ratio

Correct answer: C

EXAM SHORTCUT

Neyman structure: conditional on sufficient statistic for nuisance parameter, test has constant conditional size α ; this gives similar tests

Q23.

UMPU test for $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$ in exponential family:

- A. **Rejects for $T < c_1$ or $T > c_2$ with c_1, c_2 chosen for size and unbiasedness**
- B. Does not exist
- C. Is the same as the two-sided z-test always
- D. Rejects for $|T - \theta_0| > c$

Correct answer: A

EXAM SHORTCUT

UMPU: two critical values $c_1 < c_2$; conditions: $E\theta_0[\phi] = \alpha$ (size) and $E\theta_0[T\phi] = \alpha E\theta_0[T]$ (unbiasedness); for normal, reduces to equal-tails z-test

Q24.

To achieve power $1 - \beta$ at $\mu = \mu_1$ for one-sided z-test at level α , required n is:

- A. **$n = \sigma^2(z_\alpha + z_\beta)^2 / (\mu_1 - \mu_0)^2$**
- B. $n = \sigma^2(z_\alpha - z_\beta)^2 / (\mu_1 - \mu_0)^2$
- C. $n = \sigma^2 z^2_{1-\alpha} / (\mu_1 - \mu_0)^2$
- D. $n = \sigma^2(z_{\alpha/2} + z_\beta)^2 / (\mu_1 - \mu_0)^2$

Correct answer: A

EXAM SHORTCUT

Power = $\Phi(\delta/\sigma - z_\alpha) = 1 - \beta$; so $\delta/\sigma = z_\alpha + z_\beta$; squaring: A; D uses $z_{\alpha/2}$ (two-sided)

Q25.

When sufficient statistic T exists, NP lemma critical region can be expressed in terms of T because:

- A. T is complete
- B. T has minimum variance
- C. **Likelihood ratio is a function of T only**
- D. T is ancillary

Correct answer: C

EXAM SHORTCUT

By factorization theorem, $f(x; \theta) = g(T; \theta)h(x)$; LR = $g(T; \theta_1)/g(T; \theta_0)$; depends on x only through T

Q26.

For $X \sim \text{Beta}(\theta, 1)$, testing $H_0: \theta = 1$ vs $H_1: \theta = 2$, power of MP test at level α :

- A. α^2
- B. $2\alpha - \alpha^2$
- C. $1 - \alpha^2$
- D. **$1 - (1 - \alpha)^2$**

Correct answer: D

EXAM SHORTCUT

$f(x; \theta) = \theta x^{\theta-1}$; LR = $2x/1 = 2x$; reject for large x ; $P(X > c | \theta = 1) = 1 - c = \alpha$; $c = 1 - \alpha$; power = $P(X > 1 - \alpha | \theta = 2) = 1 - (1 - \alpha)^2$

Q27.

For scale family $f(x)/(\theta f(x/\theta))$ with f log-concave, UMP test of $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$:

- A. Rejects for large $X(n)$ (if f is uniform)
- B. **Depends on specific f**
- C. Rejects for large $\sum X_i$ (if f is exponential)
- D. Does not exist in general

Correct answer: B

EXAM SHORTCUT

Sufficient statistic and rejection region depend on the specific scale family; exponential: $\sum X_i$; uniform: $X(n)$; no universal answer

Q28.

For composite $H_0: \mu \geq \mu_0$ vs $H_1: \mu < \mu_0$ (normal, σ known), level- α test:

- A. Rejects for $\bar{X} < \mu_0 - z_{\alpha/2}\sigma/\sqrt{n}$
- B. Rejects for $\bar{X} > \mu_0 + z_{\alpha}\sigma/\sqrt{n}$
- C. **Rejects for $\bar{X} < \mu_0 - z_{\alpha}\sigma/\sqrt{n}$**
- D. Rejects for $|\bar{X} - \mu_0| > z_{\alpha/2}\sigma/\sqrt{n}$

Correct answer: C

EXAM SHORTCUT

One-sided left test; size achieved at $\mu = \mu_0$ (boundary of H_0); use z_{α} not $z_{\alpha/2}$; B is right-sided; D is two-sided

Q29.

Power function $\beta(\theta)$ of a test based on continuous data is:

- A. Continuous only for exponential families
- B. May be discontinuous for discrete distributions
- C. Always differentiable
- D. **Always continuous in θ**

Correct answer: D

EXAM SHORTCUT

For continuous data, power function is continuous (and often differentiable) in θ ; for discrete data, power function may have jumps; D is correct for continuous case

Q30.

For multiparameter family, NP lemma:

- A. **Applies directly to simple vs simple hypotheses**
- B. Gives UMP test always
- C. Does not apply
- D. Requires sufficient statistic

Correct answer: A

EXAM SHORTCUT

NP lemma applies to any simple vs simple hypothesis regardless of dimension; the LR test is MP; extension to composite is harder in multiparameter case

Q31.

For $X_1, \dots, X_n \sim \text{Exp}(\theta)$ (rate), UMP test of $H_0: \theta = \theta_0$ vs $H_1: \theta > \theta_0$:

- A. UMP does not exist
- B. Reject for large $\sum X_i$
- C. **Reject for small $\sum X_i$ (large rate = small mean)**
- D. Reject for large $\bar{X}$

Correct answer: C

EXAM SHORTCUT

Large rate θ means small mean $1/\theta$; LR increasing in $-\sum X_i$ (or decreasing in $\sum X_i$); reject for SMALL $\sum X_i$; common sign error

Q32.

Power function of two-sided z-test for $H_0: \mu = 0$ vs $H_1: \mu \neq 0$:

- A. **$\beta(\mu) = \Phi(-z\alpha/2 - \mu\sqrt{n}/\sigma) + 1 - \Phi(z\alpha/2 - \mu\sqrt{n}/\sigma)$**
- B. $\beta(\mu) = 1 - \Phi(z\alpha/2 - \mu\sqrt{n}/\sigma)$
- C. $\beta(\mu) = \Phi(\mu\sqrt{n}/\sigma - z\alpha/2)$
- D. $\beta(\mu) = 2\Phi(\mu\sqrt{n}/\sigma - z\alpha/2) - 1$

Correct answer: A

EXAM SHORTCUT

Two-sided: power = P(reject left) + P(reject right); for $\mu > 0$: left tail negligible; A is exact; C and B are one-sided approximations

Q33.

In exponential family $f(x; \theta, \lambda) = h(x) \exp[\eta_1(\theta)T_1 + \eta_2(\lambda)T_2 - B]$, similar test for $H_0: \theta = \theta_0$ (nuisance λ) conditions on:

- A. (T_1, T_2) jointly
- B. Neither
- C. T_1 (sufficient for θ)
- D. **T_2 (sufficient for λ)**

Correct answer: D

EXAM SHORTCUT

Condition on sufficient statistic for nuisance parameter λ ; conditional distribution of $T_1|T_2$ is free of λ ; gives similar test

Q34.

For $X_1, \dots, X_n \sim \text{Laplace}(\theta, 1)$, UMP test of $H_0: \theta = 0$ vs $H_1: \theta > 0$:

- A. Rejects for large sample median
- B. Rejects for large $\sum X_i$
- C. **UMP does not exist**
- D. Rejects for large $\sum |X_i|$

Correct answer: C

EXAM SHORTCUT

Laplace is NOT exponential family (absolute value in exponent); no MLR in simple statistic; UMP does not exist; sign test or Wilcoxon are used

Q35.

For $X \sim \text{Bin}(10, p)$, test $H_0: p = 0.5$ vs $H_1: p > 0.5$, reject if $X \geq 8$. Power at $p = 0.7$:

- A. $1 - P(X \leq 7 | p = 0.7)$
- B. **Both A and C (equivalent)**
- C. $P(X \geq 8 | p = 0.5)$
- D. $P(X \geq 8 | p = 0.7) = C(10, 8)(0.7)^8(0.3)^2 + C(10, 9)(0.7)^9(0.3) + C(10, 10)(0.7)^{10}$

Correct answer: B

EXAM SHORTCUT

Power at $p = 0.7$: compute under H_1 ; D and A are equivalent expressions; C is the size (under H_0)

Q36.

For $X_1, \dots, X_n \sim \Gamma(\alpha, \theta)$ with α known, MP test of $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1 < \theta_0$:

- A. Both A and C (equivalent)
- B. Reject for large $\sum X_i$
- C. Reject for large $\bar{X}$
- D. **Reject for small $\sum X_i$**

Correct answer: D

EXAM SHORTCUT

$\Gamma(\alpha, \theta)$ with scale θ : $f \propto \theta^{-\alpha} \exp(-\sum x_i / \theta)$; LR = $(\theta_0 / \theta_1)^\alpha \exp[-\sum x_i (1/\theta_1 - 1/\theta_0)]$; since $\theta_1 < \theta_0$: $1/\theta_1 > 1/\theta_0$; LR decreasing in $\sum x_i$; reject for SMALL $\sum X_i$

Q37.

For $N(\mu, \sigma^2)$ with σ^2 unknown, UMP test of $H_0: \mu \leq \mu_0$ vs $H_1: \mu > \mu_0$:

- A. UMP does not exist (nuisance σ^2)
- B. **Rejects for large $t = (X - \mu_0)/(S/\sqrt{n})$**
- C. Rejects for large $\bar{X}$
- D. Rejects for large $\bar{X}/S$

Correct answer: B**EXAM SHORTCUT**

With nuisance σ^2 , UMP doesn't exist in general; BUT the t -test is UMPU (uniformly most powerful unbiased); B is UMPU, not UMP; A is technically correct for UMP

Q38.

For one-sided test in exponential family, power function is strictly increasing because:

- A. **MLR property implies power is monotone in θ**
- B. Sufficient statistic is complete
- C. MLE is consistent
- D. CRLB is attained

Correct answer: A**EXAM SHORTCUT**

MLR $\Rightarrow$ monotone power function; this is the fundamental connection between MLR and UMP tests; completeness is separate

Q39.

For composite $H_1: \theta > \theta_0$ in exponential family, UMP test is found by:

- A. Rao-Blackwell theorem
- B. Maximizing power at $\theta_1 = \theta_0 + 1$
- C. **Applying NP lemma to each simple $H_1: \theta = \theta_1$ and noting the MP test is the same for all $\theta_1 > \theta_0$**
- D. Using LRT

Correct answer: C**EXAM SHORTCUT**

Key insight: MLR ensures the NP critical region (reject for large T) is the same for ALL $\theta_1 > \theta_0$; this common test is UMP

Q40.

For $X \sim \text{Bin}(5, p)$, testing $H_0: p = 0.5$ at level $\alpha = 0.05$, exact size requires:

- A. Reject if $X \geq 4$ (size = $6/32 > 0.05$, liberal)
- B. Reject if $X \geq 5$ (size = $1/32 < 0.05$, conservative)
- C. **Randomization at boundary since $P(X \geq 5 | p = 0.5) = 1/32 < 0.05 < P(X \geq 4 | p = 0.5) = 6/32$**
- D. Use normal approximation

Correct answer: C

EXAM SHORTCUT

Exact size 0.05 requires randomization: reject if $X = 5$; reject with prob γ if $X = 4$; $\gamma = (0.05 - 1/32)/(6/32 - 1/32) = (0.05 - 0.03125)/0.15625$

Q41.

For $X_1, \dots, X_n \sim \text{NegBin}(r, p)$ with r known, UMP test of $H_0: p \geq p_0$ vs $H_1: p < p_0$:

- A. Reject for small $\sum X_i$
- B. Reject for large $\bar{X}$
- C. Reject for large $\sum X_i$
- D. **Both A and C (equivalent)**

Correct answer: D

EXAM SHORTCUT

NegBin: $E[X] = r(1-p)/p$; smaller p means larger mean; LR in $\sum X_i$; for $p_1 < p_0$: reject for large $\sum X_i$; C and B equivalent

Q42.

Power envelope at θ_1 for testing $H_0: \theta = \theta_0$ at level α is:

- A. **All of the above (equivalent)**
- B. Power of NP MP test at θ_1
- C. $1 - \beta$ where β is Type II error of best test
- D. $\sup_{\theta \in H_0} \text{level-}\alpha \text{ tests } \beta(\theta_1)$

Correct answer: A

EXAM SHORTCUT

All three definitions are equivalent; NP lemma gives the test achieving the envelope; power envelope is the theoretical maximum

Q43.

For $N(\mu, \sigma^2)$ with both unknown, similar test of $H_0: \mu = \mu_0$ conditions on:

A. $X_{(1)}$ B. $\sum X_i^2$ C. $\bar{X}$ D. S^2 (sufficient for σ^2 under H_0)**Correct answer: D****EXAM SHORTCUT**

Under $H_0: \mu = \mu_0$, S^2 is sufficient for σ^2 ; conditioning on S^2 gives t -statistic with $n-1$ distribution free of σ^2 ; this gives the t -test as UMPU

Q44.

For $X_1, \dots, X_n \sim \text{Pareto}(\theta)$, $f(x; \theta) = \theta/x^{\theta+1}$, $x > 1$, UMP test of $H_0: \theta \leq \theta_0$ vs $H_1: \theta > \theta_0$:

A. Reject for large $\prod X_i$ B. Both B and C (equivalent since $\log \prod = \sum \log$)C. Reject for large $\sum \log X_i$ D. ~~Reject for small~~ $\sum \log X_i$ **Correct answer: D****EXAM SHORTCUT**

$\sum \log X_i \sim \Gamma(n, 1/\theta)$; larger θ means smaller $\sum \log X_i$; MLR in $-\sum \log X_i$; reject for small $\sum \log X_i$ (= large $-\sum \log X_i$)

Q45.

For $X \sim \text{Beta}(\theta, 1)$, UMP test of $H_0: \theta \leq 1$ vs $H_1: \theta > 1$ rejects for:

A. Small $\sum \log X_i$ (more negative)B. Large $\sum X_i$ C. ~~Large~~ $\sum \log X_i$ (less negative, closer to 0)D. Large $X(n)$ **Correct answer: C****EXAM SHORTCUT**

$\log X_i < 0$ for $X_i \in (0, 1)$; $-\log X_i \sim \text{Exp}(\theta)$; large θ means small $E[-\log X_i] = 1/\theta$; so large θ means $\sum \log X_i$ closer to 0 (less negative); reject for large (less negative) $\sum \log X_i$

Q46.

For location family $f(x-\theta)$ with f symmetric and log-concave, MP test of $H_0: \theta = 0$ vs $H_1: \theta = \delta > 0$:

A. ~~Depends on specific f~~

B. Rejects for large $\sum X_i$ (if f is normal)

C. Always rejects for large $\bar{X}$

D. Rejects for large $\sum \text{sgn}(X_i)$ (if f is Laplace)

Correct answer: A

EXAM SHORTCUT

MP test depends on f ; normal: reject for large $\bar{X}$; Laplace: reject for large $\sum \text{sgn}(X_i)$ (sign test); Cauchy: complex; C is wrong in general

Q47.

For Hypergeometric(N, M, n) with N, n known, UMP test of $H_0: M \leq M_0$ vs $H_1: M > M_0$:

A. Reject for small X

B. ~~Reject for large X~~ (number of successes in sample)

C. Reject for $X > M_0 n / N$

D. UMP does not exist (not exponential family)

Correct answer: B

EXAM SHORTCUT

Hypergeometric has MLR in X ; UMP exists and rejects for large X ; not exponential family but MLR still holds

Q48.

For composite $H_0: \theta \in [0, 1]$ vs $H_1: \theta > 1$, size of test is:

A. $\beta(1)$

B. $\inf_{\theta \in [0, 1]} \beta(\theta)$

C. $\beta(0)$

D. ~~$\sup_{\theta \in [0, 1]} \beta(\theta)$~~

Correct answer: D

EXAM SHORTCUT

Size = supremum of power function over H_0 ; for monotone power function, supremum achieved at boundary $\theta = 1$; so $D = \beta(1)$ in this case

Q49.

If T is sufficient and NP critical region is $\{x: \Lambda(x) > k\}$, then equivalently:

A. $\nexists x: g(T(x)) > k'$ for some function g and constant k'

B. The critical region cannot be simplified

C. $\{x: X > k'\}$ always

D. $\{x: T(x) > k'\}$ always

Correct answer: A

EXAM SHORTCUT

LR is a function of T (by factorization); so critical region is a function of T ; D is true only if LR is monotone in T ; A is the general statement

Q50.

UMP test exists for $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$ (two-sided) when:

A. Distribution is from exponential family

B. $\nexists$ Never (for two-sided alternatives in general)

C. Distribution is normal

D. Distribution has MLR

Correct answer: B

EXAM SHORTCUT

UMP for two-sided alternatives generally does NOT exist; UMPU exists for exponential families; this is a fundamental limitation of UMP theory

Likelihood Ratio Tests, SPRT & Decision Theory

55 forecast questions in this topic

Hypothesis Testing

LRT Definition

Medium

Q1.

Likelihood ratio test statistic Λ is:

A. ~~Both A and C (equivalent)~~

B. $\Lambda = \sup_{\theta \in \Theta_0} L(\theta) / \sup_{\theta \in \Theta} L(\theta)$

C. $\Lambda = \sup_{\theta \in \Theta} L(\theta) / \sup_{\theta \in \Theta_0} L(\theta)$

D. $\Lambda = L(\theta_0) / L(\hat{\theta})$ where θ_0 is restricted MLE

Correct answer: A

EXAM SHORTCUT

$\Lambda \in [0, 1]$; reject for SMALL Λ (equivalently large $-2 \log \Lambda$); C is ≥ 1 always; B and D are equivalent

Hypothesis Testing

LRT Wilks Theorem

Medium

Q2.

Under H_0 and regularity conditions, $-2 \log \Lambda \xrightarrow{d} :$

A. ~~χ^2_r where $r = \dim(\Theta) - \dim(\Theta_0)$~~

B. χ^2_n

C. $F_{r, n-r}$

D. $N(0, 1)$

Correct answer: A

EXAM SHORTCUT

Wilks' theorem: $r =$ number of restrictions (difference in parameter dimensions); for testing k parameters: $r = k$; requires regularity conditions

Hypothesis Testing

LRT Normal Mean Variance

Hard

Q3.

For $N(\mu, \sigma^2)$ with both unknown, LRT of $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$:

A. Reduces to z-test

B. Has no closed form

C. ~~Reduces to t-test: reject for large $|t| = |\bar{X} - \mu_0| / (S/\sqrt{n})$~~

D. Reduces to χ^2 test

Correct answer: C

EXAM SHORTCUT

LRT statistic: $-2 \log \Lambda = n \log[1 + t^2/(n-1)]$ which is monotone in $|t|$; LRT is equivalent to t -test; $r = 1$ df

Q4.

Wald's SPRT continues sampling while:

- A. $\Lambda m > A$
- B. $\Lambda m < B$
- C. $B < \Lambda m < A$ where $\Lambda m = \prod_{i=1}^m f(X_i; \theta_1)/f(X_i; \theta_0)$
- D. $m < n_0$ (fixed sample size)

Correct answer: C

EXAM SHORTCUT

SPRT: continue if $B < \Lambda m < A$; stop and reject H_0 if $\Lambda m \geq A$; stop and accept H_0 if $\Lambda m \leq B$

Q5.

Wald's approximate SPRT boundaries are:

- A. $A = 1 - \beta, B = \alpha$
- B. $A = 1/\alpha, B = \beta$
- C. $A \approx (1 - \beta)/\alpha, B \approx \beta/(1 - \alpha)$
- D. $A \approx (1 - \alpha)/\beta, B \approx \alpha/(1 - \beta)$

Correct answer: C

EXAM SHORTCUT

 $A = (1 - \beta)/\alpha$ (upper boundary for rejecting H_0); $B = \beta/(1 - \alpha)$ (lower boundary for accepting H_0); these are approximate (ignore overshoot)

Q6.

For $X_i \sim N(\theta, 1)$, SPRT of $H_0: \theta = 0$ vs $H_1: \theta = \delta$, log-LR after m observations:

- A. $\delta X_m - \delta^2/2$
- B. $\delta \sum_{i=1}^m X_i - m\delta^2/2$
- C. $\delta \sum_{i=1}^m X_i$
- D. $\sum_{i=1}^m X_i^2$

Correct answer: B

EXAM SHORTCUT

 $\log \Lambda m = \sum [\log f(X_i; \delta) - \log f(X_i; 0)] = \delta \sum X_i - m\delta^2/2$; linear in $\sum X_i$; SPRT reduces to boundary on $\sum X_i$

Q7.

Operating Characteristic (OC) function $L(\theta)$ of SPRT is:

- A. $P_{\theta}(\text{reject } H_0)$
- B. $1 - P_{\theta}(\text{accept } H_0)$
- C. $E_{\theta}[N]$ (expected sample size)
- D. Power function

Correct answer: B

EXAM SHORTCUT

OC function = probability of accepting H_0 ; $L(\theta_0) = 1 - \alpha$; $L(\theta_1) = \beta$; power function = $1 - L(\theta)$; C is ASN function

Q8.

ASN (Average Sample Number) function $E_{\theta}[N]$ for SPRT satisfies Wald's identity:

- A. $E_{\theta}[N] = n(\text{fixed})$
- B. $E_{\theta}[N] \approx \frac{1}{L(\theta) \log(B/A) + (1-L(\theta)) \log(A/B)} E_{\theta}[\log f(X; \theta_1)/f(X; \theta_0)]$ (Wald's approximation)
- C. $E_{\theta}[N] = \log(A/B)/E_{\theta}[\log \Lambda_1]$
- D. $E_{\theta}[N] = 1/\alpha$

Correct answer: B

EXAM SHORTCUT

Wald's approximation for ASN uses OC function $L(\theta)$ and expected log-LR per observation; C is a special case; exact formula involves overshoot correction

Q9.

Wald-Wolfowitz theorem states SPRT is optimal in that:

- A. SPRT minimizes maximum expected sample size
- B. Among all tests with error probabilities $\leq \alpha$ and $\leq \beta$, SPRT minimizes $E_{\theta_0}[N]$ and $E_{\theta_1}[N]$ simultaneously
- C. SPRT is UMP among sequential tests
- D. SPRT minimizes total error probability

Correct answer: B

EXAM SHORTCUT

Wald-Wolfowitz: SPRT minimizes ASN at BOTH θ_0 and θ_1 simultaneously; no other test with same error bounds can have smaller ASN at either point

Q10.

LRT for $H_0: \theta \in \Theta_0$ vs $H_1: \theta \in \Theta_1 = \Theta \setminus \Theta_0$:A. $\Lambda = L(\theta_0)/L(\theta)$; reject for small Λ B. $\Lambda = \sup_{\Theta_1} L / \sup_{\Theta_0} L$; reject for large Λ C. **Both A and B (equivalent)**D. $\Lambda = \sup_{\Theta_0} L / \sup_{\Theta} L$; reject for small Λ

Correct answer: C

EXAM SHORTCUT

D and A are equivalent; $\sup_{\Theta} L = \max(\sup_{\Theta_0} L, \sup_{\Theta_1} L)$; when $\theta \notin \Theta_0$, $\sup_{\Theta} L = L(\theta)$; reject for small Λ (large $-2 \log \Lambda$)

Q11.

In decision theory, risk function $R(\theta, d)$ is:A. $\sup_{\theta} L(\theta, d(X))$ B. $L(\theta, d(X))$ — loss functionC. **$E_{\theta}[L(\theta, d(X))]$ — expected loss**D. $P_{\theta}(\text{wrong decision})$

Correct answer: C

EXAM SHORTCUT

Risk = expected loss; loss function is random (depends on X); risk is its expectation; D is risk for 0-1 loss only

Q12.

Estimator d_1 dominates d_2 if:A. **$R(\theta, d_1) \leq R(\theta, d_2)$ for all θ with strict inequality for some θ** B. $\sup_{\theta} R(\theta, d_1) < \sup_{\theta} R(\theta, d_2)$ C. $R(\theta, d_1) \leq R(\theta, d_2)$ for all θ D. $R(\theta, d_1) < R(\theta, d_2)$ for all θ

Correct answer: A

EXAM SHORTCUT

Dominance: $\leq$ everywhere AND $<$ somewhere; D is too strong (strict everywhere); C allows equality everywhere (no dominance); B is minimax comparison

Q13.

Minimax estimator minimizes:

- A. $\int R(\theta, d)\pi(\theta)d\theta$ — Bayes risk
- B. $R(\theta_0, d)$ at a specific θ_0
- C. $E\theta[d(X)]$ — expected estimate
- D. $\sup_{\theta} R(\theta, d)$ — worst-case risk

Correct answer: D

EXAM SHORTCUT

Minimax: minimize maximum risk; Bayes: minimize average risk (B); minimax estimator is often Bayes w.r.t. least favorable prior

Q14.

Bayes estimator under squared error loss is:

- A. Posterior median
- B. Posterior mode (MAP)
- C. Prior mean
- D. Posterior mean $E[\theta|X]$

Correct answer: D

EXAM SHORTCUT

Squared error: posterior mean; absolute error: posterior median; 0-1 loss: posterior mode; these are fundamental results

Q15.

For $X_i \sim \text{Poisson}(\lambda)$, SPRT of $H_0: \lambda = \lambda_0$ vs $H_1: \lambda = \lambda_1 > \lambda_0$, log-LR:

- A. $-X \log(\lambda_1/\lambda_0)$
- B. $m(\lambda_1 - \lambda_0)$
- C. $\sum X_i \log(\lambda_1/\lambda_0)$
- D. $-\sum X_i \log(\lambda_1/\lambda_0) - m(\lambda_1 - \lambda_0)$

Correct answer: D

EXAM SHORTCUT

$\log \Lambda = \sum [x_i \log(\lambda_1/\lambda_0) - (\lambda_1 - \lambda_0)] = \sum X_i \log(\lambda_1/\lambda_0) - m(\lambda_1 - \lambda_0)$; linear in $\sum X_i$; SPRT boundary on $\sum X_i$

Q16.

LRT for independence in $r \times c$ contingency table: $-2 \log \Lambda \xrightarrow{d} \chi^2_{(r-1)(c-1)}$ because:)

A. Degrees of freedom = rc

B. Wilks theorem gives χ^2_{-1} always

C. $\dim(\Theta) = rc - 1$, $\dim(\Theta_0) = (r-1) + (c-1)$; $r = (rc-1) - (r+c-2) = (r-1)(c-1)$

D. Degrees of freedom = $r+c-2$

Correct answer: C

EXAM SHORTCUT

Saturated model: $rc-1$ free parameters; independence model: $(r-1) + (c-1)$ free parameters; difference = $(r-1)(c-1)$; this is the df for χ^2

Q17.

Bayes risk of estimator d w.r.t. prior π is:

A. $r(\pi, d) = E[L(\theta, d(X))]$ over joint distribution

B. $r(\pi, d) = \int R(\theta, d) \pi(\theta) d\theta = E\pi[R(\theta, d)]$

C. $r(\pi, d) = R(\theta_0, d)$ at prior mean

D. $r(\pi, d) = \sup_{\theta} R(\theta, d)$

Correct answer: A

EXAM SHORTCUT

B and A are equivalent (both are expectation of loss over joint distribution of (θ, X)); D is minimax risk; Bayes estimator minimizes B=A

Q18.

For $X_i \sim \text{Exp}(\theta)$ (rate), SPRT of $H_0: \theta = \theta_0$ vs $H_1: \theta = \theta_1 > \theta_0$, SPRT boundary on $\sum X_i$:

A. Accept H_0 if $\sum X_i > b$, reject if $\sum X_i < a$ (for some $a < b$)

B. Reject for small $\sum X_i$

C. Reject for large $\sum X_i$

D. Accept H_0 if $\sum X_i < a$, reject if $\sum X_i > b$

Correct answer: A

EXAM SHORTCUT

Large θ (rate) means small mean; log-LR = $n \log(\theta_1/\theta_0) - (\theta_1 - \theta_0) \sum X_i$; decreasing in $\sum X_i$; large $\sum X_i$ favors H_0 (small rate); accept H_0 for large $\sum X_i$

Q19.

LRT for $H_0: \sigma^2 = \sigma_0^2$ vs $H_1: \sigma^2 \neq \sigma_0^2$ (normal, μ unknown):

- A. Has no closed form
- B. Reduces to Z-test
- C. Reduces to F -test
- D. **Reduces to χ^2 test: reject for large or small $(n-1)S^2/\sigma_0^2$**

Correct answer: D**EXAM SHORTCUT**

LRT statistic: $-2 \log \Lambda = n[\log(\hat{\sigma}^2/\sigma_0^2) + \hat{\sigma}^2/\sigma_0^2 - 1]$ which is monotone function of $((n-1)S^2/\sigma_0^2)$; equivalent to χ^2 test

Q20.

For SPRT of $H_0: \theta = 0$ vs $H_1: \theta = \delta$ with $X_i \sim N(\theta, 1)$, OC function $L(\theta)$ satisfies:

- A. $L(\theta) = \Phi(-\theta/\sqrt{n})$
- B. **$L(\theta) = A h(\theta) - 1$ where $h(\theta)$ solves $E[h(\theta)] = 1$**
- C. $L(\theta) = \beta$ for all $\theta \in H_1$
- D. $L(\theta) = 1 - \Phi(\theta/\sqrt{n})$

Correct answer: B**EXAM SHORTCUT**

Wald's OC approximation uses moment generating function of log-LR; $h(\theta)$ is the non-trivial root of $E[\Lambda(h)] = 1$; complex but standard formula

Q21.

Complete class theorem states:

- A. Every admissible estimator is Bayes (or limit of Bayes)
- B. Admissible estimators form a complete class
- C. Every Bayes estimator is admissible
- D. **Both A and B**

Correct answer: D**EXAM SHORTCUT**

C: Bayes estimators with proper priors are admissible; A: admissible estimators are Bayes (or generalized Bayes); together they characterize admissibility via Bayes

Q22.

LRT for $H_0: \beta_1 = \dots = \beta_k = 0$ in multiple regression (normal errors):

- A. Has no closed form
- B. Reduces to χ^2 -test
- C. Reduces to t -test
- D. **Reduces to F -test: $F = (RSS_0 - RSS_1) / (k \sigma^2)$**

Correct answer: D

EXAM SHORTCUT

LRT = $-2 \log \Lambda = n \log(RSS_0 / RSS_1)$; for normal errors, this is monotone in F -statistic; LRT and F -test are equivalent

Q23.

For $X_i \sim \text{Bernoulli}(p)$, SPRT of $H_0: p = p_0$ vs $H_1: p = p_1 > p_0$, log-LR:

- A. $m \cdot \log[(1-p_1)/(1-p_0)]$
- B. **$-\sum X_i \log[p_1(1-p_0)/(p_0(1-p_1))] + m \cdot \log[(1-p_1)/(1-p_0)]$**
- C. $\bar{X} \cdot \log(p_1/p_0) + \log[(1-p_1)/(1-p_0)]$
- D. $\sum X_i \log(p_1/p_0)$

Correct answer: B

EXAM SHORTCUT

 $\log \Lambda_m = \sum X_i \log(p_1/p_0) + \sum (1-X_i) \log[(1-p_1)/(1-p_0)] = \sum X_i \log[p_1(1-p_0)/(p_0(1-p_1))] + m \cdot \log[(1-p_1)/(1-p_0)]$; linear in $\sum X_i$.

Q24.

For $X \sim N(\theta, 1)$, minimax estimator of θ under squared error loss is:

- A. 0 (constant)
- B. **$\bar{X}$ (since it is admissible and has constant risk)**
- C. James-Stein estimator
- D. $\bar{X}$ (sample mean)

Correct answer: B

EXAM SHORTCUT

For $p = 1$: $\bar{X}$ is minimax (constant risk = $1/n$, admissible); for $p \geq 3$: James-Stein dominates $\bar{X}$ in MSE but $\bar{X}$ is still minimax; C is wrong for $p = 1$

Q25.

LRT for goodness of fit (multinomial): $G^2 = -2 \log \Lambda = 2 \sum O_i \log(O_i/E_i)$ $df = k - 1 - p$:

- A. χ^2_k
- B. $F_{k-1, n-k}$
- C. χ^2_{k-1-p} where k = cells, p = estimated parameters
- D. χ^2_{k-1}

Correct answer: C

EXAM SHORTCUT

G^2 (log-likelihood ratio) and Pearson χ^2 are asymptotically equivalent; $df = k - 1 - p$ (subtract 1 for total constraint and p for estimated parameters)

Q26.

Minimax estimator equals Bayes estimator w.r.t. least favorable prior π^* when:

- A. Prior is uniform
- B. Bayes risk equals minimax risk: $r(\pi^*, d^*) = \sup_{\theta} R(\theta, d^*)$
- C. Loss function is squared error
- D. Sample size is large

Correct answer: B

EXAM SHORTCUT

Least favorable prior: prior that maximizes Bayes risk; if Bayes estimator has constant risk, it is minimax; key connection between Bayes and minimax

Q27.

SPRT has smaller ASN than fixed sample test because:

- A. It uses sufficient statistics
- B. It has smaller error probabilities
- C. It stops early when evidence is strong (in either direction)
- D. It uses more information per observation

Correct answer: C

EXAM SHORTCUT

SPRT's efficiency comes from early stopping; when data strongly favors one hypothesis, no need to continue; fixed sample test always uses n observations

Q28.

For exponential family, LRT of $H_0: \eta = \eta_0$ vs $H_1: \eta \neq \eta_0$:

A. $-2 \log \Lambda = 2 \sum \log f(X_i; \hat{\eta}) / f(X_i; \eta_0)$

B. **All equivalent asymptotically**

C. $-2 \log \Lambda = n(\hat{\eta} - \eta_0)' I(\eta_0) (\hat{\eta} - \eta_0)$

D. $-2 \log \Lambda = 2n[B(\hat{\eta}) - B(\eta_0) - (\hat{\eta} - \eta_0)' \nabla B(\eta_0)]$ where B is log-partition

Correct answer: B

EXAM SHORTCUT

All three are asymptotically equivalent (Wald, LRT, Score); D is exact LRT; C is Wald; they differ in finite samples

Q29.

Regret of decision d at θ is:

A. $R(\theta, d)$ — total risk

B. $E[L(\theta, d(X))]$

C. **$R(\theta, d) - \inf_{d'} R(\theta, d')$ — excess risk over best possible**

D. $\sup_{\theta} R(\theta, d) - R(\theta, d)$

Correct answer: C

EXAM SHORTCUT

Regret = how much worse than oracle (who knows θ); minimax regret minimizes worst-case regret; different from minimax risk

Q30.

LRT for $H_0: \mu_1 = \mu_2$ (two independent normal samples, equal σ^2 unknown):

A. **Reduces to two-sample t -test**

B. Has no closed form

C. Reduces to Z-test

D. Reduces to F -test

Correct answer: A

EXAM SHORTCUT

LRT statistic is monotone in t^2 ; equivalent to two-sample t -test with pooled variance; $r = 1$ df for Wilks

Q31.

For $X_i \sim \text{Geometric}(p)$, SPRT of $H_0: p = p_0$ vs $H_1: p = p_1 < p_0$, log-LR:

A. $-\sum X_i \log[(1-p_1)/(1-p_0)] + m \cdot \log(p_1/p_0)$

B. $\sum X_i \log(p_1/p_0)$

C. $m \cdot \log(p_1/p_0)$

D. $\bar{X} \cdot \log[(1-p_1)/(1-p_0)]$

Correct answer: A

EXAM SHORTCUT

$\log f(x;p) = \log p + (x-1)\log(1-p)$; $\log \Lambda_m = m \cdot \log(p_1/p_0) + \sum (X_i - 1) \log[(1-p_1)/(1-p_0)] = \sum X_i \log[(1-p_1)/(1-p_0)] + m \cdot \log(p_1/p_0)$.

Q32.

James-Stein estimator $\hat{\theta}_{JS} = (1 - \frac{p-2}{n\|\bar{X}\|^2})\bar{X}$ for $p \geq 3$:

A. Is minimax

B. Dominates $\bar{X}$ and is admissible

C. Is admissible

D. Dominates $\bar{X}$ in total MSE but is inadmissible itself

Correct answer: D

EXAM SHORTCUT

James-Stein dominates $\bar{X}$ for $p \geq 3$ (Stein's paradox); but JS itself is inadmissible — the positive-part JS estimator $\hat{\theta}_{JS+} = \max(0, 1 - (p-2)/(n\|\bar{X}\|^2))\bar{X}$ dominates JS.

Q33.

LRT for homogeneity of k binomial proportions $H_0: p_1 = \dots = p_k$:

A. $G^2 \xrightarrow{d} \chi^2_{2k-2}$

B. $G^2 \xrightarrow{d} F_{k-1, n-k}$

C. $G^2 = 2 \sum_{i,j} O_{ij} \log(O_{ij}/E_{ij}) \xrightarrow{d} \chi^2_{k-1}$

D. $G^2 \xrightarrow{d} \chi^2_k$

Correct answer: C

EXAM SHORTCUT

k proportions, 1 constraint (common p): $r = k-1$ df; same as Pearson χ^2 for $2 \times k$ table

Q34.

Truncated SPRT (with maximum sample size $n_{\max}$):

- A. Is always more efficient than SPRT
- B. Is equivalent to fixed sample test
- C. Has same error probabilities as SPRT
- D. **Stops at $n_{\max}$ and makes decision based on accumulated evidence**

Correct answer: D

EXAM SHORTCUT

Truncated SPRT: if boundaries not crossed by $n_{\max}$, make decision at $n_{\max}$; error probabilities change slightly; practical compromise between SPRT and fixed sample

Q35.

Posterior expected loss (posterior risk) of decision d given $X = x$ is:

- A. $\rho(\pi, d|x) = \int L(\theta, d)\pi(\theta)d\theta$
- B. $\rho(\pi, d|x) = R(\theta, d)$
- C. $\rho(\pi, d|x) = \int L(\theta, d)\pi(\theta|x)d\theta$
- D. **$\rho(\pi, d|x) = E[L(\theta, d)|X = x]$**

Correct answer: D

EXAM SHORTCUT

C and D are equivalent (posterior expectation of loss); A uses prior not posterior; Bayes estimator minimizes posterior risk for each x

Q36.

For nested models $M_0 \subset M_1$, LRT statistic $-2 \log \Lambda$:

- A. **All of the above**
- B. Is always non-negative
- C. Equals $2[\ell(\theta_1) - \ell(\theta_0)]$ where ℓ is log-likelihood
- D. Has χ^2 distribution under M_0

Correct answer: A

EXAM SHORTCUT

All three: B (log-likelihood increases with more parameters); C (definition); D (Wilks theorem); all hold under regularity

Q37.

For composite hypotheses $H_0: \theta \leq \theta_0$ vs $H_1: \theta \geq \theta_1$ ($\theta_0 < \theta_1$), SPRT:

- A. Does not exist
- B. **Uses $\Lambda = \prod f(X_i; \theta_1)/f(X_i; \theta_0)$ (simple vs simple at boundaries)**
- C. Uses generalized likelihood ratio
- D. Uses MLE at each stage

Correct answer: B

EXAM SHORTCUT

For composite hypotheses with indifference zone (θ_0, θ_1) , SPRT uses simple hypotheses at boundaries; error probabilities guaranteed for all $\theta \leq \theta_0$ and $\theta \geq \theta_1$

Q38.

Invariant decision rules under group G of transformations satisfy:

- A. $d(gx) = g'd(\theta)$ for all $g \in G$ (equivariance)
- B. **Both A and C**
- C. Risk is constant over orbits of G
- D. $d(gx) = d(x)$ for all $g \in G$

Correct answer: B

EXAM SHORTCUT

Equivariant estimators have constant risk over orbits; best equivariant estimator minimizes risk within equivariant class; Pitman estimator is best location equivariant

Q39.

Bartlett correction for LRT improves χ^2 approximation by:

- A. Adding b/n to $-2 \log \Lambda$
- B. Using χ^2_{r+1} instead of χ^2_r
- C. Multiplying by $n/(n-r)$
- D. **Dividing $-2 \log \Lambda$ by $1 + b/n$ where b is Bartlett factor**

Correct answer: D

EXAM SHORTCUT

Bartlett: $(-2 \log \Lambda)/(1 + b/n) \sim \chi^2_r$ to order $O(n^{-2})$; improves from $O(n^{-1})$ error; b computed from cumulants of log-likelihood

Q40.

SPRT efficiency relative to fixed sample test (same error probabilities) is measured by:

- A. α/β
- B. Power ratio
- C. $E\theta[N]/n$
- D. $n/E\theta[N]$ — ratio of fixed to expected sequential sample size

Correct answer: D

EXAM SHORTCUT

Efficiency > 1 means SPRT uses fewer observations on average; for normal testing, SPRT efficiency $\approx 50\%$ savings at H_0 and H_1 ; efficiency < 1 would mean SPRT is worse

Q41.

Under 0-1 loss $L(\theta, d) = I(\theta \neq d)$, Bayes estimator is:

- A. Posterior mode (MAP)
- B. Prior mode
- C. Posterior median
- D. Posterior mean

Correct answer: A

EXAM SHORTCUT

0-1 loss: minimize $P(\theta \neq d|X) = 1 - P(\theta = d|X)$; maximize $P(\theta = d|X)$; this is the posterior mode; for continuous θ , 0-1 loss requires careful definition

Q42.

LRT for $H_0: \mu = \mu_0$ in $Np(\mu, \Sigma)$ with Σ known:

- A. $-2 \log \Lambda = n(X - \mu_0)^T S^{-1} (X - \mu_0) \sim \chi^2_p$
- B. $-2 \log \Lambda = (X - \mu_0)^T \Sigma^{-1} (X - \mu_0) \sim \chi^2_p$
- C. $-2 \log \Lambda = n(X - \mu_0)^T \Sigma^{-1} (X - \mu_0) \sim \chi^2_p$
- D. $-2 \log \Lambda \sim F_{p, n-p}$

Correct answer: C

EXAM SHORTCUT

$n(X - \mu_0)^T \Sigma^{-1} (X - \mu_0) = T^2$ (Hotelling's T^2 when Σ known); $\sim \chi^2_p$; A uses S (unknown Σ case) giving $T^2 \sim (p(n-1))/(n-p) F_{p, n-p}$

Q43.

For $X_i \sim N(0, \sigma^2)$, SPRT of $H_0: \sigma^2 = \sigma_0^2$ vs $H_1: \sigma^2 = \sigma_1^2 > \sigma_0^2$, log-LR:

- A. $m \cdot \log(\sigma_1/\sigma_0)$
- B. $\sum X_i^2/\sigma^2$
- C. $\sum X_i^2(1/\sigma_0^2 - 1/\sigma_1^2)$
- D. $(m/2)\log(\sigma_0^2/\sigma_1^2) + (\sum X_i^2)(1/\sigma_0^2 - 1/\sigma_1^2)$

Correct answer: D

EXAM SHORTCUT

$\log \Lambda_m = \sum [\log f(X_i; \sigma_1^2) - \log f(X_i; \sigma_0^2)] = (m/2)\log(\sigma_0^2/\sigma_1^2) + (\sum X_i^2)(\sigma_0^{-2} - \sigma_1^{-2})$; linear in $\sum X_i^2$.

Q44.

Shrinkage estimators are preferred in high dimensions because:

- A. They reduce total MSE by trading bias for variance reduction
- B. They attain CRLB
- C. They are minimax
- D. They are always unbiased

Correct answer: A

EXAM SHORTCUT

Shrinkage: introduce bias but reduce variance; total MSE decreases; Stein paradox shows this is beneficial for $p \geq 3$; C: James-Stein is minimax but not all shrinkage estimators are

Q45.

LRT for $H_0: \beta = 0$ in Poisson regression $\log \mu_i = \alpha + \beta x_i$:

- A. $-2 \log \Lambda \xrightarrow{d} N(0, 1)$
- B. $-2 \log \Lambda \xrightarrow{d} \chi^2_2$
- C. $-2 \log \Lambda = 2[\ell(\hat{\alpha}, \hat{\beta}) - \ell(\alpha_0, 0)] \xrightarrow{d} \chi^2_1$
- D. $-2 \log \Lambda \xrightarrow{d} F_{1, n-2}$

Correct answer: C

EXAM SHORTCUT

One restriction ($\beta = 0$): $r = 1$ df; Wilks theorem applies; χ^2_1 ; equivalent to Wald test $(\hat{\beta}/SE)^2$ asymptotically

Q46.

SPRT stopping time N satisfies:

- A. $N \rightarrow \infty$ as $\alpha, \beta \rightarrow 0$
- B. $E\theta[N] < \infty$ for all θ under regularity conditions
- C. ~~Both A and C~~
- D. $N = n_0$ always (fixed)

Correct answer: C

EXAM SHORTCUT

B: SPRT terminates with probability 1 (Wald); A: as error bounds tighten, more observations needed; both are true

Q47.

Empirical Bayes estimates hyperparameters of prior by:

- A. Bootstrap
- B. ~~Maximizing marginal likelihood~~ $m(x|\lambda) = \int f(x|\theta)\pi(\theta|\lambda)d\theta$
- C. Cross-validation
- D. Using flat prior

Correct answer: B

EXAM SHORTCUT

Empirical Bayes: estimate λ from data (marginal likelihood or method of moments); then use posterior with estimated prior; James-Stein is an empirical Bayes estimator

Q48.

Log-rank test for comparing two survival curves is:

- A. ~~Asymptotically equivalent to LRT under proportional hazards~~
- B. Exact LRT for exponential survival
- C. Equivalent to t -test on survival times
- D. Equivalent to Wilcoxon test

Correct answer: A

EXAM SHORTCUT

Log-rank test is score test (not LRT) under Cox model; asymptotically equivalent to LRT; exact LRT for exponential gives different statistic; D: Wilcoxon is different weighting

Q49.

Curtailed sampling (stop when outcome is determined) differs from SPRT in that:

- A. Curtailment uses likelihood ratio boundaries
- B. Curtailment and SPRT are equivalent
- C. Curtailment is optimal
- D. **Curtailment stops when final decision is certain regardless of remaining observations**

Correct answer: D

EXAM SHORTCUT

Curtailment: stop when remaining observations cannot change decision; used in clinical trials; SPRT uses probabilistic boundaries; different stopping rules

Q50.

Posterior odds ratio for H_1 vs H_0 is:

- A. $P(X|H_1)/P(X|H_0)$ (Bayes factor only)
- B. $P(H_1|X)$ alone
- C. **$P(H_1|X)/P(H_0|X) = (P(X|H_1)/P(X|H_0)) \cdot P(H_1)/P(H_0)$ (Bayes factor $\times$ prior odds)**
- D. $1 - p$ -value

Correct answer: C

EXAM SHORTCUT

Posterior odds = Bayes factor $\times$ prior odds; Bayes factor alone (B) ignores prior; p -value is frequentist (D); C is the complete Bayesian update

Q51.

For curved exponential family (e.g., $N(\theta, \theta^2)$), Wilks theorem:

- A. Requires larger sample size
- B. **Still holds: $-2 \log \Lambda \xrightarrow{d} \chi^2_r$**
- C. Fails completely
- D. Holds with modified df

Correct answer: B

EXAM SHORTCUT

Wilks theorem holds for curved exponential families under regularity; r = number of restrictions; the curvature affects finite-sample performance but not asymptotic distribution

Q52.

For testing among $k > 2$ hypotheses sequentially, SPRT generalizes to:

- A. Sequential probability ratio test with $k(k-1)/2$ pairwise boundaries
- B. **Simultaneous SPRT (Armitage)**
- C. Group sequential design
- D. Repeated significance testing

Correct answer: B

EXAM SHORTCUT

Armitage's simultaneous SPRT: continue until one hypothesis is accepted; boundaries involve all k hypotheses; more complex than pairwise

Q53.

For normal location problem with squared error loss, complete class consists of:

- A. All linear estimators
- B. All unbiased estimators
- C. **All Bayes estimators w.r.t. normal priors and their limits**
- D. Only the sample mean

Correct answer: C

EXAM SHORTCUT

For normal location, complete class = Bayes estimators w.r.t. normal priors (conjugate family) and limits; includes James-Stein type estimators; D is too restrictive

Q54.

When true parameter is on boundary of H_0 (e.g., $H_0: \theta \geq 0$, true $\theta = 0$), $-2 \log \Lambda \xrightarrow{d} :$

- A. χ^2_0 (degenerate at 0)
- B. $N(0, 1)$
- C. χ^2_1
- D. **$(1)/(2\chi^2_0) + (1)/(2\chi^2_1)$ (mixture)**

Correct answer: D

EXAM SHORTCUT

Boundary case: $-2 \log \Lambda$ is 0 with prob 1/2 (when $\hat{\theta} < 0$, constrained MLE = 0) and χ^2_1 with prob 1/2; mixture distribution

Q55.

SPRT has power one property: for any $\theta \neq \theta_0$:

- A. $P_{\theta}(\text{SPRT eventually terminates}) = 1$
- B. $P_{\theta}(\text{reject } H_0) \rightarrow 1 \text{ as } n \rightarrow \infty$
- C. **Both A and B**
- D. Neither (SPRT may not terminate)

Correct answer: C

EXAM SHORTCUT

SPRT terminates with probability 1 (A); and is consistent (B); both follow from Wald's fundamental identity and the strong law of large numbers applied to the log-likelihood ratio (incomplete in source — trap explanation cuts off here)